New Methods for Option Pricing Based on the 3/2 Volatility Model and Comparative Analysis

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ABSTRACT

The 3/2 stochastic volatility model, by virtue of its nonlinear diffusion structure, exhibits significant theoretical advantages in capturing extreme asset price volatility and fitting the volatility smile, making it a core tool for derivatives pricing in highly volatile markets. However, existing performance evaluations of pricing methods under this model are largely confined to idealized academic benchmark cases in stable markets, lacking empirical validation in actual extreme market environments. Consequently, the engineering applicability boundaries of these methods remain systematically unclarified. To bridge this gap, this paper systematically implements the Poisson-Gamma approximate stepping, Quasi-Exponential (QE) approximate stepping, Inverse Gaussian (IG) approximate simulation, Gamma approximate simulation, and Fourier-Cosine (COS) method for European option pricing under the 3/2 pure diffusion model for the first time. A unified theoretical system encompassing three paradigms and eleven pricing approaches is established, namely time-stepping conditional Monte Carlo methods, exact and nearly-exact terminal simulation methods, and Fourier-domain pricing methods. The implementation logic, error properties, and convergence behavior of each method are rigorously derived. Based on this framework, seven sets of numerical experiments spanning both normal and extreme parameter regimes are conducted to quantitatively compare pricing accuracy, computational efficiency, and cross-scenario numerical stability, yielding a transparent and hierarchical performance ranking. Furthermore, taking the extreme Bitcoin flash crash of October 2025 as a case study, this paper develops a complete empirical framework consisting of "long-term fixed-parameter time-series estimation —time-varying parameter cross-sectional calibration —pricing method re-evaluation in extreme scenarios." This validates the 3/2 model's adaptability to extreme cryptocurrency market dynamics, identifies the event-driven time-varying patterns of core model parameters throughout the crash cycle, and executes a performance re-evaluation of the pricing methods under genuine extreme market conditions. The findings indicate that the Fourier-Cosine (Fourier-Cos) method is the globally optimal scheme for European option pricing, achieving exceptional pricing precision, numerical stability, and computational efficiency across all scenarios. The Poisson-Gamma and Quasi-Exponential (QE) near-exact stepping methods are the sole robust solutions across all scenarios within the Monte Carlo framework, offering a reliable implementation pathway for pricing path-dependent derivatives and dynamic hedging. Conversely, the nominally "exact" terminal simulation methods can only be theoretically validated under idealized benchmark parameters; they prevalently experience numerical overflows and systematic pricing deviations in extreme market environments, thus lacking suitability for practical engineering applications. This research bridges the existing disconnect between theoretical simulations and real-world markets in 3/2 model pricing literature, establishes method selection guidelines for diverse application scenarios, and provides systematic theoretical support and empirical foundations for derivatives pricing, model calibration, and risk management in highly volatile markets.

KEY WORDS: Option Pricing, 3/2 Volatility Model, Exact Simulation, Almost Exact Simulation, Numerical Methods

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Chapter 1 Introduction

1.1 Research Background

1.1.1 Development Status of Global and Domestic Options Markets

Since the Chicago Board Options Exchange (CBOE) officially listed standardized options in 1973, the global financial derivatives market has undergone a profound evolution from a single product category to a multi-level, highly complex system. In mature international markets, options have evolved beyond being a basic risk hedging instrument to a core vehicle for asset price discovery, tail risk management, and strategic trading. In recent years, amid heightened uncertainty in the global macroeconomy and frequent geopolitical conflicts, institutional investors have increasingly urgent demand for extreme tail risk protection, driving global options trading volume to successive record highs. In particular, the massive popularity of ultra-short-dated products represented by zero-days-to-expiration (0DTE) options in the U.S. equity market has not only profoundly reshaped the microstructure of market liquidity, but also imposed more stringent theoretical challenges on pricing models: the need for instantaneous responsiveness over extremely short time horizons and accurate capture of extreme volatility skew. These developments have increasingly exposed the limitations of traditional constant volatility models and linear stochastic volatility models.

Meanwhile, China's options market, as a critical emerging force in the global derivatives landscape, is in a critical window of leapfrog development. In the exchange-traded market, since the successful pilot launch of the SSE 50ETF options in 2015, China has rapidly built a matrix of equity options covering core broad-based indices including the CSI 300, CSI 500, and CSI 1000, as well as characteristic indices such as the STAR Market 50 and ChiNext Index. In the commodity options space, full coverage has been achieved across non-ferrous metals, ferrous metal industrial chains, agricultural products, and new energy raw materials, providing abundant risk hedging tools for real-economy enterprises and serving as a key financial underpinning for the stable operation of the real economy. In the over-the-counter (OTC) derivatives market, the OTC options market, with the China Securities Quotation System as the core nexus, has grown steadily, with an expanding range of personalized and customized risk solutions that further improve the risk management functions of the multi-tiered capital market.

1.1.2 Evolution of Option Pricing Models and Computational Methods

The evolution of option pricing theory forms the core logical thread of the development of modern financial engineering. Since the seminal Black-Scholes-Merton (BSM) pricing model proposed by Black, Scholes and Merton, the core assumptions of geometric Brownian motion for asset prices and constant volatility have provided a concise and analytically tractable theoretical framework for derivatives pricing, laying the theoretical foundation of modern derivatives markets. However, with the increasing complexity of global financial markets, the constant volatility assumption in the BSM model has systematically diverged from real market characteristics. A large body of empirical evidence shows that the return distributions of global assets universally exhibit significant leptokurtic and fat-tailed features, while the implied volatility of options displays systematic ”volatility smile” and ”volatility skew” patterns across strike prices and maturities. These phenomena are particularly pronounced during market crises and extreme market conditions, driving a paradigm shift in volatility modeling from deterministic to stochastic frameworks.

In the early exploration of stochastic volatility models, the Heston model (also known as the 1/2 model) became the standard benchmark for global derivatives pricing, owing to its elegant characterization of the mean-reverting property of volatility and the closed-form affine expression for the characteristic function of the underlying log-price. Subsequently, to address the complex volatility dynamics in interest rate and foreign exchange markets, the SABR model further enhanced the ability to capture the shape of the implied volatility curve by introducing a stochastic volatility term and a correlation structure between volatility and the underlying price, becoming the dominant model for OTC derivatives pricing. Nevertheless, the traditional 1/2 model exposes inherent theoretical limitations when dealing with the steep implied volatility skew of very short-dated options and explosive jumps in volatility during extreme market conditions. Specifically, its linear drift term struggles to explain the non-linear accelerated mean reversion of volatility during periods of extreme market stress, and fails to fully characterize the fat-tailed distribution of volatility in highly volatile markets.

Against this backdrop, the 3/2 stochastic volatility model has gradually moved into the core research focus of academia and industry. Unlike the Heston model, the 3/2 model assumes that the instantaneous variance process follows a non-linear power-law diffusion process, with a diffusion term proportional to the 3/2 power of the instantaneous variance and a drift term exhibiting a non-linear mean-reverting structure. This non-linear specification endows the model with greater flexibility, enabling it to more responsively reflect volatility clustering effects under extreme market stress and more accurately fit the shape of the volatility smile across maturities and strike prices. As a result, it delivers significantly better fitting performance than traditional affine models in pricing practice across asset classes including equities, foreign exchange, and cryptocurrencies.

Alongside the increasing dimensionality and structural complexity of pricing models, advances in numerical computational methods have simultaneously pushed forward the boundaries of financial engineering practice. Early finite difference methods (FDM), which discretize partial differential equations (PDEs), provided a robust pricing path for American options and other path-dependent derivatives, but often face the challenge of the ”curse of dimensionality” in multi-factor stochastic volatility models, with computational complexity rising exponentially with model dimensions. Monte Carlo simulation, with its strong versatility, has become the mainstream tool for dealing with high-dimensional models and path-dependent exotic options, despite its relatively slow convergence rate and significantly rising computational cost with increasing accuracy requirements. To balance pricing efficiency and accuracy, conditional Monte Carlo methods, by introducing variance reduction techniques, decouple the underlying price process from the volatility process, greatly optimizing the simulation performance and convergence speed of stochastic volatility models. Among these methods, the Inverse Gaussian (IG) almost exact simulation, Gamma almost exact simulation, Poisson-Gamma almost exact stepping, and Quasi-Exponential (QE) almost exact stepping have demonstrated an excellent balance between accuracy and efficiency in affine models such as the Heston model, yet existing research has not systematically introduced them into the pricing framework of the 3/2 pure diffusion model. In recent years, Fourier transform pricing methods based on characteristic functions, combined with numerical algorithms such as the Fast Fourier Transform (FFT) and Fourier-Cosine (COS) method, have enabled high-speed analytical solution of option prices in the complex domain, laying a solid technical foundation for real-time pricing and full-surface calibration of non-affine stochastic volatility models including the 3/2 model. However, the existing literature still lacks a systematic performance benchmarking of the COS method and various almost exact simulation methods under the 3/2 model, especially tests of full-scenario robustness under extreme market parameters.

1.2 Research Questions

Despite the significant theoretical advantages of the 3/2 model in capturing extreme market volatility and fitting the volatility surface, its non-affine mathematical structure poses severe numerical implementation challenges for option pricing. Existing studies have proposed a variety of option pricing methods under the 3/2 model, covering three major paradigms: time-stepping Monte Carlo simulation, terminal exact/almost exact simulation, and Fourier frequency-domain pricing, forming a total of eleven mainstream implementation schemes. However, four core unresolved issues remain in existing research and practice:

First, existing performance evaluations of various pricing methods under the 3/2 model are largely confined to idealized academic benchmark cases in stable market environments, lacking a systematic examination of the full-scenario numerical stability of these methods. Most nominally "exact" pricing methods can only be theoretically validated under the benchmark parameters specified in the literature. Under extreme parameter combinations such as high volatility of volatility, strong price-volatility correlation, and very short maturities, they commonly suffer from numerical overflow of special functions, distribution fitting failure, and non-numerical (nan/inf) results. The engineering applicability boundaries of these methods have not yet been clearly delineated.

Second, for the almost exact methods including Inverse Gaussian (IG), Gamma almost exact simulation, Poisson-Gamma, and QE almost exact stepping that deliver excellent performance in affine models, existing research has not completed their systematic theoretical derivation, numerical implementation, and full-scenario performance testing under the 3/2 pure diffusion model. Nor has it clarified the superiority boundaries and applicable scenarios of these methods compared to traditional discrete schemes and terminal exact simulation methods, resulting in a lack of implementable theoretical support for the application of these accuracy-and-efficiency-balanced methods in 3/2 model pricing practice.

Third, there is insufficient empirical re-validation of pricing methods in real extreme market environments, as theoretical accuracy under idealized cases does not equate to practical performance in real markets. The real parameters of highly volatile markets such as cryptocurrencies often fall outside the parameter ranges of academic benchmark cases, and structural shifts in model parameters during extreme market conditions can further amplify the performance divergence of different pricing methods. Existing research has not clarified the pricing accuracy, computational efficiency, and robustness of different methods under extreme shocks in real markets, failing to provide reliable guidance for method selection in derivatives pricing and risk management during extreme market conditions.

Fourth, research on the empirical adaptability of the 3/2 model in highly volatile markets remains inadequate, lacking analysis of the dynamic evolution of model parameters over the full cycle of extreme events. Existing studies focus on static parameter calibration and pricing formula derivation of the 3/2 model, and have not yet constructed a complete analytical framework of ”long-term fixed parameter estimation - time-varying parameter cross-sectional calibration” for extreme crash events. They also have not verified the ability of the 3/2 model under the pure diffusion framework to capture the evolution of the volatility smile during extreme market conditions, leaving the practical value of the model in highly volatile markets insufficiently empirically supported.

Accordingly, the core research questions of this paper focus on three dimensions: (1) For the first time, to systematically clarify the theoretical logic, error sources, and convergence properties of eleven mainstream option pricing methods under the 3/2 pure diffusion model, including the IG, Gamma, Poisson-Gamma, QE almost exact methods and the COS method, and to form a clear hierarchical system of method performance through multi-scenario numerical experiments that quantify the pricing accuracy, computational efficiency, and full-scenario numerical stability of different methods; (2) To conduct extreme-scenario re-validation of pricing methods based on model parameters calibrated from real extreme market events, compare the performance differences of methods between idealized cases and real markets, and define the engineering applicability boundaries of different methods; (3) To verify the market adaptability of the 3/2 model in extreme cryptocurrency market conditions, identify the time-varying patterns of the model’s core parameters over the full cycle of extreme events, and provide theoretical support and empirical evidence for derivatives pricing and risk management in highly volatile market environments.

1.3 Research Objectives and Main Content

In response to the core research questions above, this paper takes European option pricing under the 3/2 stochastic volatility model as the core research object. Through systematic research across three dimensions: theoretical construction, numerical experiments, and empirical analysis, it comprehensively evaluates the full-scenario performance of different pricing methods, clarifies their applicable boundaries, and verifies the empirical adaptability of the 3/2 model in extreme market environments. The core research objectives and main content of this paper are divided into the following three aspects:

First, for the first time, to systematically derive and implement the European option pricing schemes of the Inverse Gaussian (IG) almost exact simulation, Gamma almost exact simulation, Poisson-Gamma almost exact stepping, Quasi-Exponential (QE) almost exact stepping method, and the Fourier-Cosine (COS) pricing method under the 3/2 pure diffusion model, and to construct a complete methodological system for European option pricing under the 3/2 model. This paper categorizes the eleven mainstream pricing methods into three major paradigms: time-stepping conditional Monte Carlo methods, terminal exact and almost exact simulation methods, and Fourier frequency-domain pricing methods. It systematically derives the core theoretical logic, numerical implementation procedures, error characteristics, and convergence properties of each method, addressing the issues of fragmented methodological systems, unclear theoretical boundaries, and the application gap of core almost exact methods in the 3/2 model in existing research. This provides a unified theoretical framework and reproducible implementation scheme for subsequent numerical comparison and empirical testing.

Second, to conduct a systematic performance evaluation of different pricing methods through multi-scenario numerical experiments. This paper designs seven sets of test cases covering both conventional market and extreme parameter environments, with numerical stability, pricing accuracy, and computational efficiency as the core evaluation indicators. It focuses on benchmarking the performance differences between the IG, Gamma, Poisson-Gamma, QE almost exact methods and the COS method (first introduced into the 3/2 model in this paper) and traditional Euler/Milstein discrete schemes and terminal exact simulation methods. It quantitatively compares the performance of the eleven methods across different parameter combinations, identifies the advantageous scenarios and core limitations of each method, and forms a clear hierarchical system of method performance. This addresses the problems of one-sided performance evaluation and insufficient robustness testing in extreme scenarios in existing research, and provides quantitative evidence for optimal method selection across different application scenarios.

Third, to complete the empirical calibration of the 3/2 model and extreme-scenario revalidation of pricing methods based on real extreme market events. Taking the Bitcoin market flash crash event in October 2025 as the research sample, this paper constructs a two-step empirical framework of "long-term fixed parameter time-series estimation - time-varying parameter cross-sectional calibration". It identifies the time-varying patterns of the 3/2 model's core parameters over the full crash cycle, and verifies the model's ability to fit the evolution of the volatility smile during extreme market conditions. Meanwhile, based on the real market parameters obtained from calibration, it constructs extreme scenario test cases to conduct performance re-validation of the pricing methods, with a focus on verifying the robustness of the various almost exact methods and the COS method introduced in this paper in real extreme markets. It also conducts a horizontal comparison with the results of academic benchmark cases in Chapter 5, clarifies the core differences in method performance between idealized environments and real markets, and finally forms guidance for method selection in extreme market environments, bridging the gap between theoretical cases and real markets in existing 3/2 model pricing research.

1.4 Structure of the Thesis

This paper consists of seven chapters, organized along the logical thread of ”Theoretical Foundation - Method Construction - Numerical Experiments - Empirical Testing - Conclusion and Outlook”. The core content and structural arrangement of each chapter are as follows:

Chapter 1 is the Introduction. This chapter systematically elaborates the research background and significance of this paper, sorts out the development context of global and domestic options markets and option pricing theory, defines the core research questions, research objectives, and main content of this paper for the 3/2 stochastic volatility model, and introduces the chapter structure of the full thesis.

Chapter 2 is the Literature Review. This chapter systematically reviews the theoretical evolution of stochastic volatility models, the research status of exact simulation pricing methods, summarizes the application research of the 3/2 model in financial derivatives markets and related domestic research status, and finally identifies the gaps in existing research through literature review, establishing the academic positioning and marginal contributions of this study.

Chapter 3 presents the Specification of the 3/2 Stochastic Volatility Model. This chapter details the core theoretical specification of the 3/2 stochastic volatility model, derives the stochastic differential equations for the underlying asset price and instantaneous variance process under the risk-neutral measure, deduces the conditional distribution properties of the model's integrated variance and the moment generating function expression of the underlying price, and clarifies the economic implications of the model's core parameters, laying the theoretical foundation for the construction of subsequent pricing methods and empirical calibration.

Chapter 4 focuses on Option Pricing Methods and Algorithm Implementation. Under the unified theoretical framework of the 3/2 pure diffusion model, this chapter systematically constructs a complete implementation system of eleven option pricing methods across three paradigms. The core innovation lies in the first systematic theoretical derivation and numerical implementation of four almost exact methods (Inverse Gaussian (IG), Gamma almost exact simulation, Poisson-Gamma almost exact stepping, QE almost exact stepping) and the COS pricing method under the 3/2 model. It separately derives the core theoretical formulas of each method, presents the standardized numerical implementation procedure for each method, and conducts theoretical analysis of their error characteristics, convergence properties, and applicable scenarios, providing a unified methodological benchmark for subsequent numerical experiments.

Chapter 5 covers Numerical Experiment Design and Comparative Analysis of Results. This chapter sets a unified numerical experiment environment and evaluation indicator system, designs seven sets of test cases covering conventional market and extreme parameter environments, and systematically completes numerical testing of the eleven pricing methods. It quantitatively compares the numerical stability, pricing accuracy, and computational efficiency of different methods, and finally forms a hierarchical performance system and scenario-based selection recommendations for the methods.

Chapter 6 is Empirical Calibration and Pricing Validation of the 3/2 Stochastic Volatility Model in Extreme Cryptocurrency Market Conditions. Taking the Bitcoin market flash crash event in October 2025 as the research sample, this chapter constructs a complete empirical framework of ”long-term fixed parameter time-series estimation - time-varying parameter cross-sectional calibration - extreme-scenario re-validation of pricing methods”. It identifies the time-varying patterns of the model's core parameters and the model fitting performance over the full crash cycle, verifies the model's ability to capture the evolution of the volatility smile during extreme market conditions. Meanwhile, it completes extreme-scenario performance re-validation of the pricing methods based on real market calibrated parameters, conducts a horizontal comparison with the results of academic benchmark cases in Chapter 5, and defines the applicable boundaries of different pricing methods in extreme market environments, with a focus on verifying the engineering applicability of the various almost exact methods and the COS method introduced in this paper in real extreme markets.

Chapter 7 is Conclusion and Outlook. This chapter systematically summarizes the core research findings of the full thesis, objectively analyzes the limitations of this study, and provides an outlook on future research directions from the dimensions of model extension, method optimization, empirical scope expansion, and risk management applications.

Chapter 2 Literature Review

In the developmental history of stochastic volatility models, the 3/2 model has gradually become a research focal point in the fields of option pricing and risk management within financial engineering, owing to its unique nonlinear diffusion structure and superior ability to capture extreme market volatility. This chapter follows a logical thread of "model paradigm evolution —theoretical framework development —pricing method iteration —market application expansion —domestic research status." It systematically reviews the domestic and international research achievements related to the 3/2 model, ultimately clarifying the core gaps in existing research through a literature summary to establish the academic positioning and marginal contributions of this study.

2.1 Evolution of Stochastic Volatility Models

The evolution of option pricing theory is essentially a process of market participants continuously deepening their understanding of the "non-normalit" of financial asset returns and the "time-varying risk" of volatility. Since the Black-Scholes-Merton (BSM) model Black et al. (1973), Merton et al. (1971) established the basic paradigm of arbitrage-free pricing, its core assumptions—constant volatility and the underlying price following geometric Brownian motion—have provided a concise and efficient analytical framework for derivatives pricing, laying the foundation of modern financial engineering. However, as the complexity of global financial markets continues to escalate, the core assumptions of the BSM model have systematically diverged from true market characteristics. A large number of empirical studies have shown that the return distributions of various global assets universally exhibit significant "heavy-tailed" characteristics. Meanwhile, implied volatility in the options market presents a "volatility smile" varying with strike prices and a "volatility skew" varying with time to maturity Rubinstein (1985). These phenomena are particularly prominent during market crises and extreme conditions, revealing the nonlinear pricing characteristics of the market's tail risk premium and driving a paradigm shift in volatility modeling from determinism to stochasticity.

To correct the theoretical deficiencies of the BSM model, academia first developed local volatility models. Derman et al. (1994) and Dupire (1994) constructed deterministic local volatility frameworks dependent on the underlying price and time from the perspectives of discrete trees and continuous partial differential equations, respectively, which can perfectly fit the implied volatility surface at a specific moment. However, local volatility models fundamentally remain within the realm of deterministic volatility; they cannot characterize the stochastic fluctuation features of volatility itself and possess weak predictive power for future volatility surfaces, making it difficult to meet the practical demands of dynamic hedging and long-term derivatives pricing.

Against this backdrop, Stochastic Volatility (SV) models emerged. Their core concept is to define volatility itself as a stochastic diffusion process, thereby simultaneously characterizing the stochastic fluctuations of the underlying price and the time-varying features of volatility itself. Affine SV Models, represented by Heston (1997), introduced a square-root diffusion process (also known as the 1/2 model) to characterize the mean-reverting property of volatility, deriving an affine closed-form solution for the characteristic function of the underlying log-price. This greatly reduced the computational complexity of model calibration and option pricing, becoming a standard pricing cornerstone in the global derivatives industry. Subsequently, targeting the complex volatility dynamics in interest rate and foreign exchange markets, the SABR model proposed by Hagan et al. (2002) further enhanced the ability to capture the shape of implied volatility curves by introducing a stochastic volatility term and a correlation structure between volatility and the underlying price, becoming the mainstream model for OTC interest rate derivatives pricing.

However, with the evolution of market microstructure and the frequent occurrence of extreme market conditions, the endogenous limitations of the traditional 1/2 model have gradually become apparent. Scholars have found that the Heston model possesses theoretical shortcomings in fitting the steep implied volatility of ultra-short-term options, and its linear drift term struggles to explain the explosive growth and nonlinear mean-reverting characteristics of volatility during extreme crises (Jones, 2003). This drove academia to transition from affine frameworks to non-affine frameworks. As a typical representative of non-affine SV models, the 3/2 model features a diffusion term proportional to the instantaneous variance raised to the power of 3/2, and a drift term exhibiting a nonlinear mean-reverting structure. This stronger nonlinearity endows the model with a more flexible kurtosis characterization and tail risk capture capability. Consequently, when dealing with scenarios involving severe underlying price jumps and highly clustered volatility, it demonstrates a deeper theoretical foundation and superior empirical performance compared to the traditional 1/2 model.

2.2 Development of the 3/2 Model and Its Extended Frameworks

Since Lewis (2000) systematically derived the fundamental analytical properties of the 3/2 stochastic volatility model, academic research has extensively explored the theoretical framework optimization, analytical property expansion, and numerical implementation schemes of the 3/2 model, forming a relatively complete system of theoretical achievements.

In terms of constructing the model's fundamental analytical framework, Carr et al. (2007) developed a novel pricing framework by modeling the variance swap rate directly. This bypassed the complex process of modeling the market price of instantaneous volatility risk, providing a completely new implementation path for the risk-neutral pricing of the 3/2 model. Addressing the option pricing and calibration challenges brought by the 3/2 model's non-affine nature, Medvedev et al. (2007) proposed a comprehensive analytical framework based on short-maturity asymptotic expansions. They derived a closed-form expansion for implied volatility under non-affine local volatility models, providing a core theoretical tool for the rapid cross-sectional calibration of non-affine models, including the 3/2 model, and serving as a crucial foundation for subsequent empirical research on the 3/2 model.

Regarding the extension and optimization of the model structure, to achieve consistent modeling of spot equities and VIX derivatives, Baldeaux et al. (2012b) introduced a simultaneous jump term into the pure diffusion 3/2 model, constructing the "3/2 plus jumps model" to further enhance the model's ability to characterize extreme volatility in severe markets. Grasselli (2017), in their pioneering research, proposed the "4/2 stochastic volatility model," which defines the diffusion term of the instantaneous variance as a linear combination of a 1/2 term and a 3/2 term. This organically integrated the Heston model and the 3/2 model within the same framework, forming a generalized SV model framework that effectively combines the advantages of both models, greatly expanding the theoretical application boundaries of the 3/2 model. To tackle the simulation difficulties caused by the 3/2 model's non-affine characteristics, Kouarfate et al. (2021) developed a novel simulation framework based on an explicit solution for the 3/2 model's variance process. This framework ingeniously leverages the core mathematical property that the variance process of the 3/2 model is the reciprocal of a CIR process, providing a new implementation approach for Monte Carlo simulations of the 3/2 model.

2.3 Evolution of Exact Simulation Methods

In the pricing practice of stochastic volatility models, Monte Carlo simulation is the core tool for handling path-dependent derivatives and high-dimensional model pricing. The central developmental direction of simulation methods is to eliminate time-discretization errors while balancing computational efficiency and full-scenario numerical stability. Since exact simulation methods were introduced, academia has conducted continuous optimization research on exact simulation schemes for both affine models and the non-affine 3/2 model.

In foundational research on the exact simulation of affine SV models, Broadie et al. (2006) first proposed an unbiased exact simulation method for the underlying stock price and variance process under the Heston model and other affine jump-diffusion processes, known as the Exact MC scheme. By sampling the conditional distribution of the integrated variance, this method completely eliminated the systematic errors caused by time discretization, laying the theoretical foundation for the exact simulation of SV models. Subsequently, Glasserman et al. (2011) significantly optimized the Broadie-Kaya algorithm. By utilizing Pitman et al. (1982)'s conclusions on the decomposition of integral series of diffusion processes, they decomposed the conditional integrated variance into an infinite series of Gamma random variables. This avoided the computationally expensive numerical inverse Laplace transform in the original algorithm, markedly improving the computational efficiency of exact simulations.

Focusing on the exact simulation of the non-affine 3/2 model, Baldeaux (2012a) conducted pioneering research, exploring the exact simulation scheme for the stock price process under the 3/2 model for the first time. Utilizing conclusions regarding the transition density of diffusion processes derived via Lie symmetry analysis by Craddock et al. (2009), this study successfully adapted and applied Broadie et al. (2006)'s exact simulation algorithm for affine processes to the 3/2 model. They derived a closed-form solution for the conditional Laplace transform of the integrated variance under the 3/2 model, providing the first theoretically viable implementation scheme for its unbiased simulation.

In recent years, exact simulation algorithms for the 3/2 model have achieved further breakthroughs in numerical stability and computational efficiency. Choi et al. (2024) proposed a novel exact simulation scheme that does not require the use of highly oscillatory modified Bessel functions. Their core finding was that when conditioning on the Poisson variable used to simulate the terminal variance, the conditional integrated variance of the 3/2 model can be significantly simplified, fundamentally resolving the core pain point of modified Bessel functions easily causing numerical overflow under extreme parameters in the original Baldeaux algorithm. Additionally, Brignone et al. (2026) proposed a new exact simulation framework utilizing the conditional Fourier-Cosine (COS) method. By constructing the conditional cumulative distribution function of the integrated variance through cosine series expansion, they achieved highly efficient and accurate sampling of the integrated variance. This resolved the long-standing implementation difficulties of error control and high computational overhead in exact simulations since Broadie et al. (2006), further expanding the engineering application boundaries of exact simulation methods.

2.4 Research on Fourier Transform Methods

The Fourier transform pricing method is the core technical pathway for the rapid pricing and comprehensive calibration of European options under SV models. Its central idea is to convert the risk-neutral expectation of the option price into a complex-domain integral of the underlying log-price's characteristic function, achieving rapid solutions through numerical integration algorithms and completely circumventing the statistical and time-discretization errors of Monte Carlo simulations.

In foundational research on Fourier transform pricing methods, Carr et al. (1999) proposed an option pricing method based on the Fast Fourier Transform (FFT). By introducing a damping factor to resolve the non-integrability issue of the option payoff function's Fourier transform, they derived a closed-form inverse Fourier transform expression for European option prices. Combined with the FFT algorithm, this enabled the simultaneous calculation of option prices across a full range of strike prices within an $O(N \log N)$ time complexity, becoming the mainstream engineering method for option pricing under SV models. Addressing the limitations of the FFT method, such as the picket-fence effect and the restriction to equally spaced strike prices, Fang et al. (2009) proposed the COS pricing method based on Fourier cosine series expansions. By expanding the probability density function of underlying returns into a cosine series over an effective support interval, they derived a closed-form series summation for option prices. This achieved exponential convergence rates while allowing flexible calculations for arbitrary strike prices, demonstrating extremely strong applicability and computational efficiency advantages in non-affine SV models.

For the 3/2 model, due to its non-affine structural nature, the underlying log-price lacks the simple exponential-affine characteristic function found in affine models. The analytical expression of its characteristic function involves complex special functions, posing certain challenges to the application of Fourier transform methods. Lewis (2000)'s early research on complex-domain integral pricing provided the core theoretical framework for exploring the analytical properties of the 3/2 model; Dufresne (2001)'s research further pointed out that the integral distribution of the 3/2 process is closely related to Whittaker functions and modified Bessel functions. He fully derived the distributional properties of the integrated variance under the 3/2 model, laying the mathematical foundation for the closed-form solution of the characteristic function. In practical applications, the FFT pricing method is widely used for parameter calibration and option pricing of the 3/2 model, but it still faces challenges of insufficient numerical stability and excessive computation time under extremely short maturities and extreme volatility parameters Drimus (2012). In contrast, the COS method, with its exponential convergence rate and strong numerical stability, has demonstrated significant performance advantages in the pricing and calibration of the 3/2 model, becoming a primary focus of research in this field in recent years.

2.5 Research on Stochastic Volatility Models in the Cryptocurrency Options Market

As crypto assets represented by Bitcoin (BTC) and Ethereum (ETH) enter the era of institutional trading, the scale of the cryptocurrency options market has grown exponentially, and its unique market dynamics have become a frontier research direction in financial engineering. Unlike traditional equity and foreign exchange markets, the cryptocurrency market is characterized by 7×24 continuous trading, no price limits, extremely high endogenous volatility, and frequent extreme market events. Its return distribution exhibits significant heavy-tailed features and an extremely high volatility of volatility (Vol-of-Vol, VoV), posing severe challenges to traditional option pricing models.

Hou et al. (2020)'s research points out that the volatility of crypto assets exhibits stronger persistence and a higher Vol-of-Vol than traditional stock assets, with extreme tail risks significantly higher than those in traditional financial markets. This frequently causes severe valuation deviations in crypto option pricing when using the BSM model or even the basic Heston model; especially under extreme conditions, the models' ability to fit the volatility smile sharply declines. Addressing this phenomenon, academia primarily explores model frameworks suitable for the crypto market from two directions: one stream of research attempts to capture the instantaneous extreme jumps in cryptocurrency prices by introducing jump-diffusion processes into SV models Scaillet et al. (2020); the other focuses on improving the nonlinear structure of the volatility process, exploring the adaptability of non-affine SV models in the crypto market.

In recent years, due to the 3/2 model's natural advantages in characterizing implied volatility skew and capturing nonlinear mean-reverting features, research on its adaptability to the cryptocurrency market has significantly increased. Studies show that when crypto assets face regulatory policy shifts or macro liquidity shocks, the mean reversion of their volatility exhibits obvious nonlinear acceleration, which aligns highly with the mathematical properties of the 3/2 model (Fassas et al., 2022). Furthermore, given the crypto market's unique 7×24 continuous trading mechanism, utilizing the 3/2 model combined with high-frequency data for real-time volatility forecasting and dynamic option hedging has become a current research hotspot in the field. However, overall, existing research on the 3/2 model in the crypto market mostly focuses on static parameter calibration and model fit comparisons, lacking dynamic evolutionary analysis of model parameters throughout the full cycle of extreme crash events. Moreover, there has been no systematic empirical testing of the performance of different pricing methods under extreme market conditions. This research gap provides critical space for the empirical analysis conducted in this thesis.

2.6 Literature Review Summary and Research Positioning of this Paper

In summary, international academia has conducted extensive research around the theoretical construction, pricing method iteration, and market application expansion of the 3/2 stochastic volatility model. Significant achievements have been made in exploring analytical properties, constructing extended frameworks, optimizing exact simulation methods, and improving Fourier pricing methods, laying a solid theoretical foundation for this study. Concurrently, domestic research largely focuses on traditional SV models like the Heston and SABR models, while systematic research on the 3/2 model remains relatively scarce. There are still prominent research gaps concerning performance evaluations of pricing methods in extreme market environments and empirical adaptability tests in high-volatility crypto asset markets.

Despite the relative abundance of existing research, three core shortcomings remain. First, performance evaluations of existing 3/2 model pricing methods are largely confined to idealized cases using benchmark parameters from literature. There is a lack of full-scenario numerical stability testing covering both conventional markets and extreme parameter combinations. Some methods experience issues like numerical overflow and computational failure under real-world extreme parameters, such as extreme Vol-of-Vol, strong correlation coefficients, and extremely short maturities, leaving their engineering applicability boundaries vaguely defined.

Second, existing research has not yet conducted empirical re-evaluations of pricing methods using parameters calibrated from genuinely extreme markets. Performance in theoretical cases cannot fully equate to practical efficacy in real markets, and empirical evidence is still lacking regarding the differences in accuracy, efficiency, and robustness among various pricing methods during extreme conditions. Third, empirical research on the 3/2 model in highly volatile cryptocurrency markets remains imperfect, lacking dynamic evolutionary analysis of model parameters and volatility smile fitting tests over the full cycle of extreme crash events; the pure diffusion 3/2 model's adaptability to extreme markets has not yet been fully validated.

Based on the deficiencies in existing research, this paper clarifies its core research positioning: taking European option pricing under the 3/2 stochastic volatility model as the research subject, this study systematically constructs a unified implementation framework for eleven mainstream pricing methods across three paradigms. Through multi-scenario numerical experiments and empirical testing in extreme markets, it comprehensively evaluates the numerical stability, pricing accuracy, and computational efficiency of each method to clarify their engineering applicability boundaries. Concurrently, using a severe Bitcoin crash event as a sample, it validates the 3/2 model's adaptability to extreme markets and identifies the time-varying patterns of its model parameters. This research aims to bridge the gaps in existing studies regarding performance testing in extreme scenarios and empirical analysis of real markets, providing reliable method selection and empirical support for 3/2 model pricing practices in high-volatility markets.

Chapter 3 Specification of the 3/2 Volatility Model

3.1 Introduction to the 3/2 Model

As an advanced stochastic volatility framework, the 3/2 model is a complex evolutionary form of the Square-Root Process (popularized by Cox-Ingersoll-Ross and later further developed by Steven Heston).

In the current financial industry, the Heston model (i.e., the 1/2 model) has long been regarded as the industry standard. However, when facing extreme market stress, the Heston model often fails to accurately capture the "volatility of volatilit" observed in the market and struggles to perfectly fit the steepness of the short-term volatility skew.

To overcome this limitation, researchers proposed a completely new approach in the late 1990s and early 2000s: modifying the exponent of the diffusion term from 1/2 to 3/2. This specification change allows the model to generate more explosive volatility dynamics and a nonlinear mean reversion speed. Consequently, the 3/2 model allows volatility to spike and presents a steeper skew under market stress, which highly aligns with actual empirical data. However, it should be noted that under certain normal market conditions, the 3/2 model may lack the stability inherent in the Heston model.

3.2 Stochastic Differential Equation (SDE) Specification

Under the risk-neutral measure, the asset price process and the instantaneous variance/volatility process of the 3/2 model can be characterized by the following system of stochastic differential equations:

The dynamics of the underlying asset price process S_t satisfy:

$$dS_t = rS_t dt + S_t \sqrt{V_t}(\rho dW_t^1 + \sqrt{1-\rho^2} dW_t^2) \tag{3.1}$$

Where r represents the risk-free interest rate, W_t^1 and W_t^2 are two independent standard Brownian motions, and ρ is the correlation coefficient between the two Brownian motions. Meanwhile, the dynamic evolution of the variance process V_t satisfies:

$$dV_t = \kappa V_t(\theta - V_t)dt + \epsilon V_t^{3/2} dW_t^1 \tag{3.2}$$

In the above equations:

- κ controls the speed of mean reversion.

- θ represents the long-term average variance level.
- ϵ is the volatility of volatility parameter, controlling the degree of fluctuation of the variance process.
- W_t^1 and W_t^2 are two independent standard Brownian motions driving the randomness in the variance process and the price process, respectively.

It is worth noting that this variance process can also be equivalently expressed in terms of dV_t/V_t, thereby more intuitively reflecting the drift and diffusion characteristics of its relative changes.

3.3 Integrated Variance and Conditional Distribution

In the option pricing framework of the 3/2 model, the integrated variance V_T over the time interval $[0, T]$ plays a crucial role. It is defined as follows:

$$V_T = \int_0^T V_t dt \tag{3.3}$$

Since the randomness of the asset price is partially driven by volatility, when we condition on the integrated variance V_T, the terminal price of the underlying asset S_T follows a lognormal distribution. This important property means that, given V_T, we can borrow the pricing framework of the Black-Scholes (BS) model.

According to Itô's isometry principle:

$$\int_0^T \rho^* \sigma_t dX_t \sim \rho^* \sqrt{V_T} X_1 \sim N(0, (\rho^*)^2 V_T)$$

Accordingly, the effective volatility σ between time 0 and T can be expressed as:

$$\sigma = \rho^* \sqrt{V_T/T}$$

Through further stochastic calculus derivation, the log-return formula of the terminal price relative to the initial price can be expanded as:

$$\log\left(\frac{S_T}{S_0}\right) = \frac{\rho}{\epsilon}\log\left(\frac{V_T}{V_0}\right) + \left(\kappa + \frac{\epsilon^2}{2}\right)V_T - \kappa\theta T - \frac{1}{2}V_T + \rho^*\sqrt{V_T}X_1 \tag{3.4}$$

This formula is the core foundation for pricing via subsequent exact simulation methods.

3.4 Moment Generating Function of the 3/2 Model

Under the risk-neutral measure $\mathbb{Q}$, the stochastic differential equations (SDEs) of the 3/2 model are defined by Equation (3.1) and Equation (3.2).

To utilize Fourier transform methods for option pricing, the core lies in solving the moment generating function (MGF) of the log-asset price $x_T = \ln S_T$.

According to the derivations by Lewis (2000) and Carr and Sun (2007), let $\phi(u, \tau) = \mathbb{E}[e^{ux_T} | \mathscr{F}_t]$ be defined as the moment generating function of the log-price. Within the 3/2 model framework, because the reciprocal of its variance process V_t follows a CIR process (square-root process), this MGF does not possess the common exponential-affine form found in affine models. Instead, it manifests as an analytical form involving Kummer's Confluent Hypergeometric Function (${}_1F_1$).

Specifically, given the initial state $V_0 = \sigma_0^2$, the auxiliary variables are defined as follows: Parameter transformation terms:

$$\mu = \frac{1}{2} + \frac{\kappa - u\rho\epsilon}{\epsilon^2}, \quad \tilde{c} = \frac{u(1-u)}{\epsilon^2}$$

Characteristic parameters:

$$\delta = \sqrt{\mu^2 + \tilde{c}}, \quad \alpha = -\mu + \delta, \quad \beta = 1 + 2\delta$$

Time-dependent term: define

$$X = \frac{2\kappa\theta}{\epsilon^2\sigma^2(e^{\kappa\theta\tau} - 1)}$$

This term characterizes the nonlinear impact of the mean reversion strength and the maturity τ on the volatility structure. In summary, the moment generating function of the log-price under the 3/2 model can be expressed as:

$$M(u, \tau) = \frac{\Gamma(\beta - \alpha)}{\Gamma(\beta)} X^{\alpha} \, {}_1F_1(\alpha, \beta, -X)$$

In the process of numerical computation, the calculation of ${}_1F_1(\alpha, \beta, -X)$ often involves the processing of special functions in the complex domain. In code implementation, to prevent numerical overflow in large-parameter scenarios (such as extreme volatility or long maturities), the log-gamma function (Log-Gamma) is used for reconstruction. For the MGF pre-factor:

$$M(u, \tau) = \frac{\Gamma(\beta - \alpha)}{\Gamma(\beta)} X^{\alpha} \, {}_1F_1(\alpha, \beta, -X)$$

Use

$$\ln P = \ln \Gamma(\beta - \alpha) - \ln \Gamma(\beta) + \alpha \ln X, \qquad P = \exp(\ln P),$$

then

$$M = P \cdot {}_1F_1(\alpha, \beta, -X)$$

to replace direct multiplication/exponentiation. This approach guarantees the numerical stability of the operations.

Chapter 4 Option Pricing Methods and Algorithmic Implementation

Under the unified framework of the 3/2 stochastic volatility model, this chapter systematically constructs a methodological system of eleven classes for European option pricing. All methods utilize the arbitrage-free risk-neutral pricing principle as their theoretical benchmark. Complete theoretical derivations, convergence analyses, and error characteristic characterizations are developed across three major paradigms: time-discretization approximation, exact/near-exact terminal simulation, and Fourier frequency-domain integration. This establishes a rigorous theoretical foundation for subsequent numerical comparisons and empirical testing.

Under the risk-neutral measure $\mathbb{Q}$ in a complete market, the underlying asset price and variance processes of the 3/2 stochastic volatility model satisfy the equations given by (3.1) and (3.2).

For a European call option with time to maturity T and strike price K, its arbitrage-free price satisfies the risk-neutral pricing formula:

$$C(S_0, v_0; K, T) = e^{-rT}\mathbb{E}^{\mathbb{Q}}\Big[\max(S_T - K, 0) \,\Big|\, S_0, v_0\Big], \tag{4.1}$$

where S_0 and v_0 are the initial price and initial instantaneous variance of the underlying asset, respectively. Unlike the affine structure of the Heston model, the 3/2 model belongs to the class of non-affine stochastic volatility models; thus, its European option price lacks a closed-form analytical solution. Consequently, numerical methods must be employed to approximate the aforementioned conditional expectation. The eleven classes of pricing methods constructed in this chapter cover the complete technical spectrum from time-domain path simulation to frequency-domain integral transformation, effectively approximating option prices from various dimensions.

4.1 Time-Stepping Monte Carlo Pricing Methods

The core idea of time-stepping methods is to discretize the continuous-time diffusion process, generating sample paths of the variance process through step-by-step iteration. The log-price of the underlying asset is then generated using the closed-form analytical solution (3.4) given the variance path, thereby circumventing the discretization error of the price process and reducing the statistical variance of the Monte Carlo simulation. This section sequentially derives the theoretical construction of the Euler discretization scheme, the Milstein discretization scheme, the Exact Stepping method, the Poisson-Gamma Near-Exact Stepping method, and the Quasi-Exponential (QE) Near-Exact Stepping method, and analyzes the convergence characteristics and error sources of each method.

4.1.1 Euler and Milstein Discretization Schemes

4.1.1.1 Euler Discretization Scheme for the Variance Process

The Euler discretization scheme is the most fundamental discretization method for numerically solving stochastic differential equations. Its core lies in performing a first-order Taylor expansion approximation on the increments of the diffusion process. For a general Itô diffusion process

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t,$$

the discrete form of the Euler scheme is

$$X_{t+\Delta t} \approx X_t + \mu(X_t)\Delta t + \sigma(X_t)\Delta W_t,$$

where Δt is the discrete time step, and ΔW_t is the Brownian motion increment following a normal distribution $N(0, \Delta t)$ with mean 0 and variance Δt. Applying this scheme to the variance process of the 3/2 model, the drift term is $\mu(v_t) = \kappa(\theta - v_t)v_t$, and the diffusion term is $\sigma(v_t) = \epsilon v_t^{3/2}$. From this, the Euler discrete iterative formula for the variance process can be obtained. The specific discretization scheme is as follows:

Divide the maturity interval $[0, T]$ equally into M time steps, with a time step length of $\Delta t = T/M$, and a time grid of $\{t_0 = 0, t_1, \dots, t_M = T\}$. For the variance process of the 3/2 model, the Euler iterative formula at each time step is:

$$v_{t_{n+1}} = v_{t_n} + \kappa(\theta - v_{t_n})v_{t_n}\Delta t + \epsilon v_{t_n}^{3/2}\Delta W_n^1, \tag{4.2}$$

where $\Delta W_n^1 \sim N(0, \Delta t)$ is the Brownian motion increment at the n-th time step, and the increments at each time step are mutually independent.

After completing the iteration for all time steps, the integrated variance V_T is calculated using the trapezoidal numerical integration method to improve integration accuracy:

$$I_T = \int_0^T v_t dt \approx \sum_{n=0}^{M-1} \frac{v_{t_n} + v_{t_{n+1}}}{2} \cdot \Delta t. \tag{4.3}$$

By substituting the iterated terminal variance $v_T = v_{t_M}$ and the numerically integrated V_T

into Equation (3.4), the logarithmic terminal price corresponding to that path can be generated, subsequently allowing for the calculation of the option's payoff at maturity.

4.1.1.2 Milstein Discretization Scheme for the Variance Process

The Milstein discretization scheme is a second-order correction to the Euler scheme. By introducing the first-order derivative term of the diffusion term, it eliminates the first-order discrete bias caused by the state dependence of the diffusion term, significantly improving the simulation accuracy of the variance process.

For the variance process of the 3/2 model, its diffusion term is $\sigma_v(v_t) = \epsilon v_t^{3/2}$. Taking the first derivative with respect to v_t yields:

$$\sigma_v'(v_t) = \frac{\partial \sigma_v(v_t)}{\partial v_t} = \frac{3}{2}\epsilon v_t^{1/2}. \tag{4.4}$$

Substituting this into the general form of the Milstein scheme yields the Milstein discrete iterative formula for the variance process:

$$v_{t_{n+1}} = v_{t_n} + \kappa(\theta - v_{t_n})v_{t_n}\Delta t + \epsilon v_{t_n}^{3/2}\Delta W_n^1 + \frac{1}{2}\sigma_v(v_{t_n})\sigma_v'(v_{t_n})\left((\Delta W_n^1)^2 - \Delta t\right). \tag{4.5}$$

After substituting the diffusion term and its derivative, the final Milstein iterative formula is:

$$v_{t_{n+1}} = v_{t_n} + \kappa(\theta - v_{t_n})v_{t_n}\Delta t + \epsilon v_{t_n}^{3/2}\Delta W_n^1 + \frac{3}{4}\epsilon^2 v_{t_n}^2\left((\Delta W_n^1)^2 - \Delta t\right). \tag{4.6}$$

Consistent with the Euler scheme, after completing the iteration of the variance path, the integrated variance V_t is calculated via Equation (4.3), which is then substituted into Equation (3.4) to generate the logarithmic terminal price.

4.1.1.3 Error Characteristics and Convergence Analysis

Based on the Euler/Milstein schemes under the conditional Monte Carlo framework, the error sources can be decomposed into three independent parts. The convergence characteristics of each part are as follows:

1. Time discretization error of the variance process: The Euler scheme has a strong order of convergence of 1/2 and a weak order of convergence of 1 for the diffusion process; the Milstein scheme elevates the strong order of convergence to 1 through a second-order correction, while the weak order remains 1. The discrete biases of both schemes

monotonically decay as the time step Δt decreases. Given the same step size, the discrete bias of the Milstein scheme is significantly lower than that of the Euler scheme.

2. Numerical integration error of the integrated variance: Calculating the integrated variance using the trapezoidal method has a convergence order of 2, which is significantly higher than the discrete convergence order of the variance process. Therefore, in practical calculations, the integration error is negligible and will not become the dominant term in the overall error.
3. Statistical error of the Monte Carlo simulation: Because the conditional Monte Carlo method is employed, the discrete noise of the price process is eliminated through an analytical mapping. The statistical variance of the simulation is significantly lower than that of fully discretized standard Monte Carlo methods. The statistical error decays at a rate of $O(N^{-1/2})$ as the number of simulated paths N increases.

4.1.2 Exact Stepping Method

The Exact Stepping method provides the highest precision among time-stepping conditional Monte Carlo methods. Its core relies on utilizing the specific mathematical structure of the variance process in the 3/2 model. Through a variable transformation, it maps the process to an affine process with a known transition distribution, thereby achieving exact sampling of the variance process without discrete bias and completely eliminating systematic errors caused by time discretization.

4.1.2.1 Reciprocal Transformation and Exact Distribution of the Variance Process

Applying a reciprocal transformation to the variance process of the 3/2 model, let $x_t = 1/v_t$. The dynamic process of x_t can be derived via Itô's Lemma. First, compute the first and second derivatives of x_t with respect to v_t:

$$\frac{\partial x_t}{\partial v_t} = -\frac{1}{v_t^2}, \quad \frac{\partial^2 x_t}{\partial v_t^2} = \frac{2}{v_t^3}. \tag{4.7}$$

Substituting this into the Itô Lemma formula $dx_t = \frac{\partial x_t}{\partial v_t} dv_t + \frac{1}{2}\frac{\partial^2 x_t}{\partial v_t^2}(dv_t)^2$, and inserting the SDE of the 3/2 model's variance $dv_t = \kappa v_t(\theta - v_t)dt + \epsilon v_t^{3/2} dW_t^1$, where the quadratic variation term is $(dv_t)^2 = \epsilon^2 v_t^3 dt$. Rearranging yields:

$$dx_t = \left(\kappa - \kappa\theta x_t\right) dt - \epsilon\sqrt{x_t} dW_t^1. \tag{4.8}$$

Equation (4.8) demonstrates that the transformed process x_t is a classical CIR (Cox-Ingersoll-Ross) square-root diffusion process, characterized by an affine drift structure and a square-root diffusion structure. A core property of the CIR process is that its transition probability distribution has an explicit analytical form, following a non-central chi-squared distribution. Specifically, given y_{t_n} at the initial time t_n, after a time step Δt, the conditional distribution of $x_{t_{n+1}}$ at time t_{n+1} is:

$$x_{t_{n+1}} \mid x_{t_n} \sim \frac{1}{c} \cdot \chi'^2 (d, \lambda), \tag{4.9}$$

where $\chi'^2(d, \lambda)$ represents a non-central chi-squared random variable with degrees of freedom d and a non-centrality parameter λ. The distribution parameters are determined by the 3/2 model parameters and the time step:

$$d = \frac{4\kappa\theta}{\epsilon^2}, \quad c = \frac{\epsilon^2 \left(1 - e^{-\kappa\Delta t}\right)}{4\kappa}, \quad \lambda = c \cdot y_{t_n} e^{-\kappa\Delta t}. \tag{4.10}$$

Through the inverse transformation $v_{t_{n+1}} = 1/x_{t_{n+1}}$, exact samples of the variance process for the 3/2 model can be obtained. This sampling procedure introduces no time discretization bias.

4.1.2.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation procedure of the Exact Stepping method is as follows:

Divide the maturity interval $[0, T]$ equally into M time steps, with a time step length of $\Delta t = T/M$, and a time grid of $\{t_0 = 0, t_1, \ldots, t_M = T\}$. At each time step:

1. Given the current variance v_{t_n}, calculate $y_{t_n} = 1/v_{t_n}$;
2. Generate a non-central chi-squared random variable $\chi'^2(d, \lambda)$ according to the distribution parameters in Equation (4.9);
3. Calculate $x_{t_{n+1}} = \chi'^2(d, \lambda)/c$;
4. Apply the inverse transformation to obtain $v_{t_{n+1}} = 1/x_{t_{n+1}}$.

After completing the iterations for all time steps, calculate the integrated variance V_T using the trapezoidal numerical integration method:

$$V_T = \int_0^T v_t dt \approx \sum_{n=0}^{M-1} \frac{v_{t_n} + v_{t_{n+1}}}{2} \cdot \Delta t. \tag{4.11}$$

Finally, substitute the iterated terminal variance $v_T = v_{t_M}$ and the numerically integrated V_T into the core closed-form mapping (3.4).

4.1.2.3 Error Characteristics and Convergence Analysis

The sources of error in the Exact Stepping method consist of only two components, completely eliminating the time discretization error of the variance process:

1. Numerical integration error of the integrated variance: Using the trapezoidal rule to calculate the integrated variance results in a convergence order of 2. In practical applications, even with a small number of time steps, this error is negligible and will not dominate the overall error. If further elimination of the integration error is desired, one could combine the integral moment generating function of the CIR process to analytically compute the conditional distribution of V_T.
2. Statistical error of the Monte Carlo simulation: Due to the use of the conditional Monte Carlo method and exact sampling of the variance process, the statistical variance of the simulation is minimized. The statistical error decays at a rate of $O(N^{-1/2})$ as the number of simulated paths N increases.

Among all time-stepping methods, the Exact Stepping method exhibits the smallest bias.

4.1.3 Poisson-Gamma Near-Exact Stepping Method

The core of the Poisson-Gamma Near-Exact Stepping method lies in leveraging the classical statistical properties of the CIR process derived from the reciprocal transformation of the 3/2 model's variance process. Its transition distribution can be precisely represented as an infinite mixture series of Gamma distributions weighted by Poisson probabilities. By truncating this series at a finite number of terms, an efficient, near-exact sampling of the variance process is achieved. This maintains extremely low bias while avoiding the high numerical computation costs associated with non-central chi-squared distribution sampling. This method is developed entirely based on the reciprocal-transformed CIR process and is an equivalent numerical implementation of the CIR transition distribution.

4.1.3.1 Reciprocal Transformation and Poisson-Gamma Mixture Representation of the CIR Process

Consistent with the Exact Stepping method, we first apply a reciprocal transformation to the variance process of the 3/2 model, let $x_t = 1/v_t$, which via Itô's lemma yields Equation (4.8). According to the classic mixture property of the non-central chi-squared distribution, a non-central chi-squared random variable $\chi'^2(d, \lambda)$ with d degrees of freedom and non-centrality parameter λ can be exactly decomposed into an infinite mixture of Poisson and

Gamma distributions:

$$\chi'^2(d,\lambda) \mid N \sim \text{Gamma}\left(\frac{d}{2}+N,\frac{1}{2}\right), \quad N \sim \text{Poisson}\left(\frac{\lambda}{2}\right), \tag{4.12}$$

where N is a latent variable following a Poisson distribution, and Gamma(α,β) is a Gamma distribution with shape parameter α and rate parameter β, whose probability density function is:

$$f_{\text{Gamma}}(x;\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0. \tag{4.13}$$

Combining this with the conclusion regarding the transition distribution of the CIR process in the Exact Stepping method, given an initial value x_{t_n}, the conditional distribution of $x_{t_{n+1}}$ after a time step Δt is:

$$y_{t_{n+1}} \mid y_{t_n} \sim \frac{1}{c} \cdot \chi'^2(d,\lambda), \tag{4.14}$$

where the distribution parameters are:

$$d = \frac{4\kappa\theta}{\epsilon^2}, \quad c = \frac{\epsilon^2\left(1-e^{-\kappa\Delta t}\right)}{4\kappa}, \quad \lambda = c \cdot y_{t_n} e^{-\kappa\Delta t}. \tag{4.15}$$

Substituting Equation (4.12) into the above yields the direct Poisson-Gamma mixture representation of $y_{t_{n+1}}$:

$$y_{t_{n+1}} \mid N \sim \frac{1}{c} \cdot \text{Gamma}\left(\frac{d}{2}+N,\frac{1}{2}\right), \quad N \sim \text{Poisson}\left(\frac{\lambda}{2}\right). \tag{4.16}$$

Equation (4.16) represents the core of this method: the transition distribution of the CIR process is inherently an infinite mixture of Gamma distributions weighted by Poisson probabilities. There is zero distribution fitting error; one only needs to apply a finite truncation to the Poisson distribution values to achieve near-exact sampling.

4.1.3.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation procedure of the Poisson-Gamma Near-Exact Stepping method aligns closely with the Exact Stepping method. It merely replaces the non-central chi-squared sampling with the more efficient Poisson-Gamma mixture sampling. The specific procedure is as follows:

Divide the maturity interval $[0,T]$ equally into M time steps, with a time step length of $\Delta t = T/M$, and a time grid of $\{t_0 = 0, t_1, \ldots, t_M = T\}$. At each time step:

1. Given the current variance v_{t_n}, calculate $x_{t_n} = 1/v_{t_n}$;

2. Pre-calculate the fixed parameters d, c of the CIR transition distribution, and the non-centrality parameter $\lambda = c \cdot x_{t_n} e^{-\kappa\Delta t}$;
3. Generate a Poisson latent variable $N \sim \text{Poisson}\left(\frac{\lambda}{2}\right)$. In practical calculations, an upper bound truncation is applied to N (setting $N_{\max} = \lfloor \lambda/2 + 5\sqrt{\lambda/2} \rfloor$, ensuring the truncation probability is less than 10^{-6});
4. Generate a Gamma random variable $G \sim \text{Gamma}\left(\frac{d}{2} + N, \frac{1}{2}\right)$;
5. Calculate $x_{t_{n+1}} = G/c$, then apply the inverse transformation to obtain the variance at the next time step $v_{t_{n+1}} = 1/x_{t_{n+1}}$.

After completing the iterations for all time steps, compute the integrated variance V_T using the trapezoidal numerical integration method. Then, substitute v_T and V_T into Equation (3.4) to generate the logarithmic terminal price.

4.1.3.3 Error Characteristics and Convergence Analysis

The error sources for the Poisson-Gamma Near-Exact Stepping method contain only three parts, and its bias level is substantially lower than that of the Euler/Milstein discrete schemes:

1. Truncation error of the Poisson distribution: This is the sole inherent bias of the method. Because the probability mass of the Poisson distribution decays exponentially as N increases, the truncation error decreases exponentially with the increase of the truncation upper bound $N_{\max}$. Under normal parameters, setting $N_{\max}$ to the Poisson mean plus five standard deviations can control the truncation error to below 10^{-6}, making it negligible.
2. Numerical integration error of the integrated variance: Using the trapezoidal method yields a convergence order of 2. In practical applications, this error is far smaller than the truncation error and will not dominate the overall error.
3. Statistical error of the Monte Carlo simulation: Since the sampling bias of the variance process is minuscule, and the conditional Monte Carlo framework eliminates the discrete noise of the price process, the statistical error decays at a rate of $O(N^{-1/2})$ as the number of paths N increases, placing it on the same level as the Exact Stepping method.

The core advantage of this method is the balance between computational efficiency and precision. Compared to generating non-central chi-squared variables in the Exact Stepping method, generating Poisson and Gamma random numbers incurs extremely low computational costs, offering a significant speed boost. Compared to the Euler/Milstein schemes, this method entirely eliminates the discrete bias of the diffusion process, leaving only a negligible truncation error, improving precision by several orders of magnitude.

4.1.4 Quasi-Exponential (QE) Near-Exact Stepping Method

The Quasi-Exponential (QE) Near-Exact Stepping method was initially designed by Andersen (2007) for the Heston model. Here, we extend it to the 3/2 model. Its core logic lies in matching the conditional first and second moments of the variance process to construct a simple exponential-type approximate distribution. This avoids sampling from complex distributions and maintains excellent numerical stability even in extreme scenarios featuring high volatility and thick tails.

4.1.4.1 Moment Matching of the Variance Process and QE Distribution Construction

The central idea of the QE method is "moment matching". No matter how complex the true transition distribution is, one only needs to match its first two moments to construct a sufficiently precise approximate distribution. For the variance process of the 3/2 model, given the initial variance v_t, let its reciprocal process be $x_t = 1/v_t$. After a time step Δt, its conditional first moment $\mathbb{E}[x_{t+\Delta t} \mid x_t]$ and conditional second moment $\mathbb{E}[x^2_{t+\Delta t} \mid x_t]$ can be derived analytically by solving the moment differential equations of the variance process. Let $m_1 = \mathbb{E}[v_{t+\Delta t} \mid v_t]$ and $m_2 = \mathbb{E}[v^2_{t+\Delta t} \mid v_t]$ denote the conditional first and second moments, respectively. Compute the conditional variance $\sigma^2 = m_2 - m_1^2$. The QE method selects two different approximate distributions based on the magnitude of the variance:

1. Low variance scenario: When the conditional variance is relatively small, a Shifted Log-Normal Distribution is used as an approximation;
2. High variance scenario: When the conditional variance is large, an Exponential-Two-Point Mixture distribution is used as an approximation.

By satisfying the moment matching conditions $m_1 = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{Var}(X)$, all parameters of the QE distribution can be solved analytically, enabling efficient sampling.

4.1.4.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation procedure of the QE Near-Exact Stepping method is as follows: Divide the maturity interval $[0, T]$ equally into M time steps, with a time step length of $\Delta t = T/M$, and a time grid of $\{t_0 = 0, t_1, \dots, t_M = T\}$. At each time step:

1. Given the current reciprocal variance x_{t_n}, analytically compute the conditional mean m_1 and conditional variance σ^2 via the moment equations;
2. Evaluate the current scenario against a predefined threshold to determine the appropriate form of the QE distribution;
3. Based on whether the conditional variance is high or low, select the corresponding type of approximate distribution, and calculate the distribution parameters via moment matching;
4. Sample from the constructed QE distribution to obtain the approximate sample $x_{t_{n+1}}$, and take its reciprocal to obtain $v_{t_{n+1}}$.

After completing the iterations for all time steps, compute the integrated variance V_T using the trapezoidal numerical integration method. Then, substitute v_T and V_T into Equation (3.4) to generate the logarithmic terminal price.

4.1.4.3 Error Characteristics and Convergence Analysis

The sources of error in the QE Near-Exact Stepping method consist of three parts:

1. Approximation bias of moment matching: This error is the inherent bias of the method, featuring a weak convergence order of 1. Because only the first two moments are matched, omitting information from higher-order moments, slight deviations may occur in extreme heavy-tailed scenarios. Nonetheless, it remains significantly superior to the Euler/Milstein schemes.
2. Threshold error in distribution selection: This error can be controlled to a minimal level by setting an appropriate threshold (typically based on the variance coefficient σ/m_1).
3. Numerical integration error of the integrated variance and Monte Carlo statistical error: These are consistent with the previously discussed methods.

The greatest advantage of the QE method is its exceptionally high computational efficiency and robust numerical stability, though the reciprocal operation here may mildly impact stability.

4.2 Exact and Near-Exact Simulation Pricing Methods

The terminal simulation class of methods discussed in this section completely circumvents the step-by-step iterative discretization process seen in time-stepping methods. Instead, it directly samples the joint distribution of the terminal value v_T of the variance process and the integrated variance $V_T = \int_0^T v_s ds$ over the maturity interval $[0, T]$. It then directly generates the terminal price of the underlying asset using the closed-form logarithmic price mapping derived in Section 4.1, thoroughly eradicating the discretization error caused by time steps. All methods strictly follow the conditional Monte Carlo framework: core statistics of the variance process are sampled first, followed by the generation of the underlying price through a conditional normal distribution, maximizing the reduction of statistical variance in the simulation.

4.2.1 Exact Simulation Method

The exact simulation method by Baldeaux (2012a) is the first unbiased terminal exact simulation method under the 3/2 model. Its core exploits the reciprocal transformation property of the 3/2 model's integrated variance process, combined with modified Bessel functions and inverse Laplace transforms, to achieve the joint exact sampling of v_T and V_T, completely eliminating all systematic biases.

4.2.1.1 Theoretical Derivation

Consistent with time-stepping methods, a reciprocal transformation is first applied to the variance process of the 3/2 model, let $x_t = 1/v_t$, leading to the CIR square-root diffusion process (4.8). The conditional distribution of the terminal value $x_T = 1/v_T$ of this process is a non-central chi-squared distribution, which can be exactly sampled, with the specific form aligning with Section 4.2.2.

The core contribution of Baldeaux (2012a) lies in deriving the closed-form expression of the conditional Laplace transform of the integrated variance $V_T = \int_0^T v_s ds$, given the initial value y_0 and the terminal value y_T:

$$\mathscr{L}(\beta) = \mathbb{E}\left[e^{-\beta V_T} \mid y_0, y_T\right] = \frac{I_\nu\left(\sqrt{\frac{4\kappa\theta}{\epsilon^2} + \frac{8\beta}{\epsilon^2}} \cdot \frac{2\sqrt{y_0 y_T} e^{-\kappa T/2}}{1-e^{-\kappa T}}\right)}{I_\nu\left(\sqrt{\frac{4\kappa\theta}{\epsilon^2}} \cdot \frac{2\sqrt{y_0 y_T} e^{-\kappa T/2}}{1-e^{-\kappa T}}\right)}, \tag{4.17}$$

where $\nu = \frac{2\kappa\theta}{\epsilon^2} - 1$ is the order of the modified Bessel function, and $I_\nu(\cdot)$ is the modified Bessel function of the first kind of order ν.

Based on this closed-form Laplace transform, the conditional probability density function of V_T can be obtained via the inverse Fourier transform, which then allows for the exact sampling of V_T through inverse transform sampling. Ultimately, the unbiased simulation of the underlying asset price is achieved by the joint sampling of V_T and v_T in the core mapping that generates the underlying asset's terminal price, as per Equation (3.4).

4.2.1.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation flow of Baldeaux's Exact Simulation method completely avoids time discretization, proceeding as follows:

1. Exact sampling of terminal variance v_T: Based on the non-central chi-squared transition distribution of the CIR process, directly sample to obtain y_T at maturity, then take the reciprocal to get $v_T = 1/x_T$. This sampling incurs zero systematic bias.
2. Exact sampling of integrated variance V_T: Given $y_0 = 1/v_0$ and $y_T = 1/v_T$, and based on the Laplace transform in Equation (4.17), derive the conditional cumulative distribution function of V_T via a numerical inverse Fourier transform, then draw an exact sample of V_T using uniform inverse transform sampling.
3. Finally, substitute $v_T = 1/x_T$ and V_T into Equation (3.4) to generate the underlying asset's terminal price samples.

4.2.1.3 Error Characteristics and Convergence Analysis

Baldeaux's Exact Simulation method is the only unbiased terminal simulation method under the 3/2 model. Its error sources are limited to two parts:

1. Numerical truncation error of the inverse Laplace transform: This error decays exponentially as the number of truncated terms in the Fourier cosine series increases. Under typical parameters, truncating at 256 terms can restrain the error to below 10^{-8}, making it negligible.
2. Statistical error of the Monte Carlo simulation: Since both the variance process and integrated variance are sampled exactly, and the conditional Monte Carlo framework eliminates discrete noise from the price process, the statistical variance of the simulation is at the theoretical minimum. The statistical error decays at a rate of $O(N^{-1/2})$ as the number of simulated paths N increases.

4.2.2 Conditional Fourier-Cosine Simulation Method

The Conditional Fourier-Cosine simulation method was proposed by Brignone et al. (2026). Its core approach involves constructing the conditional cumulative distribution function of the integrated variance V_T via a Fourier cosine series expansion based on the conditional characteristic function of the underlying logarithmic price. This enables highly efficient and accurate sampling, massively boosting computational efficiency compared to Baldeaux's exact simulation method while maintaining comparable accuracy.

4.2.2.1 Theoretical Derivation

Consistent with the Baldeaux method, the terminal variance v_T is first exactly sampled via the non-central chi-squared distribution of the CIR process. The core problem then shifts to how to efficiently sample the integrated variance V_T given v_0 and v_T.

Let the conditional characteristic function of I_T be denoted as $\varphi(u) = \mathbb{E}\left[e^{iuI_T} \mid v_0, v_T\right]$. This characteristic function can be obtained through the analytical continuation of the Laplace transform in Equation (4.17), possessing a definitive closed-form expression. According to Fourier cosine expansion theory, the conditional probability density function of I_T can be expanded into a cosine series over the effective support interval $[a, b]$:

$$f_{I_T|v_0,v_T}(x) = \frac{2}{b-a}\sum_{n=0}^{\infty} \mathrm{Re}\left[\varphi\left(\frac{n\pi}{b-a}\right)e^{-i\frac{n\pi a}{b-a}}\right]\cos\left(\frac{n\pi(x-a)}{b-a}\right), \tag{4.18}$$

where $[a, b]$ is the effective support interval of I_T, usually set to $a = 0$ and $b = \mathbb{E}[I_T] + 10\sqrt{\mathrm{Var}(I_T)}$ to cover over 99.99% of the probability mass.

Integrating the density function yields the closed-form series expansion of the conditional cumulative distribution function of V_T:

$$F_{V_T|v_0,v_T}(x) = \int_a^x f(z)dz = \frac{2}{b-a}\sum_{n=0}^{\infty} \mathrm{Re}\left[\varphi\left(\frac{n\pi}{b-a}\right)e^{-i\frac{n\pi a}{b-a}}\right]\cdot \Psi_n(x), \tag{4.19}$$

where $\Psi_n(x)$ is the closed-form integral of the cosine term, which can be computed analytically without numerical integration:

$$\Psi_n(x) = \begin{cases} x-a, & n = 0, \\ \dfrac{b-a}{n\pi}\sin\left(\dfrac{n\pi(x-a)}{b-a}\right), & n \geq 1. \end{cases} \tag{4.20}$$

4.2.2.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation flow of the Conditional Fourier-Cosine simulation method is as follows:

1. Exact sampling of terminal variance v_T: As in the Baldeaux method, v_T is exactly sampled through the non-central chi-squared distribution of the CIR process;
2. Efficient sampling of integrated variance V_T: Given v_0 and v_T, calculate the conditional characteristic function of V_T, construct the conditional cumulative distribution function using the truncated cosine series, and draw a sample of V_T via uniform inverse transform sampling;

3. Generation of underlying asset terminal price: Finally, substitute $v_T = 1/x_T$ and V_T into Equation (3.4) to generate the underlying asset terminal price samples.

4.2.2.3 Error Characteristics and Convergence Analysis

The error sources for this method primarily consist of three components:

1. Truncation error of the cosine series: This error decays exponentially with an increasing number of truncated terms. Under typical parameters, truncating at merely 128 terms reaches an accuracy of 10^{-6}, converging much faster than ordinary polynomials;
2. Truncation error of the support interval: By appropriately setting the $[a, b]$ interval, this error can be kept below 10^{-8}, rendering it negligible;
3. Statistical error of the Monte Carlo simulation: Because the variance process sampling is unbiased, and the conditional Monte Carlo framework eliminates the discrete noise of the price process, the statistical error decays at a rate of $O(N^{-1/2})$ as the number of simulated paths N increases, performing at the same level as the Baldeaux method.

The core advantage of this method lies in its computational efficiency. Compared to the path-by-path inverse Laplace transform of the Baldeaux method, the coefficients of the cosine series can be pre-computed in batches, lifting speed by more than an order of magnitude.

4.2.3 Conditional Moment Matching Simulation Method

The core of the conditional moment matching simulation method involves matching the first few moments of the integrated variance V_T to construct an analytical approximate distribution for sampling. This avoids complex characteristic function calculations and series expansions, achieving extreme computational efficiency while ensuring a baseline of precision.

4.2.3.1 Theoretical Derivation

The first four moments of the integrated variance V_T in the 3/2 model can be analytically derived via the recursive moment differential equations of the CIR process:

1. First moment (mean): $\mathbb{E}[V_T] = \int_0^T \mathbb{E}[v_s]ds$, where $\mathbb{E}[v_s]$ is the unconditional mean of the 3/2 model variance process, which possesses a closed-form expression;
2. Second moment: $\mathbb{E}[V_T^2] = 2\int_0^T \int_0^s \mathbb{E}[v_s v_t]dtds$, where the covariance $\mathbb{E}[v_s v_t]$ can be analytically computed through the transition moments of the CIR process;
3. Third and fourth moments: These can be derived analytically through the recursive relations of the moment equations, subsequently enabling the calculation of the skewness

and excess kurtosis of V_T.

Based on these first four moments, an approximate probability density function of V_T can be constructed via the Gram-Charlier Type A expansion:

$$f(x) = \phi_{\mu,\sigma}(x)\left(1 + \frac{\kappa_3}{6}H_3\left(\frac{x-\mu}{\sigma}\right) + \frac{\kappa_4}{24}H_4\left(\frac{x-\mu}{\sigma}\right)\right), \tag{4.21}$$

where $\phi_{\mu,\sigma}(x)$ is the normal density function with mean μ and variance σ^2; κ_3 and κ_4 are the standardized third and fourth moments of V_T; and $H_k(\cdot)$ represents the k-th order Hermite polynomial, whose first four orders take the forms:

$$H_0(x) = 1, \quad H_1(x) = x, \quad H_2(x) = x^2 - 1, \quad H_3(x) = x^3 - 3x, \quad H_4(x) = x^4 - 6x^2 + 3. \tag{4.22}$$

Integrating this density function provides an approximate cumulative distribution function, yielding a sample of V_T through inverse transform sampling.

4.2.3.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation procedure for the Conditional Moment Matching simulation method is as follows:

1. Exact sampling of terminal variance v_T: Exact sampling of v_T is accomplished through the non-central chi-squared distribution of the CIR process;
2. Moment-matching sampling of integrated variance V_T: Given v_0 and v_T, analytically calculate the first four moments of V_T, construct the Gram-Charlier approximate distribution, and obtain a sample of V_T via inverse transform sampling;
3. Generation of terminal price: Substitute v_T and V_T into the core closed-form mapping to generate the logarithmic terminal price.

4.2.3.3 Error Characteristics and Convergence Analysis

1. Approximation error of moment matching: This is the method's inherent bias. Matching only the first four moments disregards higher-order moment information. The bias is minimal under standard parameters but increases in extreme heavy-tailed scenarios, displaying a weak convergence order of $O(N^{-2})$;
2. Statistical error of Monte Carlo simulation: The convergence speed is $O(N^{-1/2})$.

The central advantage of this method is extreme computational efficiency. All moments can be pre-computed, circumventing path-by-path complex numerical calculations, making it well-suited for massive-scale path simulations and parameter calibration scenarios.

4.2.4 Inverse Gaussian (IG) Near-Exact Simulation Method

Proposed by Choi et al. (2024), the Inverse Gaussian Near-Exact Simulation method capitalizes on the Inverse Gaussian distribution's exceptional ability to fit positively skewed, heavy-tailed non-negative random variables. By matching the first two moments of the integrated variance V_T to construct an approximate distribution, it establishes a prime balance between accuracy and efficiency.

4.2.4.1 Theoretical Derivation

The Inverse Gaussian distribution is a class of two-parameter continuous distributions defined on the domain of positive real numbers, with a probability density function of:

$$f_{\mathrm{IG}}(x; \mu, \lambda) = \sqrt{\frac{\lambda}{2\pi x^3}} \exp\left(-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right), \quad x > 0, \tag{4.23}$$

where $\mu > 0$ is the mean of the distribution, and $\lambda > 0$ is the shape parameter. Its variance satisfies $\mathrm{Var}(x) = \mu^3/\lambda$.

The integrated variance I_T of the 3/2 model is a non-negative random variable that exhibits significant positive skewness and thick tails, which is highly consistent with the statistical properties of the Inverse Gaussian distribution. Through moment matching conditions, the parameters of the Inverse Gaussian distribution can be solved analytically:

1. Given v_0 and v_T, analytically calculate the conditional mean $\mu = \mathbb{E}[I_T \mid v_0, v_T]$ and conditional variance $\sigma^2 = \mathrm{Var}(I_T \mid v_0, v_T)$ of I_T;
2. Force the mean and variance of the Inverse Gaussian distribution to match the target values, solving for:

$$\mu_{\mathrm{IG}} = \mu, \quad \lambda_{\mathrm{IG}} = \frac{\mu^3}{\sigma^2}. \tag{4.24}$$

4.2.4.2 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the execution steps for the Inverse Gaussian Near-Exact Simulation method are:

1. Exact sampling of terminal variance v_T: Exact sampling of v_T is accomplished through the non-central chi-squared distribution of the CIR process;
2. IG distribution sampling of integrated variance V_T: Given v_0 and v_T, analytically evaluate the conditional mean and variance of V_T, calibrate the parameters of the IG distribution, and generate random numbers from the IG distribution to obtain the V_T sample;

3. Generation of terminal price: Substitute v_T and V_T into the core closed-form mapping to yield the logarithmic terminal price.

4.2.4.3 Error Characteristics and Convergence Analysis

The primary error sources of this approach entail two components:

1. Distribution fitting error: This represents the method's inherent bias, as the IG distribution fits solely the first two moments, neglecting the influence of higher-order moments;
2. Statistical error of Monte Carlo simulation: The convergence speed is $O(N^{-1/2})$.

The principal asset of this method lies in bridging accuracy with efficiency. The generation of IG distribution random numbers is exceptionally swift, and its fitting accuracy verges on that of exact simulation methods, cementing it as the most widely used simulation scheme for the 3/2 model in practical applications.

4.2.5 Gamma Near-Exact Simulation Method

The Gamma distribution is a two-parameter continuous distribution defined on the positive real numbers, whose probability density function is:

$$f_{\text{Gamma}}(x;\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0, \tag{4.25}$$

where $\alpha > 0$ is the shape parameter, and $\beta > 0$ is the rate parameter. Its mean and variance satisfy $\mathbb{E}[x] = \alpha/\beta$ and $\mathrm{Var}(x) = \alpha/\beta^2$.

Via moment matching conditions, the parameters of the Gamma distribution can be solved analytically:

1. Given v_0 and v_T, analytically calculate the conditional mean μ and conditional variance σ^2 of I_T;
2. Force the mean and variance of the Gamma distribution to align with the target values, solving for:

$$\alpha = \frac{\mu^2}{\sigma^2}, \quad \beta = \frac{\mu}{\sigma^2}. \tag{4.26}$$

4.2.5.1 Implementation Under the Conditional Monte Carlo Framework

Under the conditional Monte Carlo framework, the implementation of the Gamma Near-Exact Simulation method mimics the IG method exactly, simply substituting the IG distribution with the Gamma distribution for sampling. Ultimately, v_T and I_T are plugged into the core closed-form mapping to generate the terminal underlying price.

4.2.5.2 Error Characteristics and Convergence Analysis

The error characteristics of this method parallel the IG method. The essential difference is that the Gamma distribution's fit accuracy for the thick tails of I_T is slightly inferior to the IG distribution, but its random number generation exhibits higher computational efficiency, offering a notable advantage in ultra-large-scale path simulation scenarios. Its error sources remain the distribution fitting error and the Monte Carlo statistical error, scaling at a convergence speed of $O(N^{-1/2})$.

4.3 Fourier Frequency-Domain Pricing Methods

Fourier frequency-domain pricing methods entirely bypass the path generation process characteristic of Monte Carlo simulations. The core technique translates the risk-neutral expectation of the option price into a Fourier integral of the characteristic function of the underlying logarithmic price, securing swift solutions through numerical integration algorithms. This approach commands core advantages—high computational efficiency, zero statistical error, and aptitude for batch pricing across strike prices—establishing it as the dominant method for 3/2 model option pricing and parameter calibration.

4.3.1 Fast Fourier Transform (FFT) Pricing Method

Proposed by Carr et al. (1999), the FFT pricing method centers on expressing the European option price as a Fourier transform of the characteristic function. Employing the Fast Fourier Transform algorithm, it simultaneously computes option prices for all equidistant strike prices within a time complexity of $O(N \log N)$.

4.3.1.1 Theoretical Derivation

For a European call option with a time to maturity T and strike price K, its risk-neutral price can be represented as:

$$C(K) = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\max(S_T - K, 0)\right] = e^{-rT}\int_{-\infty}^{\infty} \max(S_0 e^x - K, 0) f(x)\, dx, \tag{4.27}$$

where $x = \log(S_T/S_0)$ is the logarithmic return of the underlying asset, and $f(x)$ is the probability density function of x.

Let $k = \log(K/S_0)$ denote the logarithmic strike price. Rewriting the option price as a function $C(k)$ of k, one can derive the closed form of $C(k)$'s Fourier transform through the convolution property of Fourier transforms:

$$\hat{C}(u) = \int_{-\infty}^{\infty} e^{iuk} C(k)\, dk = e^{-rT} \frac{\phi(u-i)}{iu - u^2}, \tag{4.28}$$

where $\phi(u) = \mathbb{E}^{\mathbb{Q}}\left[e^{iux}\right]$ is the characteristic function of the underlying logarithmic return, serving as the nucleus of frequency-domain pricing methods.

For the 3/2 model, the characteristic function of the logarithmic return can be expressed in closed form via the confluent hypergeometric function ${}_1F_1$:

$$\phi(u) = \exp\left(iurT\right) \cdot \frac{\Gamma(\beta - \alpha)}{\Gamma(\beta)} \cdot \left(\frac{2\kappa\theta}{\epsilon^2 v_0}\left(e^{\kappa\theta T} - 1\right)\right)^{\alpha} \cdot {}_1F_1\left(\alpha, \beta, -\frac{2\kappa\theta}{\epsilon^2 v_0}\left(e^{\kappa\theta T} - 1\right)\right), \tag{4.29}$$

where parameters α, β are analytically determined by the model parameters and the Fourier variable u:

$$\alpha = -\frac{1}{2} + \frac{\kappa - \rho\epsilon u}{\epsilon^2} + \sqrt{\left(\frac{1}{2} - \frac{\kappa - \rho\epsilon u}{\epsilon^2}\right)^2 + \frac{u(iu+1)}{\epsilon^2}}, \quad \beta = 1 + 2\alpha. \tag{4.30}$$

Through the inverse Fourier transform, the integral expression for the option price emerges:

$$C(k) = \frac{e^{-rT}}{\pi} \int_0^{\infty} \mathrm{Re}\left[e^{-iuk} \frac{\phi(u-i)}{iu - u^2}\right] du. \tag{4.31}$$

Discretizing this continuous integral into a Riemann sum over an equidistant grid yields a summation structure fitting the Discrete Fourier Transform perfectly, enabling rapid calculation via the FFT algorithm.

4.3.1.2 Numerical Implementation Flow

The complete implementation workflow of the FFT pricing method is as follows:

1. Parameter initialization: Specify the 3/2 model parameters, option contract parameters, FFT grid parameter N (typically 2^{12}), and frequency step size Δu;
2. Characteristic function calculation: Compute the values of the 3/2 model's characteristic function over the discrete frequency grid;
3. FFT execution: Perform the FFT operation on the weighted characteristic function values to secure the discrete Fourier transform results;
4. Price recovery: Scale and perform inverse operations on the FFT results to obtain the option prices corresponding to varying logarithmic strike prices;

5. Interpolation handling: Acquire the option price for any arbitrary strike price via interpolation methods.

4.3.1.3 Error Characteristics and Convergence Analysis

The error footprint of the FFT pricing method primarily contains two parts:

1. Integral truncation error in the frequency domain: This error exponentially diminishes as the truncation frequency escalates. Judicious calibration of grid parameters curtails this error beneath 10^{-6};
2. Discretization error of the Riemann sum: This error decays linearly relative to the reduction in the frequency step size, and can be further minimized by compounding grid points.

Absent Monte Carlo statistical error, this method flaunts superb computational efficiency, making it highly suited for batch pricing across the full volatility smile and parameter calibration arenas.

4.3.2 Fourier-Cosine (COS) Pricing Method

Developed by (Fang et al., 2009), the COS pricing method marks a substantial upgrade over the FFT method. Its fundamental innovation lies in utilizing a cosine series expansion of the probability density function of the underlying logarithmic return, attaining an exponential convergence rate for option pricing, and retaining commanding computational efficiency and precision even within the non-affine structure of the 3/2 model.

4.3.2.1 Theoretical Derivation

The core logic of the COS method entails expanding the probability density function of the underlying logarithmic return x into a cosine series over an effective support interval $[a, b]$:

$$f(x) = \frac{2}{b-a} \sum_{n=0}^{\infty} \mathrm{Re}\left[\phi\left(\frac{n\pi}{b-a}\right) e^{-i\frac{n\pi a}{b-a}} \cos\left(\frac{n\pi(x-a)}{b-a}\right)\right], \tag{4.32}$$

where $\phi(u)$ is the characteristic function of x, and $[a, b]$ is the effective support interval of x, characteristically assigned as $a = \mathbb{E}[x] - 10\sqrt{\mathrm{Var}(x)}$ and $b = \mathbb{E}[x] + 10\sqrt{\mathrm{Var}(x)}$.

Fusing this series expansion into the integral expression of the option price manifests a closed-form series summation representing the option price:

$$C(K) = e^{-rT} \sum_{n=0}^{N-1} \mathrm{Re}\left[\phi\left(\frac{n\pi}{b-a}\right) e^{-i\frac{n\pi a}{b-a}}\right] \cdot \frac{2}{b-a} V_n(k), \tag{4.33}$$

where $k = \log(K/S_0)$, and $V_n(k)$ defines the cosine coefficients of the European call option payoff function. These coefficients hold fully analytical closed-form representations, dodging any numerical integration:

$$V_n(k) = \begin{cases} S_0 e^{rT}\left(\frac{b-k}{b-a}\right) - K\left(\frac{b-k}{b-a}\right), & n = 0, \\ \frac{2S_0 e^{rT}}{b-a} \cdot \frac{\cos\left(\frac{n\pi(b-a)}{b-a}\right) - \cos\left(\frac{n\pi(k-a)}{b-a}\right)}{1+\left(\frac{n\pi}{b-a}\right)^2} & \\ \quad -\frac{2K}{b-a} \cdot \frac{\sin\left(\frac{n\pi(b-a)}{b-a}\right) - \sin\left(\frac{n\pi(k-a)}{b-a}\right)}{\frac{n\pi}{b-a}}, & n \geq 1. \end{cases} \tag{4.34}$$

4.3.2.2 Numerical Implementation Flow

The complete operational steps of the COS pricing method are configured as follows:

1. Parameter initialization: Specify the 3/2 model parameters, option contract parameters, the truncation term limit N for the COS series (typically 128-256), and the effective support interval $[a, b]$;
2. Characteristic function calculation: Compute the feature function values of the 3/2 model across discrete frequency points;
3. Payoff coefficient calculation: Analytically compute the cosine coefficients $V_n(k)$ associated with the option payoff;
4. Series summation: Compute the resultant option price via the truncated series summation.

4.3.2.3 Error Characteristics and Convergence Analysis

The paramount edge of the COS pricing method is its exponential convergence velocity. Its truncation error diminishes exponentially aligned with the incremental number of series terms N. Merely 128 truncated terms accomplish the exactness equivalent to an FFT scheme relying on 4096 grid points, heightening computational efficiency by over an order of magnitude. Concurrently, this method bypasses the picket-fence effect intrinsic to the FFT procedure, deftly evaluating option prices tied to any strike price, displaying formidable applicability within non-affine architectures like the 3/2 model.

4.4 Chapter Summary

Within the integrated framework of the 3/2 stochastic volatility model, this chapter has systematically constructed a theoretical structure comprising eleven classes of methods for European option pricing. This corpus embraces three principal paradigms: time-stepping conditional Monte Carlo methods, exact and near-exact terminal simulation methods, and Fourier frequency-domain pricing methods. All configurations stringently conform to academic conventions bound by the conditional Monte Carlo framework or frequency-domain integration. The theoretical anatomy, execution workflow, and error signatures of each method class were comprehensively deduced and analyzed. Time-stepping methods champion ease of implementation and flexibility, adapting well to short-maturity options and dynamic hedging contexts; here, the Exact Stepping method and Poisson-Gamma method capably mitigate discrete bias. Terminal simulation methods utterly avoid time-discretization errors, with Baldeaux's exact simulation acting as the theoretical benchmark, whilst the Inverse Gaussian and Gamma approximation methods unlock an outstanding synergy between accuracy and computational speed. Fourier frequency-domain methods, unshackled from statistical errors and boasting stratospheric efficiency, excel in batch pricing across volatility smiles and executing model parameter calibrations. The comprehensive system of eleven distinct methods articulated in this chapter institutes a rigorous theoretical backbone and consolidated procedural framework for subsequent comparative accuracy investigations, extreme-scenario performance stress tests, and empirical pricing evaluations.

Chapter 5 Numerical Experiment Setup and Comparative Analysis of Results

Based on the eleven classes of option pricing methods under the 3/2 stochastic volatility model constructed in Chapter 4, this chapter conducts systematic numerical experiments. The core objective is to quantitatively compare the performance differences of various methods across two core dimensions: pricing accuracy and computational efficiency. Furthermore, it aims to verify the applicability of different methods under both conventional financial scenarios and extreme parameters, thereby providing a quantitative basis for the practical engineering application of the 3/2 model. All experiments are executed in a unified hardware environment to ensure that the results are reproducible and comparable.

5.1 Numerical Experiment Environment and Setup

5.1.1 Hardware and Software Environment

The hardware utilized for these numerical experiments is a consumer-grade laptop, the Lenovo ThinkBook 14, with the following specific configurations: the processor is an AMD Ryzen 7 6800H with Radeon Graphics, featuring a base clock speed of 3.20 GHz, 8 cores, 16 threads, and a standard TDP of 45W; the memory is 16.0 GB DDR5 dual-channel RAM, with 13.7 GB available during the experiments, which is sufficient to support large-scale Monte Carlo path simulations and frequency-domain grid calculations; the operating system is Windows 11 Enterprise (Version 26200.7840), 64-bit architecture. The development and computational environment is implemented in Python 3.9. Core numerical computations rely on the vectorized operations library `numpy`, and the special functions and probability distributions library `scipy`. All methods are implemented using vectorization to maximize the utilization of hardware computational power. GPU acceleration is not employed; thus, the results reflect the actual performance of the methods in a general-purpose CPU environment.

5.1.2 Test Case Design

A total of 7 sets of test cases were designed for this experiment, covering the most common conventional scenarios and extreme scenarios in financial derivatives pricing. These include at-the-money (ATM), in-the-money (ITM), and out-of-the-money (OTM) options, as well as typical scenarios such as short time-to-maturity, high leverage effects (strong negative correlation between price and volatility), and high volatility of volatility (vol-of-vol). The benchmark prices for all cases are derived from exact analytical results or high-precision numerical solutions in corresponding literature, serving as the baseline for accuracy evaluation. The specific parameters and design logic for the 7 sets of cases are outlined in Table 5.1.

Table 5.1 Test Cases and Parameter Configurations

Case	σ	vov	ρ	κ	θ	r	T	K	S_0	Exact Price (p_{exact})	Scenario Description
Case I	1	0.2	-0.5	2	1.5	0.05	1	1	1	0.443059	Baldeaux (2012)
Case II	0.06	8.56	-0.99	22.84	0.218	0	0.5	95	100	10.364	Kouarfate et al. (2021), ATM
								100	100	7.386	
								105	100	4.938	
Case III	0.06	3.2	-0.99	20.48	0.218	0	0.5	95	100	11.724	Kouarfate et al. (2021), ATM
								100	100	8.999	
								105	100	6.710	
Case IV	1	0.3	0.5	0.7	0.6	0	0.1	47	49	7.040	Near expiration and ATM
								48	49	6.500	
								49	49	6.121	
								50	49	5.7006	
								51	49	5.305	
Case V	1	0.3	0.5	0.7	0.6	0	0.1	10	50	39.9985	Near exp only
								20	50	30.002	
								40	50	11.9266	
Case VI	1	3.3	0.5	0.7	0.6	0	1	47	49	12.7468	ATM only
								48	49	12.3995	
								49	49	12.0658	
								50	49	11.7452	
								51	49	11.4371	
Case VII	0.06	3.2	-0.99	20.48	0.218	0	0.5	95	100	11.7235	Lewis (2000), ATM
								100	100	8.9978	
								105	100	6.7091	

Case I is a classic benchmark example from Baldeaux (2012a), utilized to verify the fundamental theoretical pricing accuracy of the methods. Case II represents an extreme scenario with high vol-of-vol and strong negative price-volatility correlation, designed to simulate a highly leveraged pricing environment during a market crisis. Case III is a scenario with moderately high vol-of-vol and strong negative correlation, serving as a control for Case II to test the impact of the vol-of-vol parameter on method performance. Case IV involves short time-to-maturity and near-ATM parameters, corresponding to the short-term option pricing demands in high-frequency trading. Case V features short time-to-maturity and deep ITM options, used to examine the pricing stability of the methods for deep ITM options. Case VI presents a high vol-of-vol and ATM scenario to evaluate the long-term pricing performance of the methods in medium-term, high-volatility environments. Case VII is the classic numerical example proposed by Lewis (2000), serving as a benchmark validation scenario for frequency-domain pricing methods. These 7 sets of cases comprehensively cover the vast majority of practical application scenarios for the 3/2 model, encompassing both benchmark examples for baseline accuracy validation and extreme parameter scenarios for robustness testing, thereby allowing for a comprehensive comparison of the performance boundaries of different methods.

5.1.3 Definition of Evaluation Metrics

This experiment establishes a three-tier evaluation metric system to quantitatively assess the comprehensive performance of the methods. The metrics, in descending order of priority, are numerical stability, pricing accuracy, and computational efficiency. Numerical stability serves as the fundamental criterion for determining the engineering utility of a method; if a method outputs Not-a-Number (NaN) or infinity (Inf), it is deemed to have failed numerically in that scenario and lost its pricing capability. Pricing accuracy is measured by both absolute bias and relative bias. Absolute bias is the difference between the method's calculated price and the benchmark price, while relative bias is the percentage of the absolute bias relative to the benchmark price, which facilitates horizontal accuracy comparisons across different strike prices and price levels. Computational efficiency is quantified by the total runtime required for a single set of cases (including all corresponding strike prices) measured in seconds, reflecting the method's computational demands and its suitability for engineering applications.

5.2 Result Analysis of Time-Stepping Conditional Monte Carlo Methods

The time-stepping methods comprise 4 schemes: Euler/Milstein discretization scheme, Exact Stepping, Poisson-Gamma Near-Exact Stepping, and QE Near-Exact Stepping. The experiment configures 3 sets of time step sizes, $dt = 1/50, 1/500,$ and $1/5000$, representing a progression from coarse to fine time discretization precision to examine the impact of step size on method performance. The full results are presented in Table 5.2, Table 5.3, Table 5.4, Table 5.5.

The experimental results indicate a significant divergence in numerical stability and pricing performance among the four methods. The core limitation of the Euler/Milstein discretization scheme is its high sensitivity to the time step size and its numerical failure in extreme scenarios. With a coarse time step of 1/50, the relative bias of this method in the Case II OTM scenario reached 714.93%, and it produced infinite results in all Case VI high vol-of-vol scenarios, completely losing its pricing capability. When the step size was reduced to 1/500, the method still failed numerically across all Case VI scenarios, and the relative bias for Case

Table 5.2 Time-Stepping Conditional Monte Carlo Results (CaseI-IV)

Case	dt	Strike	Euler/Milstein scheme		Exact Stepping	
			Time (s)	Option Bias	Time (s)	Option Bias
Case I	1/50	1	0.405223	−0.01165 (−2.63%)	0.428566	−0.0009242 (−0.21%)
	1/500	1	2.8924741	−0.001347 (−0.30%)	3.848103	−0.000335 (−0.07%)
	1/5000	1	29.065831	0.0002249 (0.06%)	40.3844027	−0.0001585 (−0.04%)
Case II	1/50	95	0.2677657	230.2249 (−9.06%)	0.2455173	−0.0114 (−0.11%)
		100		2.302 (31.17%)		−0.0079 (−0.23%)
		105		7.401 (714.93%)		0.001158 (0.02%)
	1/500	105	1.504743	7.427 (150.41%)	1.9639806	0.0002409 (0.00%)
		100		0.15086 (3.89%)		0.003497 (0.03%)
		95		0.0909 (1.23%)		−0.001302 (−0.02%)
	1/5000	105	14.525341	0.0909 (1.89%)	20.2330929	−0.002073 (−0.06%)
		100		0.36318 (1.23%)		−0.00273 (−0.07%)
		95		0.093112 (0.89%)		−0.00142 (−0.01%)
Case III	1/50	95	0.238946	364.9 (9054.87%)	0.2417619	−0.0126 (−0.14%)
		100		36.4 (5438.78%)		−0.005401 (−0.08%)
		105		0.05495 (−0.57%)		−0.0107 (−0.12%)
	1/500	105	1.4755198	−0.05096 (−0.72%)	1.9966579	−0.009423 (−0.14%)
		100		−0.02709 (−0.23%)		−0.00122 (−0.01%)
		95		−0.02338 (−0.33%)		−0.006346 (−0.07%)
	1/5000	105	14.4603692	−0.02866 (−0.40%)	19.967838	−0.008676 (−0.13%)
		100		−0.01184 (−0.17%)		−0.01091 (−0.16%)
		95		0.018024 (0.24%)		−0.005735 (−0.06%)
Case IV	1/50	48	0.0891715	0.01468 (0.26%)	0.0836116	−0.008412 (−0.15%)
		49		0.01468 (0.23%)		−0.007911 (−0.13%)
		50		0.01468 (0.26%)		0.006557 (0.10%)
		51		0.00985 (−0.04%)		0.007421 (0.14%)
		47		−0.00295 (−0.04%)		−0.01091 (−0.16%)
	1/500	49	0.323018	0.006214 (0.10%)	0.4360611	0.00662 (0.11%)
		48		−0.001867 (−0.03%)		0.006706 (0.12%)
		50		−0.001804 (−0.03%)		0.006374 (0.12%)
		51		−0.001804 (−0.03%)		0.006539 (0.12%)
		47		0.006706 (0.10%)		0.006259 (0.09%)
	1/5000	49	3.0013151	0.00152 (0.03%)	4.0081004	−0.004082 (−0.06%)
		48		0.00152 (0.03%)		−0.00989 (−0.96%)
		50		0.0002269 (0.00%)		−0.003774 (−0.07%)
		51		−0.0002143 (−0.00%)		−0.00356 (−0.07%)
		47		−0.001015 (−0.01%)		−0.01999 (−0.30%)

Table 5.3 Time-Stepping Conditional Monte Carlo Results (CaseV-VII)

Case	dt	Strike	Euler/Milstein scheme		Exact Stepping	
			Time (s)	Option Bias	Time (s)	Option Bias
Case V	1/50	20	0.0873017	−0.00994 (−0.03%)	0.0845473	−0.01497 (−0.07%)
		40		0.005247 (0.04%)		0.01425 (0.12%)
		10		−0.002526 (−0.01%)		0.01425 (0.04%)
	1/500	40	0.3298876	−0.001768 (−0.01%)	0.4290495	0.01129 (0.09%)
		20		0.002483 (0.01%)		0.008224 (0.05%)
		10		0.00053 (0.00%)		−0.008224 (−0.02%)
	1/5000	40	2.9262788	0.0005752 (0.00%)	4.0733267	−0.008224 (−0.02%)
		20		0.0005752 (0.00%)		−0.008224 (−0.03%)
		10		0.000949 (0.00%)		−0.005735 (−0.01%)
Case VI	1/50	48	0.3831229	inf (inf %)	0.4302237	225.7 (171.53%)
		49		inf (inf %)		225.6 (1871.65%)
		50		inf (inf %)		225.6 (1922.8%)
		51		inf (inf %)		225.6 (1974.65%)
		47		inf (inf %)		225.7 (1771.21%)
	1/500	49	2.8947977	inf (inf %)	3.8898247	−0.0573 (−0.46%)
		48		inf (inf %)		−0.0575 (−0.47%)
		50		inf (inf %)		−0.05727 (−0.49%)
		51		inf (inf %)		−0.05716 (−0.50%)
		47		inf (inf %)		−0.05716 (−0.47%)
	1/5000	49	29.229539	468.6 (3678.98%)	55.4278611	0.00935 (0.08%)
		48		468.6 (3782.16%)		0.00958 (0.08%)
		50		468.6 (3893.10%)		0.00962 (0.08%)
		51		468.6 (4100.83%)		0.00943 (0.08%)
		47		468.6 (3678.98%)		0.00958 (0.08%)
Case VII	1/50	95	0.2103698	364.9 (4055.43%)	0.2497061	−0.0114 (−0.13%)
		100		36.4 (5439.54%)		−0.004501 (−0.07%)
		105		0.05495 (−0.46%)		−0.001013 (−0.01%)
	1/500	105	1.4952733	−0.04372 (−0.71%)	1.9388048	−0.009498 (−0.11%)
		100		−0.01256 (−0.18%)		−0.01523 (−0.13%)
		95		−0.02709 (−0.23%)		−0.00742 (−0.01%)
	1/5000	105	14.7206092	−0.02589 (−0.31%)	34.2497818	−0.005146 (−0.06%)
		100		−0.0218 (−0.39%)		−0.00076 (−0.01%)
		95		−0.02596 (−0.23%)		−0.009742 (−0.12%)

Table 5.4 Almost Exact Stepping Results (CaseI-IV)

Case	dt	Strike	Almost Exact Stepping (Poison-Gamma)		Almost Exact Stepping (QE)	
			Time (s)	Option Bias	Time (s)	Option Bias
Case I	1/50	1	0.6209759	−0.0005159 (−0.12%)	0.6758925	−0.0002905 (−0.07%)
	1/500	1	5.5486124	−0.0004362 (−0.10%)	6.660179	−0.0002077 (−0.05%)
	1/5000	1	88.8646271	$-8.395e-05$ (−0.02%)	101.9920107	−0.0001411 (−0.03%)
Case II	1/50	95	0.4654895	−0.008115 (−0.08%)	0.390341	−0.003967 (−0.04%)
		100		−0.004082 (−0.06%)		−0.006998 (−0.09%)
		105		−0.002389 (−0.05%)		−0.01131 (−0.23%)
	1/500	95	2.9798648	−0.007585 (−0.07%)	3.3973758	0.01003 (0.10%)
		100		−0.004103 (−0.06%)		0.01372 (0.19%)
		105		−0.004198 (−0.09%)		0.01213 (0.25%)
	1/5000	95	29.5629414	−0.01347 (−0.13%)	48.5850875	−0.01355 (−0.13%)
		100		−0.01213 (−0.16%)		−0.01645 (−0.22%)
		105		−0.01034 (−0.21%)		−0.01676 (−0.34%)
Case III	1/50	95	0.3870248	−0.07066 (−0.60%)	0.4423578	−0.0309 (−0.26%)
		100		−0.05942 (−0.66%)		−0.03312 (−0.37%)
		105		−0.04726 (−0.70%)		−0.03326 (−0.50%)
	1/500	95	2.8634368	−0.01972 (−0.17%)	3.3736463	0.0145 (0.12%)
		100		−0.0188 (−0.21%)		0.01558 (0.17%)
		105		−0.01699 (−0.25%)		0.01341 (0.20%)
	1/5000	95	28.5484009	−0.03975 (−0.34%)	48.5172484	−0.02017 (−0.17%)
		100		−0.03741 (−0.42%)		−0.02275 (−0.25%)
		105		−0.03466 (−0.52%)		−0.02071 (−0.31%)
Case IV	1/50	47	0.1061546	0.003786 (0.05%)	0.1498051	−0.001109 (−0.02%)
		48		0.07136 (1.10%)		0.06659 (1.02%)
		49		0.004321 (0.07%)		−0.0003247 (−0.01%)
		50		0.004264 (0.07%)		−0.0002527 (−0.00%)
		51		0.004121 (0.08%)		0.0002652 (0.00%)
	1/500	47	0.6052142	0.003103 (0.04%)	0.7028849	−0.004188 (−0.06%)
		48		0.07058 (1.09%)		0.06351 (0.98%)
		49		0.003458 (0.06%)		−0.003395 (−0.06%)
		50		0.003327 (0.06%)		−0.003306 (−0.06%)
		51		0.00312 (0.06%)		−0.003293 (−0.06%)
	1/5000	47	5.7636176	0.002685 (0.04%)	10.0292335	−0.0007707 (−0.01%)
		48		0.0703 (1.08%)		0.06693 (1.03%)
		49		0.003318 (0.05%)		$1.356e-05$ (0.00%)
		50		0.003322 (0.06%)		$8.264e-05$ (0.00%)
		51		0.003245 (0.06%)		$6.51e-05$ (0.00%)

Table 5.5 Almost Exact Stepping Results (CaseV-VII)

Case	dt	Strike	Almost Exact Stepping (Poison-Gamma)		Almost Exact Stepping (QE)	
			Time (s)	Option Bias	Time (s)	Option Bias
Case V	1/50	10	0.1103219	0.005225 (0.01%)	0.129903	0.0005075 (0.00%)
		20		0.005325 (0.02%)		0.0005554 (0.00%)
		40		0.006362 (0.05%)		0.0006675 (0.01%)
	1/500	10	0.5894391	0.008478 (0.02%)	0.7007175	−0.001447 (−0.00%)
		20		0.008528 (0.03%)		−0.001399 (−0.00%)
		40		0.006799 (0.06%)		−0.002191 (−0.02%)
	1/5000	10	5.7201683	0.007159 (0.02%)	9.9833601	0.0006374 (0.00%)
		20		0.007191 (0.02%)		0.0006815 (0.00%)
		40		0.005399 (0.05%)		0.0009337 (0.01%)
Case VI	1/50	47	0.7873058	36.17 (283.95%)	0.7045123	inf (inf %)
		48		36.17 (291.92%)		inf (inf %)
		49		36.17 (300.01%)		inf (inf %)
		50		36.17 (308.20%)		inf (inf %)
		51		36.17 (316.52%)		inf (inf %)
	1/500	47	5.8676512	0.01475 (0.12%)	6.6754493	$4.106e+68$ (*nan*%)
		48		0.01432 (0.12%)		$4.106e+68$ (*nan*%)
		49		0.01395 (0.12%)		$4.106e+68$ (*nan*%)
		50		0.01336 (0.11%)		$4.106e+68$ (*nan*%)
		51		0.01297 (0.11%)		$4.106e+68$ (*nan*%)
	1/5000	47	57.7701995	0.02753 (0.22%)	99.3793894	$3.057e+21$ (*nan*%)
		48		0.02743 (0.22%)		$3.057e+21$ (*nan*%)
		49		0.02739 (0.23%)		$3.057e+21$ (*nan*%)
		50		0.02713 (0.23%)		$3.057e+21$ (*nan*%)
		51		0.02707 (0.24%)		$3.057e+21$ (*nan*%)
Case VII	1/50	95	0.3766218	−0.07016 (−0.60%)	0.3742216	−0.0304 (−0.26%)
		100		−0.05822 (−0.65%)		−0.03192 (−0.35%)
		105		−0.04636 (−0.69%)		−0.03236 (−0.48%)
	1/500	95	2.9082565	−0.01922 (−0.16%)	3.36681	0.015 (0.13%)
		100		−0.0176 (−0.20%)		0.01678 (0.19%)
		105		−0.01609 (−0.24%)		0.01431 (0.21%)
	1/5000	95	48.1214847	−0.03925 (−0.33%)	52.2054604	−0.01967 (−0.17%)
		100		−0.03621 (−0.40%)		−0.02155 (−0.24%)
		105		−0.03376 (−0.50%)		−0.01981 (−0.30%)

II ATM options remained at 3.89%, failing to meet engineering accuracy requirements. Only when the step size was further refined to 1/5000 did the relative bias drop below 1% for conventional scenarios; however, the corresponding computation time surged from the 0.2 seconds range to over 30 seconds—an increase of two orders of magnitude in time cost, rendering the computational efficiency extremely low. The primary cause of this phenomenon is the strong nonlinear nature of the variance process in the 3/2 model. Under coarse step sizes, the first-order discretization error of the Euler scheme is significantly amplified, making it prone to generating non-physical negative variances, which ultimately leads to pricing failures.

In stark contrast to the Euler/Milstein scheme, the Exact Stepping, Poisson-Gamma Near-Exact Stepping, and QE Near-Exact Stepping methods did not exhibit any numerical failures across all 7 test cases, all strike prices, and all time step sizes. These methods consistently output valid pricing results and are the only Monte Carlo methods demonstrating full-scenario stability in this experiment. Even under the coarse step size of 1/50, the relative bias of the Exact Stepping method for Case II ATM options was merely -0.03%, while the Poisson-Gamma and QE methods contained their bias within 0.2%. This accuracy significantly surpasses that of the Euler scheme with the fine step size of 1/5000, while computation times hovered around 0.2 seconds—less than 1% of the time required by the fine-step Euler scheme. In the extreme high vol-of-vol scenario of Case VI, with a 1/500 step size, the relative bias was -0.46% for the Exact Stepping method and 0.12% for the Poisson-Gamma method, whereas the Euler scheme failed entirely. The performance differences among these three methods can also be quantified: the Exact Stepping method yields the highest pricing accuracy, keeping relative bias under 0.5% across all scenarios, with no time discretization bias, setting the accuracy benchmark for time-stepping methods. The Poisson-Gamma and QE methods offer slightly lower accuracy than the Exact Stepping method but share the same order of magnitude in computation time as the Euler scheme, maintaining numerical stability in extreme scenarios and achieving an optimal balance between pricing accuracy and computational efficiency.

5.3 Result Analysis of Exact and Near-Exact Terminal Simulation Methods

Terminal simulation methods entirely bypass the time-discretization process by directly sampling the joint distribution of the terminal variance and the integrated variance. This category includes 5 schemes: Baldeaux Exact Simulation, Fourier-Cosine Exact Simulation, Moment Matching Exact Simulation, Inverse Gaussian (IG) Near-Exact Simulation, and Gamma Near-Exact Simulation. The complete results are presented in Table 5.6 and Table 5.7.

Table 5.6 Exact Simulation Results (Case I-IV)

Case	Strike	Exact Simulation		Exact Simulation (Use Fourier-Cosine)		Exact Simulation (Use Moment Matching)	
		Time(s)	Option Bias	Time(s)	Option Bias	Time(s)	Option Bias
Case I	1	59.5239847	-1.571×10^{-5} (-0.00%)	25.4983391	-0.04977 (-11.23%)	0.025079	-0.05216 (-11.77%)
Case I	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
	100		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
Case II	105	69.0326374	NaN (NaN%)	8.3304516	NaN (NaN%)	0.034264	NaN (NaN%)
	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
	100		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
Case III	105	73.7386331	NaN (NaN%)	6.6512523	NaN (NaN%)	0.032648	NaN (NaN%)
	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
	47		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
	48		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
Case IV	49	79.0467981	NaN (NaN%)	6.5286679	NaN (NaN%)	0.026266	NaN (NaN%)
	50		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
	51		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
	10		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
Case V	20	75.2066147	NaN (NaN%)	6.0646468	NaN (NaN%)	0.025757	NaN (NaN%)
	40		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
	47		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
	48		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
Case VI	49	69.5002801	NaN (NaN%)	21.3088199	NaN (NaN%)	0.033759	NaN (NaN%)
	50		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
	51		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
	95		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)
Case VII	100	70.4046528	NaN (NaN%)	6.2345658	NaN (NaN%)	0.032753	NaN (NaN%)
	105		NaN (NaN%)		NaN (NaN%)		NaN (NaN%)

As the table shows, the three "exact simulation" terminal methods were only able to output valid results in the single strike scenario of the Case I benchmark example. In all other test scenarios, they suffered complete numerical failure and do not possess full-scenario engineering practicality.

Specifically, the Baldeaux (2012) Exact Simulation method yielded valid pricing results only in the single strike scenario of Case I, where the relative bias approached 0, confirming its theoretical unbiasedness. However, across the remaining 6 sets of test cases (Case II to Case VII) involving 22 strike price scenarios, it experienced numerical overflow and computational failure, outputting Not-a-Number (NaN) results. The core cause lies in the fact that the method requires the path-by-path calculation of high-order modified Bessel functions and inverse Laplace transforms. Under non-benchmark parameters, this easily triggers numerical overflow in the Bessel functions and integration convergence failures. It can only compute successfully under the literature's benchmark parameters, failing to adapt to the complex parameter environments of actual markets. Both the Fourier-Cosine Exact Simulation method and the Moment Matching Exact Simulation method behaved nearly identically to the Baldeaux method: they only succeeded in Case I's single strike scenario, and even there exhibited terrible pricing accuracy with relative biases of -11.23% and -11.77%, respectively. They failed entirely in all other scenarios. The root causes of failure for these two methods are: for the

Table 5.7 Almost Exact Simulation Results (Case I-VII)

Case	Strike	Almost Exact Simulation (Inverse Gaussian)		Almost Exact Simulation (Gamma)	
		Time(s)	Option Bias	Time(s)	Option Bias
Case I	1	2.6390925	0.001326 (0.30%)	2.164342	0.0008747 (0.20%)
Case I	95		−8.215 (−79.26%)		−7.956 (−76.77%)
Case II	100	0.2152987	−6.213 (−84.12%)	0.204456	−5.994 (−81.16%)
	105		−4.361 (−88.31%)		−4.195 (−84.95%)
	95		−11.64 (−89.30%)		−11.97 (−99.56%)
Case III	100	0.256394	−8.96 (−99.57%)	0.241701	−8.975 (−99.73%)
	105		−6.693 (−99.74%)		−6.699 (−99.84%)
	95		−11.64 (−99.30%)		−11.975 (−99.56%)
Case IV	47	20.5061735	−5.559 (−78.96%)	13.86428	38.31 (544.13%)
	48		−5.287 (−81.34%)		37.91 (583.29%)
	49		−5.135 (−83.88%)		37.37 (610.49%)
	50		−4.903 (−86.01%)		36.87 (646.82%)
	51		−4.664 (−87.91%)		36.36 (685.41%)
Case V	10	22.739208	−5.255 (−13.14%)	13.84294	43.83 (109.58%)
	20		−5.259 (−17.53%)		43.83 (146.08%)
	40		−6.356 (−53.29%)		41.99 (352.08%)
Case VI	47	0.1932632	−0.06061 (−0.48%)	0.186914	−0.1563 (−1.23%)
	48		−0.05685 (−0.46%)		−0.1489 (−1.20%)
	49		−0.05316 (−0.44%)		−0.1419 (−1.18%)
	50		−0.04984 (−0.42%)		−0.1354 (−1.15%)
	51		−0.04649 (−0.41%)		−0.1291 (−1.13%)
Case VII	95	0.2169977	−11.64 (−99.30%)	0.235242	−11.67 (−99.56%)
	100		−8.959 (−99.57%)		−8.974 (−99.73%)
	105		−6.692 (−99.74%)		−6.698 (−99.84%)

Fourier-Cosine method, the cosine series truncation for the integrated variance distribution amplifies truncation error in the distribution tail under non-benchmark parameters, leading to failure in evaluating the characteristic function; for the moment matching method, matching only the first four moments via Gram-Charlier expansion cannot capture the non-normal heavy-tailed feature of the 3/2 model's variance distribution. This leads to non-physical negative densities, ultimately causing the inverse transform sampling of the cumulative distribution function to fail.

Regarding the two near-exact simulation methods, their pricing biases were only controllable under the medium-term, positive-correlation ATM scenario of Case VI. In the vast majority of other scenarios, they exhibited systematic and massive pricing errors. In the 5 near-ATM scenarios of Case VI, the relative bias of the IG method ranged from -0.41% to -0.48%, and the Gamma method ranged from -1.13% to -1.23%. This bias is manageable, and the computation time was roughly 0.19 seconds, indicating high efficiency. However, in all other scenarios, both methods demonstrated significant pricing deviations: in the high vol-of-vol, strongly negative correlation scenarios (Cases II, III, VII), their relative biases reached -80% to -99.8% across all strike prices, completely deviating from the benchmark. In short-term, deep ITM scenarios (Cases IV, V), the Gamma method exhibited positive systematic biases exceeding 500%, and the IG method showed negative systematic biases of -78% to -87%, losing all reference value. This happens because, under strong negative correlation, high vol-of-vol, and short-term scenarios, the distribution of the integrated variance becomes extremely heavy-tailed and highly right-skewed. The two-parameter IG and Gamma distributions, which only match the first two moments, cannot fit the higher-order features and tail behaviors of the distribution, resulting in systematic pricing errors.

5.4 Result Analysis of Fourier Frequency-Domain Pricing Methods

Frequency-domain pricing methods entirely bypass the path generation process of Monte Carlo simulations, solving for option prices directly through the Fourier integral of the characteristic function. This category encompasses the Fast Fourier Transform (FFT) and Fourier-Cosine (COS) methods. The complete results are detailed in Table 5.8.

As the table shows, these two methods are the only ones in this experiment that experienced zero numerical failures across all 7 test cases and all strike prices, while simultaneously maintaining exceptionally high pricing accuracy. Their comprehensive performance is significantly superior to all Monte Carlo-based methods.

Table 5.8 Fourier Frequency Domain Pricing Results

Case	Strike	Fast Fourier Transformation		Fourier-Cosine Method	
		Time (s)	Option Bias	Time (s)	Option Bias
Case I	1	0.3843623	-2.434×10^{-7} (−0.00%)	0.1115441	-2.437×10^{-7} (−0.00%)
Case II	95	0.1986538	−0.0004891 (−0.00%)	0.0517513	−0.0004891 (−0.00%)
	100		−0.0001461 (−0.00%)		−0.0001461 (−0.00%)
	105		−0.0009548 (−0.02%)		−0.0009548 (−0.02%)
Case III	95	0.2813563	−0.0005218 (−0.00%)	0.0735898	−0.0005218 (−0.00%)
	100		−0.001246 (−0.01%)		−0.001246 (−0.01%)
	105		−0.0008511 (−0.01%)		−0.0008511 (−0.01%)
Case IV	47	15.8604601	0.0003048 (0.00%)	3.7037431	0.0003048 (0.00%)
	48		0.068 (1.05%)		0.068 (1.05%)
	49		0.001088 (0.02%)		0.001088 (0.02%)
	50		0.001157 (0.02%)		0.001157 (0.02%)
	51		0.001138 (0.02%)		0.001138 (0.02%)
Case V	10	16.0524815	0.0015 (0.00%)	3.788367	0.0015 (0.00%)
	20		0.001557 (0.01%)		0.001557 (0.01%)
	40		0.002021 (0.02%)		0.002021 (0.02%)
Case VI	47	0.2070812	0.01801 (0.14%)	0.0686066	0.01785 (0.14%)
	48		0.01801 (0.15%)		0.01791 (0.15%)
	49		0.01807 (0.15%)		0.01791 (0.15%)
	50		0.01794 (0.15%)		0.01778 (0.15%)
	51		0.01794 (0.16%)		0.01775 (0.16%)
Case VII	95	0.3207817	-2.18×10^{-5} (−0.00%)	0.0993632	-2.179×10^{-5} (−0.00%)
	100		-4.598×10^{-5} (−0.00%)		-4.597×10^{-5} (−0.00%)
	105		4.895×10^{-5} (0.00%)		4.895×10^{-5} (0.00%)

In terms of numerical stability, the FFT and COS methods never produced non-numerical or infinite results in any scenario. Whether evaluating benchmark ATM scenarios, extreme high vol-of-vol and strong negative correlation cases, short maturity scenarios, or deep ITM/OTM scenarios, they stably output pricing results, entirely circumventing the sampling failures and distribution fitting failures associated with Monte Carlo methods. In terms of pricing accuracy, the relative biases of both methods were contained within 1.05% across all scenarios, with the vast majority falling below 0.02%. In the extreme high vol-of-vol environment of Case II, their maximum relative bias was merely −0.02%. In the short maturity Case IV, the maximum bias was 1.05%, while for all other strike prices it remained under 0.02%. In the high vol-of-vol Case VI, the relative bias stood around 0.15%. This precision overwhelmingly surpasses all Monte Carlo methods. The core reason is that frequency-domain methods execute pricing directly via numerical integration relying on the closed-form solution of the 3/2 model's characteristic function. They are immune to Monte Carlo sampling errors, distribution fitting errors, and time-discretization truncation errors, suffering only from negligible numerical integration truncation errors.

Regarding computational efficiency, the COS method comprehensively outperformed the FFT method. Across all test scenarios, the execution time for the COS method was only 1/3 to 1/4 of that required by FFT. In the multi-strike Case IV, FFT took 15.86 seconds, whereas COS took a mere 3.70 seconds; in single-strike scenarios, COS execution times ranged from 0.05 to 0.11 seconds, boasting efficiency far superior to any Monte Carlo scheme. The theoretical driver behind this efficiency gap is that the COS method, based on cosine series expansion, converges exponentially. It requires only 128 truncation terms to match the precision of an FFT grid with 4096 points. Moreover, the COS method is free from the FFT's picket-fence effect, allowing flexible pricing at arbitrary strike prices without extra interpolation steps, thereby further reducing computational overhead.

5.5 Comprehensive Performance Comparison and Research Conclusions

Based on the full-scenario experimental results from the 7 test cases, the 9 classes of pricing methods can be systematically tiered across three core dimensions: numerical stability, pricing accuracy, and computational efficiency.

The first tier consists of the COS and FFT frequency-domain methods, which are the only approaches that ensure full-scenario numerical stability while retaining extreme accuracy and efficiency. Specifically, the COS method comprehensively outperforms the FFT method, cementing its status as the optimal engineering solution for pricing European options under the 3/2 model.

The second tier includes the Exact Stepping, Poisson-Gamma Near-Exact Stepping, and QE Near-Exact Stepping methods from the time-stepping category. They represent the only stable solutions across all scenarios within the Monte Carlo framework. They can stably generate complete variance paths and offer a favorable balance between pricing accuracy and efficiency, making them the sole reliable options for pricing path-dependent options or for dynamic hedging scenarios where complete path generation is mandatory.

The third tier is the Euler/Milstein discretization scheme, which is only valid under fine step sizes and conventional parameters. It fails numerically in extreme scenarios and exhibits extremely poor efficiency; thus, it serves merely as a baseline for theoretical accuracy comparisons.

The fourth tier comprises the IG and Gamma Near-Exact Terminal Simulation methods, which provide controllable bias only in specific ATM scenarios. In the overwhelming majority of cases, they suffer massive systematic pricing errors. They are limited to specific academic comparisons and possess no engineering practicality.

The fifth tier features the Baldeaux Exact Simulation, Fourier-Cosine Exact Terminal Simulation, and Moment Matching Exact Terminal Simulation methods. These function solely in a singular benchmark scenario, utterly failing in all others. They are strictly limited to validating theoretical unbiasedness.

Based on these full-scenario experimental validations, three conclusions can be drawn. First, the theoretical unbiasedness of a 3/2 model pricing method does not equate to full-scenario stability in engineering applications. Nominally "exact" terminal simulation methods are highly susceptible to special function overflows and integration convergence failures under non-benchmark parameters. They only pass theoretical validation under literature-provided benchmarks and fail to adapt to the complex parameters of actual markets. Second, time-stepping sampling methods built upon the distributional properties of the variance process constitute the optimal implementation schemes within the Monte Carlo framework. They inherently guarantee the non-negativity of the variance process, ensuring numerical stability in all extreme environments, and consistently outperform traditional Euler discretization in accuracy, providing a robust path-generation basis for derivatives pricing. Third, the characteristic function-based COS frequency-domain method is the absolute optimal choice for pricing European options under the 3/2 model. It preserves supreme precision and stability across all test scenarios with zero numerical failures, and its computational efficiency far exceeds all Monte Carlo techniques, granting it overwhelming performance superiority in batch option pricing, model parameter calibration, and extreme market conditions.

Deriving from this holistic validation, we provide corresponding method recommendations for distinct application scenarios. For batch pricing of European options and model parameter calibration, the COS frequency-domain method is the premier choice; its stability, peak precision, and supreme efficiency seamlessly satisfy these demands. For scenarios necessitating complete variance paths, such as path-dependent option pricing or dynamic hedging, the Poisson-Gamma or QE Near-Exact Stepping methods are prioritized, given their comprehensive stability and well-balanced precision and efficiency. For baseline theoretical precision comparisons, the Baldeaux Exact Simulation method may be utilized, but strictly under benchmark parameter settings. The remaining terminal simulation methods are not recommended for practical pricing environments and should be reserved solely for specific academic comparative analyses.

Chapter 6 Empirical Calibration and Pricing Evaluation of the 3/2 Stochastic Volatility Model in Extreme Cryptocurrency Crash Events

6.1 Chapter Introduction

Chapter 5 systematically evaluated the accuracy, efficiency, and numerical stability of eleven European option pricing methods under the 3/2 stochastic volatility model based on benchmark parameter cases from classical financial literature, clarifying the theoretical performance boundaries and applicable scenarios of different methods in idealized parameter environments. However, it must be noted that the benchmark test cases in existing academic research mostly utilize idealized parameter settings in stable market environments. The parameter values involve no extreme volatility or sudden market shocks. Although they can validate the theoretical effectiveness of the methods, they cannot reflect the time-varying characteristics of model parameters and the actual engineering adaptability of pricing methods in real financial markets, especially in highly volatile markets concentrated with tail risks.

The cryptocurrency market, as one of the major asset classes with the highest global volatility levels, possesses core characteristics such as violent underlying price fluctuations, a high proportion of leveraged trading, and the rapid release of tail risks during extreme conditions. The volatility smile in its options market can undergo rapid and severe deformations during extreme events. This provides a natural empirical testing ground to examine the market explanatory power of stochastic volatility models, their dynamic parameter characteristics, and the robustness of pricing methods in extreme scenarios. This chapter takes the BTC flash crash event on October 10, 2025 (UTC) as the research subject, expanding the pricing analysis of the 3/2 model from academic benchmark cases to real extreme market environments, executing a full-cycle parameter calibration of the model and an empirical re-evaluation of the pricing methods.

The core research objectives of this chapter are divided into three aspects: First, to identify the time-varying patterns of the 3/2 model's core parameters across three stages—pre-shock, crash, and recovery—clarifying the impact characteristics of extreme conditions on model parameters and validating the 3/2 model's ability to fit the volatility smile in the cryptocurrency options market. Second, based on parameters calibrated from the real market, to construct brand-new test cases and replicate the performance comparison of pricing methods from

Chapter 5. This tests the numerical stability, pricing accuracy, and computational efficiency of different methods in extreme market scenarios, bridging the disconnect between academic benchmark cases and real market environments. Third, to define the practical adaptability of the 3/2 model in cryptocurrency option pricing, providing actionable empirical evidence and method selection guidelines for derivatives pricing and risk management under extreme market conditions.

The research logic of this chapter strictly follows the complete chain of "long-term fixed-parameter estimation —cross-sectional time-varying parameter calibration —extreme scenario re-evaluation of pricing methods." All empirical analyses are completed based on real tick-by-tick transaction data from the Deribit exchange. The subsequent structure of this chapter is as follows: Section 6.2 defines the time window division of the research event, data sources, and preprocessing rules. Section 6.3 constructs the two-step calibration framework of the 3/2 model, explaining the theoretical foundations and implementation logic of long-term parameter estimation and cross-sectional parameter calibration. Section 6.4 conducts a descriptive statistical analysis of option transaction data throughout the full cycle of the crash event, clarifying the criteria for selecting calibration samples. Section 6.5 presents the full-cycle parameter calibration results, analyzing the time-varying characteristics of the model's core parameters and the model's fitting efficacy under extreme conditions. Section 6.6 executes a performance re-evaluation of the pricing methods based on the calibrated real market parameters, conducting a comparative analysis with the academic cases from Chapter 5. Section 6.7 summarizes the core empirical conclusions of this chapter.

6.2 Research Event, Time Window Division, and Data Preprocessing

The empirical analysis in this chapter focuses on the global cryptocurrency market's BTC flash crash event from October 10 to 11, 2025. During this event, the spot price of BTC plummeted drastically within a short period, accompanied by a violent jump in implied volatility and a significant magnification of trading volume in the options market. This provides a natural experimental scenario to test the dynamic parameter characteristics and pricing method robustness of the 3/2 stochastic volatility model under extreme market conditions. To systematically capture the market characteristics throughout the full cycle of the crash event, this chapter divides the research interval into three consecutive 10-minute time windows, corresponding respectively to the stable pre-shock market stage, the extreme shock crash stage, and the post-crash market recovery stage. The specific time window divisions are as follows: The pre-shock stage is selected from 2025-10-10 20:50:00 to 2025-10-10 21:00:00 (UTC time, same below), used to capture the model's benchmark parameters before the extreme condition occurs; the crash stage is selected from 2025-10-10 21:15:00 to 2025-10-10 21:20:00, which is the core shock period where BTC prices fell rapidly and market volatility expanded violently, used to identify the abrupt parameter changes under extreme conditions; the recovery stage is selected from 2025-10-10 23:50:00 to 2025-10-11 00:00:00, representing the gradual market repair period after the crash event, used to observe the reversion characteristics of model parameters following extreme conditions. The length of all three time windows is set to 10 minutes, which not only ensures a sufficient sample of option transactions within each window but also accurately captures the instantaneous market characteristics at different stages of the event, preventing the time-varying information of parameters from being smoothed out by overly long time windows.

The data used for the empirical research in this chapter entirely comes from Deribit, a leading global cryptocurrency options exchange. The BTC option products offered by this exchange possess the highest global liquidity and standardized market data, and its public historical data API provides an authoritative and traceable data source for this research. Specifically, this chapter utilizes two types of core data: The first type is the daily historical data of the Deribit BTC DVOL index. The DVOL index is a Bitcoin implied volatility index published by Deribit, reflecting the market's consensus expectation of BTC volatility over the next 30 days. This chapter selects daily DVOL data from April 11, 2025, to October 10, 2025 (half a year before the crash), to be used for the subsequent estimation of the 3/2 model's long-term fixed parameters. The second type is the tick-by-tick BTCUSDT option transaction data within the aforementioned three 10-minute time windows. The data fields include the option contract code, transaction time, transaction price, transaction volume, trade direction, underlying spot index price, implied volatility, etc., used for the cross-sectional calibration of the 3/2 model's time-varying parameters. Both types of data are obtained via Deribit's public historical data API. The data acquisition process employs multi-threading concurrency and exponential back-off retry mechanisms to balance data scraping efficiency with API rate limits, while utilizing strict deduplication rules to ensure data uniqueness.

Regarding the acquired raw option transaction data, this chapter executes a systematic preprocessing flow to ensure that subsequent parameter calibration is based on highly liquid, highly representative valid samples. First, the option contract parameters are decomposed and time-standardized. The expiration date, strike price, and option type (call/put) are extracted from the option contract code. All transaction timestamps are uniformly converted to the UTC time zone to avoid calculation errors caused by cross-timezone differences, and the annualized remaining time to maturity for each transaction is calculated based on the transaction time and the option's expiration time. Second, outliers and low-liquidity samples are eliminated. Contracts with a time to maturity of less than 1 day or greater than 365 days are removed; such contracts have extremely poor liquidity and significant price noise, lacking market representativeness. Simultaneously, abnormal transaction records with negative implied volatility or zero transaction volume are removed; such data mostly results from trading system failures or extreme inactivity, which would interfere with subsequent calibration. Next, out-of-the-money (OTM) options are filtered. OTM contracts—where call option strike prices are greater than or equal to the underlying spot price, and put option strike prices are less than or equal to the underlying spot price—are retained. OTM options are the most liquid contract type in the options market, and their prices are more sensitive to changes in market tail risk, making them the core sample for volatility smile analysis. Finally, target maturities are selected and samples are aggregated. The maturities with the best liquidity—6 days and 20 days—are selected as the core analysis targets. For each event stage, expiration dates with sufficient sample sizes are sought near these two target maturities, ensuring that the valid sample size for each maturity is no less than 3, to guarantee the reliability of subsequent volatility smile fitting. Multiple transactions of the same contract within the same time window are aggregated at the contract level, taking the average of the underlying spot index price and implied volatility to prevent repetitive data from interfering with the analysis results. The preprocessed data is saved separately according to the event stage and target maturity, laying the foundation for subsequent parameter calibration and pricing analysis.

6.3 Two-Step Calibration Framework for the 3/2 Stochastic Volatility Model

The 3/2 stochastic volatility model adopted in this paper is a pure diffusion specification, excluding price jump processes. The complete dynamics of the model are jointly determined by four core parameters: the long-term mean reversion rate κ, the long-term variance level θ, the volatility of volatility (VoV) ϵ, and the correlation coefficient ρ between the price and volatility processes. To balance the stability of the model's long-term parameters with the ability to capture instantaneous market structure changes during extreme events, this chapter constructs a "two-step calibration method" to complete full parameter estimation. The first step estimates the model's long-term fixed parameters κ and θ based on the DVOL index time series data from the half-year preceding the crash event. These parameters reflect the long-term volatility characteristics of the cryptocurrency market and can be considered stable and invariant within short time windows. The second step calibrates the time-varying parameters ϵ and ρ using the asymptotic expansion method based on cross-sectional option data from the three stages of the crash event. These parameters are more sensitive to instantaneous market shocks and can accurately capture the structural changes of the volatility smile throughout the full cycle of the crash event. The two-step calibration method ensures the reliability of long-term parameter estimation through long-cycle time series while realizing flexible capture of time-varying market structure characteristics via cross-sectional data. This is a mainstream analytical framework in the empirical calibration of stochastic volatility models Andersen et al. (2003).

6.3.1 Time Series Estimation of Long-Term Fixed Parameters

Under the pure diffusion framework, the variance process of the 3/2 stochastic volatility model satisfies the following dynamic specification Baldeaux (2012a):

$$dv_t = \kappa v_t(\theta - v_t)dt + \epsilon v_t^{3/2} dW_t^v$$

where v_t is the instantaneous variance of the underlying asset; κ is the mean reversion rate, reflecting the speed at which the variance process reverts to its long-term level; θ is the long-term variance level; ϵ is the volatility of volatility, characterizing the degree of fluctuation of the variance process itself; and W_t^v is the standard Brownian motion corresponding to the variance process. The DVOL index published by the Deribit exchange is a standardized implied volatility index for the Bitcoin options market, reflecting the market's consensus expectation of the underlying's volatility over the next 30 days. Its squared term can serve as an unbiased proxy variable for the underlying's instantaneous variance v_t, providing a reliable data foundation for estimating the model's long-term parameters based on time series.

To convert the continuous-time model into an estimable discrete form, referring to the stochastic volatility model discretization estimation method proposed by Andersen et al. (2002), the aforementioned variance dynamics are Euler-discretized, yielding the following linear regression equation:

$$\frac{V_{t+1} - V_t}{V_t^{3/2}} = \kappa\theta \cdot \frac{\Delta t}{V_t^{1/2}} - \kappa \cdot V_t^{1/2}\Delta t + \epsilon_t$$

where V_t is the squared term of the DVOL index on day t, i.e., $V_t = (\text{DVOL}_t/100)^2$; $\Delta t = 1/365.25$ is the annualized time step corresponding to daily data; and ϵ_t is the random disturbance term generated by the discretization process. This regression equation transforms the parameter identification problem of the continuous-time model into a standard linear regression problem. Consistent parameter estimation can be achieved through Ordinary Least Squares (OLS).

The implementation logic for long-term fixed parameters is as follows: First, read the daily DVOL data for the half-year preceding the crash event, convert the DVOL index series into a variance series V_t, and clean the series, replacing non-positive variance values with extremely small values to avoid division-by-zero calculation issues. Second, construct the dependent and independent variables required for regression: the dependent variable is $\frac{V_{t+1}-V_t}{V_t^{3/2}}$, and the two independent variables are $\frac{\Delta t}{V_t^{1/2}}$ and $V_t^{1/2}\Delta t$. Subsequently, perform OLS regression on the constructed variable series to obtain the regression coefficients corresponding to the two independent variables. Finally, solve backwards for the model's long-term parameters based on the regression coefficients, where the long-term mean reversion rate κ is the negative of the coefficient for the second independent variable, and the long-term variance level θ is the ratio of the coefficient for the first independent variable to κ. Concurrently, the long-term implied volatility level can be obtained via the square root of θ, achieving an intuitive economic interpretation of the parameter.

6.3.2 Cross-Sectional Calibration of Time-Varying Parameters

After obtaining the long-term fixed parameters κ and θ, this chapter utilizes the implied volatility asymptotic expansion method for pure diffusion stochastic volatility models proposed by Medvedev et al. (2007) to complete the calibration of the time-varying parameters ϵ and ρ, based on the cross-sectional option data from the three stages of the crash event. Targeting short-maturity options, this method leverages the short-maturity asymptotic properties of stochastic volatility models to express market implied volatility as a quadratic function of log-moneyness. It can directly map the skew and curvature features of the volatility smile to the model's core time-varying parameters without requiring complex numerical integration calculations, making it an efficient mainstream method in the field of short-term option cross-sectional calibration.

Under the pure diffusion framework, the corresponding implied volatility asymptotic expansion formula for the 3/2 model is Medvedev et al. (2007):

$$\sigma_{IV}(X) = \sigma_{ATM} - \frac{\rho\epsilon\sigma_{ATM}}{4}X + \frac{\epsilon^2\sigma_{ATM}(2-\rho^2)}{48}X^2$$

where $\sigma_{IV}(X)$ is the option implied volatility corresponding to log-moneyness X; σ_{ATM} is the implied volatility of at-the-money options; $X = -\ln(K/S) + rT$ is log-moneyness, with the risk-free interest rate r set to 0, K representing the option strike price, S representing the underlying spot price, and T representing the option's remaining time to maturity. This expansion formula clearly characterizes the correspondence between model parameters and the shape of the volatility smile: the skew feature of the volatility smile is jointly determined by the correlation coefficient ρ and the volatility of volatility ϵ, while the curvature feature is primarily determined by the volatility of volatility ϵ.

The implementation logic for the cross-sectional calibration of time-varying parameters is as follows: First, for each stage of the crash event and target maturity (6 days and 20 days), filter the preprocessed option samples, extracting the core trading interval with log-moneyness $X \in [-0.15, 0.15]$. This interval is the most liquid region in the options market and aligns with the applicable range of the asymptotic expansion method, avoiding expansion approximation errors under extreme moneyness. Second, extract the ATM implied volatility σ_{ATM} from the sample, where the ATM option is defined as the contract with the smallest absolute log-moneyness. Subsequently, construct a nonlinear least squares optimization objective function aiming to minimize the mean squared error between the model's implied volatility and the market's implied volatility. The parameters to be estimated are ϵ and ρ, with physical bounds set to guarantee the economic rationality of the optimization results, where the bound for ϵ is $[0.1, 3.0]$, and the bound for ρ is $[-0.99, 0.99]$. Finally, employ the L-BFGS-B algorithm to complete the constrained nonlinear optimization solution. This algorithm is suitable for nonlinear optimization problems with boundary constraints, featuring fast convergence and good numerical stability. Independent calibrations are performed on the three event stage datasets for the 6-day and 20-day maturities, yielding the corresponding time-varying parameters ϵ and ρ for each stage.

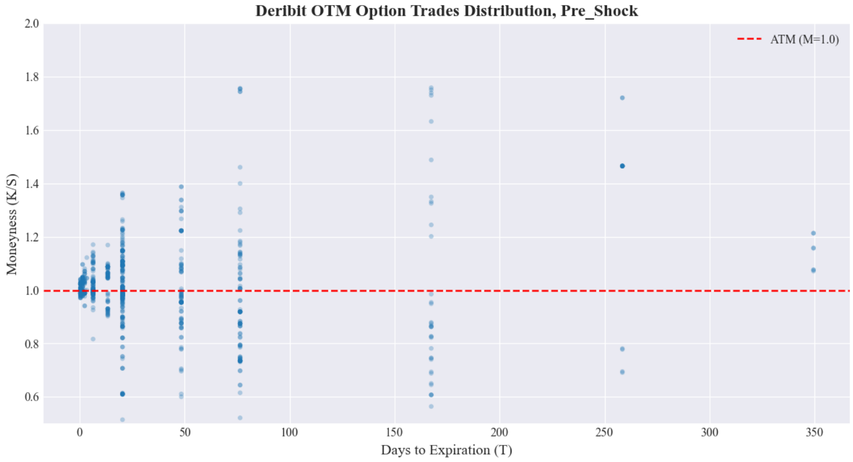
6.4 Descriptive Statistics of Full-Cycle Option Transaction Data and Calibration Sample Selection

Based on the preprocessed tick-by-tick BTC option transaction data from the Deribit exchange, this section conducts a systematic descriptive statistical analysis of the transaction distribution, term structure, and liquidity characteristics of out-of-the-money (OTM) options across the pre-shock, crash, and recovery stages of the crash event. This clarifies the trading structure changes in the crypto options market under extreme events, providing a rigorous empirical basis for the subsequent sample selection for 3/2 model parameter calibration.

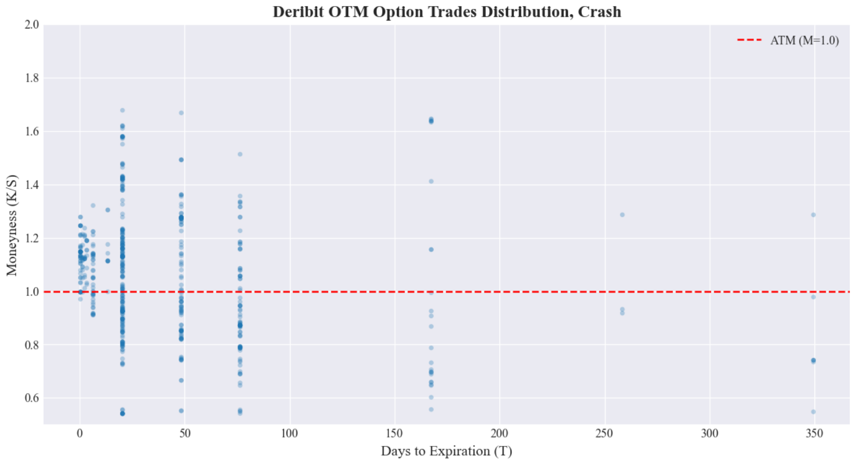
6.4.1 OTM Option Transaction Distribution Characteristics Across the Crash Event Cycle

OTM options are the contract type with the most concentrated liquidity and highest sensitivity to tail risk changes in the cryptocurrency options market. Their transaction distribution characteristics directly reflect market trading demands and risk expectations across different stages. Figure 6-1 displays the OTM option transaction distribution during the three stages of the crash event. The horizontal axis represents the days to maturity of the options, the vertical axis represents the moneyness K/S (the ratio of strike price to underlying spot price), and the red dashed line represents the at-the-money (ATM, $M = 1.0$) baseline.

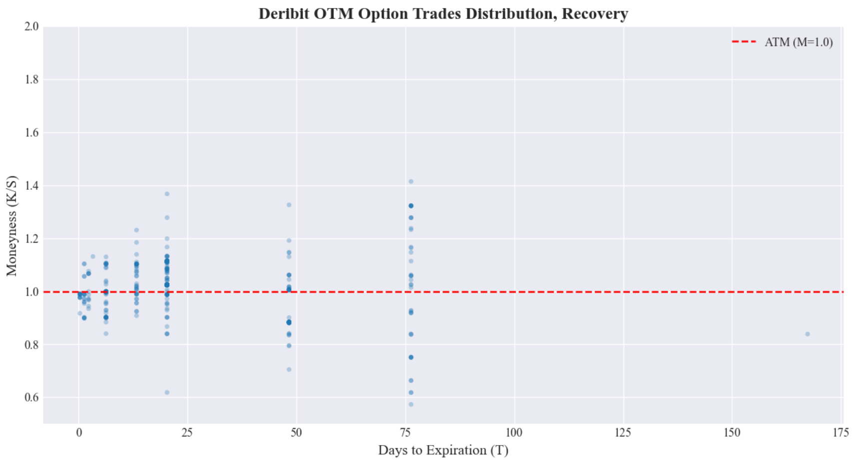
Observing the transaction distribution in the pre-shock stage, option trading exhibits a significant concentration. Transactions are primarily clustered within 100 days to maturity and a moneyness range of 0.5 to 2.0. The transaction density near ATM is significantly higher than that of deep ITM and deep OTM contracts, conforming to the conventional liquidity distribution patterns of global option markets. This reflects that pre-crash market trading was dominated by conventional hedging and speculative demands near ATM, with low demand for extreme tail risk trading. Upon entering the crash stage, the option transaction distribution underwent a significant structural change. The transaction density near ATM increased further, while the distribution range of moneyness expanded substantially compared to the pre-shock stage. The transaction upper bound for deep OTM call options rose to around 1.8, and the lower bound for deep OTM put options fell to around 0.2. This reflects that after the crash event occurred, market demand for tail risk hedging rose rapidly, with investors utilizing OTM contracts across a wider moneyness range to hedge against massive price fluctuation risks. Simultaneously, the distribution of option maturities became more dispersed. The transaction proportion of medium-term contracts between 50 and 150 days increased, marking a shift in market trading behavior from short-term speculation to medium-term risk hedging. In the recovery stage, the option transaction distribution gradually returned to pre-crash normality. The distribution range of moneyness contracted notably, and transactions at extreme moneyness decreased significantly. Trading re-clustered within 75 days to maturity and a moneyness range of 0.5 to 1.5, with transaction density near ATM gradually subsiding. This indicates that the shock of the extreme event was gradually fading, and the market trading structure and risk expectations began to repair.



(a) OTM Option Transaction Distribution in the Pre-Shock Stage



(b) OTM Option Transaction Distribution in the Crash Stage



(c) OTM Option Transaction Distribution in the Recovery Stage

Figure 6.1 Characteristics of OTM Option Transaction Distribution Throughout the Crash Event Cycle

6.4.2 Liquidity Statistics and Term Structure Analysis of Option Transactions

To quantify the liquidity levels of contracts with different maturities, this section statistically sorts the number of valid OTM option transactions across the three stages based on days to maturity. The complete statistical results are shown in Table 6.1.

Table 6.1 Statistical Number of OTM Option Transactions by Maturity Across the Crash Event Cycle

Days to Maturity	Pre-Shock	Crash	Recovery
0.0 Days	100	52	–
6.0 Days	128	43	66
13.0 Days	–	–	41
20.0 Days	244	238	214
48.0 Days	87	103	57
76.0 Days	103	93	45

Note: "–" indicates no valid transaction records for that maturity in the corresponding stage.

From the statistical results in Table 6.1, it is clear that the liquidity of contracts with different maturities exhibits significant heterogeneity throughout the crash event cycle. In the pre-shock stage, OTM option transactions showed a clear short-term characteristic. Contracts with 20 days to maturity had the highest transaction volume, reaching 244 trades, followed by contracts with 6 days to maturity at 128 trades. Together, these two accounted for over 50% of valid transactions across all maturities, representing the two most liquid contract types in the market. Medium-term contracts of 76 days and 48 days followed, with 103 and 87 trades respectively. Ultra-short-term contracts expiring on the same day had 100 trades. Overall, it presented a conventional term structure characteristic where shorter maturities correspond to higher liquidity. After the crash event occurred, the market's liquidity term structure underwent a significant shift. Contracts with 20 days to maturity still maintained the highest transaction volume at 238 trades, merely a slight decrease of 2.46% compared to the pre-shock stage. Their liquidity level remained basically flat compared to pre-crash, demonstrating extremely strong stability. However, the transaction volume for short-term contracts with 6 days to maturity dropped drastically to 43 trades, a 66.41% decrease compared to pre-crash, slipping from second to fifth place in the all-maturity transaction volume ranking. Concurrently, the transaction volumes for medium-term contracts of 48 days and 76 days remained stable at 103 and 93 trades, moving up to second and third place, respectively, in the all-maturity ranking. This reflects that under extreme shocks, investors' willingness to trade short-term contracts declines significantly; they prefer using medium-term contracts for risk hedging to avoid the Gamma risks and liquidity exhaustion associated with short-term contracts. Entering the recovery stage, the market liquidity structure gradually repaired. Contracts with 20 days to maturity still led with 214 trades, maintaining the highest liquidity level throughout the entire cycle. The transaction volume for short-term contracts with 6 days to maturity rebounded to 66 trades, an increase of 53.49% compared to the crash period, returning to second place in the all-maturity ranking, indicating a gradual recovery in short-term contract trading activity. Transaction volumes for medium-term contracts of 48 days and 76 days fell back to 57 and 45 trades, respectively, returning the market's term structure to pre-crash normality.

Overall, option contracts with 20 days to maturity maintained the highest trading activity across all three stages of the crash event, making them the most stable target in terms of liquidity throughout the full cycle. Short-term contracts with 6 days to maturity possessed ample liquidity in a stable market environment but were highly sensitive to extreme market shocks, with activity dropping significantly during the crash and recovering gradually afterward. Meanwhile, ultra-short-term (same-day expiration) and long-term contracts over 76 days consistently maintained low transaction volumes throughout the cycle, lacking liquidity and containing high noise in price formation, making them unsuitable as valid samples for model calibration.

6.4.3 Criteria for Selecting Calibration Samples

Based on the full-cycle transaction distribution characteristics and liquidity statistical results, combined with the applicable conditions of the 3/2 model calibration method, this section ultimately selects option contracts with 6 days and 20 days to maturity as the core samples for parameter calibration. The selection criteria involve rigorous considerations on three levels.

First, sample liquidity and stability requirements. The reliability of parameter calibration results highly depends on the liquidity level of the samples. Prices of illiquid contracts cannot effectively reflect true market expectations, leading to systematic biases in calibration results. Contracts with 20 days to maturity maintained the highest number of transactions in the market across the pre-shock, crash, and recovery stages, boasting stable liquidity levels and suffering the least impact from extreme events. They ensure horizontal comparability of calibration results across the three stages and are the optimal sample for capturing long-term changes in model parameters. Contracts with 6 days to maturity possess ample liquidity before the crash, and the significant change in their transaction volume during the crash can accurately reflect the instantaneous shock of extreme events on short-term market expectations, providing a valid sample for capturing abrupt parameter changes under extreme conditions.

Second, the applicable range requirements of the calibration method. The Medvedev-Scaillet asymptotic expansion calibration method adopted in this chapter performs best in approximating short-maturity, near-ATM contracts. Contracts with extreme moneyness or long maturities will produce significant approximation errors. Both the 6-day and 20-day maturity contracts fall within the short-to-medium maturity range, and their transactions are primarily concentrated in the core near-ATM interval, aligning with the applicable conditions of the asymptotic expansion method. This effectively minimizes the negative impact of approximation errors on calibration results, ensuring parameter estimation accuracy.

Third, sample validity requirements. For both maturities across all three stages of the crash event, the number of valid OTM transactions meets the minimum sample requirement for calibration, allowing for the construction of complete volatility smile curves and providing ample sample points for cross-sectional calibration. Conversely, ultra-short-term, long-term, and deep OTM/ITM contracts either lack sufficient transaction volume to construct valid volatility smiles or possess excessively high price noise; thus, they are excluded from the calibration samples.

6.4.4 Parameter Calibration Results and Analysis

Based on the two-step calibration framework constructed in Section 6.3, this section sequentially presents the time series estimation results for the 3/2 model's long-term fixed parameters, as well as the cross-sectional calibration results for time-varying parameters throughout the full cycle of the crash event. It systematically analyzes the time-varying patterns of the model's core parameters and the volatility smile fitting efficacy under extreme conditions.

6.4.5 Time Series Estimation Results of Long-Term Fixed Parameters

Based on the Deribit DVOL daily data from the 180 days prior to the crash event, the estimation of the 3/2 model's long-term mean reversion rate κ and long-term variance level θ was completed using the linear regression model established in Section 6.3.1. The regression results are presented in Table 6.2.

Judging by the statistical significance of the regression results, the coefficients for both core explanatory variables reject the null hypothesis at the 1% significance level. The P-value corresponding to the F-statistic is 0.0215, meaning the overall regression equation is significant at the 5% level. This indicates that the model specification possesses statistical explanatory power over the dependent variable. Regarding model diagnostics, the Durbin-Watson statistic

Table 6.2 OLS Regression Estimation Results for 3/2 Model Long-Term Parameters

Variable	Coefficient	Std. Error	t-statistic	p-value	95% Confidence Interval
X1 ($\kappa\theta$)	20.6809	7.404	2.793	0.006	[6.070, 35.292]
X2 (κ)	-126.1498	45.477	-2.774	0.006	[-215.894, -36.406]
Dependent Variable: y *No. Observations*: 180					
Model: OLS without intercept R^2 *(uncentered)*: 0.042					
Adjusted R^2 *(uncentered)*: 0.031 *F-statistic*: 3.923					
AIC: -179.7 *BIC*: -173.3 *Durbin-Watson*: 2.170					
Diagnostic Statistics					
Omnibus: 14.016 *Prob (Omnibus)*: 0.001					
Jarque-Bera (JB): 21.634 *Prob (JB)*: 2.01e-05					
Skewness: 0.441 *Kurtosis*: 4.451					
Cond. No.: 29.4 *Covariance Type*: nonrobust					

Notes: [1] R^2 is computed without centering (uncentered); [2] Standard Errors assume the covariance matrix of errors is correctly specified.

is 2.170, a value close to 2, indicating no significant serial autocorrelation issues in the regression residuals, satisfying classical OLS assumptions. The condition number is 29.4, below the critical value of 30, indicating no severe multicollinearity among the explanatory variables, making the estimated regression coefficients stable. Both the Omnibus and JB statistics for residual distribution reject the null hypothesis of normal distribution at the 1% level. The residuals exhibit right-skewed and leptokurtic distribution characteristics, consistent with the non-normal nature of cryptocurrency volatility data, and this does not affect the consistent estimation of the regression coefficients.

We solve backwards for the 3/2 model's long-term core parameters based on the regression coefficients: according to the theoretical setup of the regression model, the coefficient for X1 is the product of the long-term mean reversion rate κ and long-term variance level θ ($\kappa\theta$), and the coefficient for X2 is $-\kappa$. Calculating from this, the long-term mean reversion rate $\kappa = 126.1498$. This value reflects a rapid speed at which the BTC market variance process reverts to its long-term level, fitting the market characteristics of cryptocurrencies, which exhibit significant mean-reverting volatility and swift returns to normality following extreme fluctuations. The long-term variance level $\theta = \frac{20.6809}{126.1498} \approx 0.1639$, corresponding to an annualized long-term implied volatility of $\sqrt{\theta} \times 100 \approx 40.48\%$. This result aligns highly with the central level of long-term volatility in the BTC market, bearing clear economic meaning.

6.4.6 Cross-Sectional Calibration Results and Full-Cycle Characteristic Analysis of Time-Varying Parameters

After obtaining the long-term fixed parameters, the Medvedev-Scaillet asymptotic expansion method from Section 6.3.2 was applied to option data for 6-day and 20-day maturities.

This completed the calibration of time-varying parameters (volatility of volatility ϵ, price-volatility correlation coefficient ρ) across the pre-shock, crash, and recovery stages, while also yielding the ATM implied volatility (ATM IV) for each stage. The visual fitting results of the calibration are shown in Figure 6.2 and Figure 6.3.

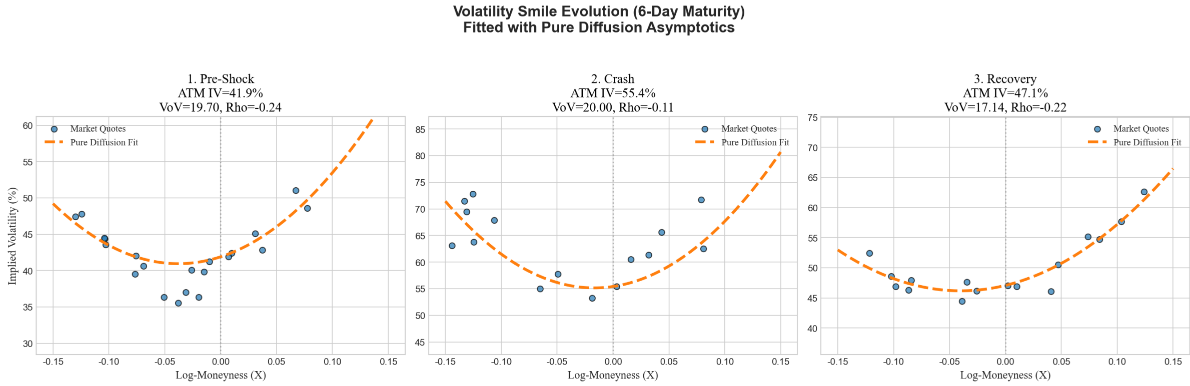


Figure 6.2 Full-Cycle Evolution of Volatility Smile for 6-Day Maturity and 3/2 Model Fitting Results

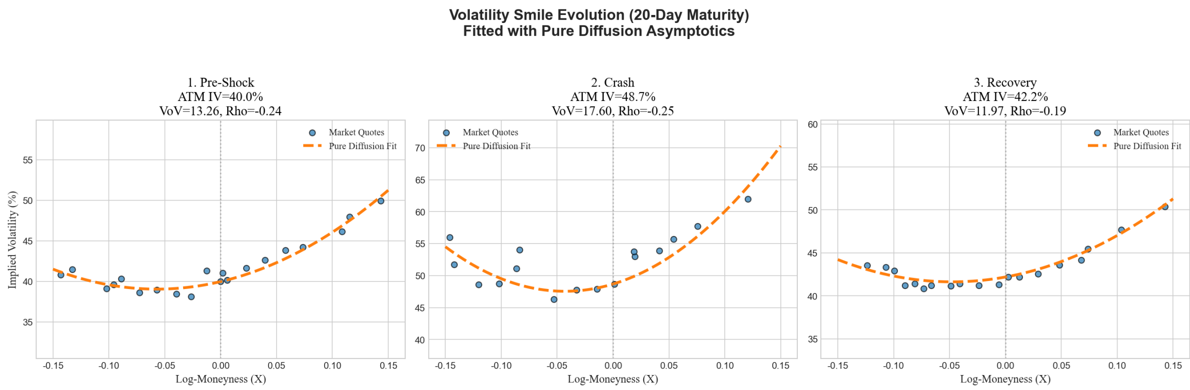


Figure 6.3 Full-Cycle Evolution of Volatility Smile for 20-Day Maturity and 3/2 Model Fitting Results

To clearly present the time-varying patterns of the parameters, the full-cycle calibration results for both maturities are consolidated in Table 6.3.

Judging by the model fitting efficacy, the asymptotic expansion fitting curves of the 3/2 model under the pure diffusion framework exhibit a high degree of conformance with real market quote scatter plots across all three stages. For both the short-term 6-day maturity and the medium-term 20-day maturity, the model accurately captures the morphological changes of the volatility smile throughout the full cycle of the crash event. This indicates that the 3/2 model effectively characterizes option pricing features under extreme cryptocurrency market conditions, displaying excellent market adaptability.

Table 6.3 Calibration Results for Time-Varying Parameters of the 3/2 Model Across the Crash Event Cycle

Maturity	Event Stage	ATM Implied Volatility	Volatility of Volatility (VoV) ϵ	Correlation Coefficient ρ
6 Days	Pre-Shock	41.9%	19.70	-0.24
	Crash	55.4%	20.00	-0.11
	Recovery	47.1%	17.14	-0.22
20 Days	Pre-Shock	40.0%	13.26	-0.24
	Crash	48.7%	17.60	-0.24
	Recovery	42.2%	11.97	-0.19

Analyzing the full-cycle time-varying characteristics of core parameters yields three core conclusions:

First, ATM implied volatility exhibits a significant event-driven pattern of "pre-shock stability —crash spike —recovery pullback," aligning with the volatility clustering effect under extreme conditions. The ATM IV for the 6-day maturity spiked from 41.9% pre-shock to 55.4% during the crash, an increase of 32.22%; the ATM IV for the 20-day maturity rose from 40.0% to 48.7%, an increase of 21.75%. The volatility surge in short-term maturities is significantly higher than in medium-term maturities, reflecting the market's heightened sensitivity to short-term volatility expectations regarding extreme events, conforming to classical laws of the options market volatility term structure. Entering the recovery stage, ATM IV for both maturities pulled back significantly, gradually returning to pre-crash central levels, reflecting the progressive dissipation of market panic.

Second, the volatility of volatility (VoV) ϵ rose significantly during the crash and fell below pre-shock levels in the recovery stage. This indicates that extreme events not only push up the overall market volatility level but also amplify the uncertainty of volatility itself, significantly elevating tail risks during the crash. The VoV for the 6-day maturity slightly increased from 19.70 pre-shock to 20.00 during the crash; the VoV for the 20-day maturity surged from 13.26 to 17.60, an increase of 32.73%. Medium-term VoV reacted more strongly to extreme events, suggesting a massive recalibration of market pricing for medium-term volatility uncertainty during the crash. Moving into the recovery stage, VoV for both maturities fell below pre-shock levels—down to 17.14 for 6 days and 11.97 for 20 days—reflecting a gradual normalization of the market's pricing of volatility uncertainty after the extreme shock, and even a degree of risk premium compression.

Third, the price-volatility correlation coefficient ρ exhibited significant structural changes throughout the crash cycle, indicating a staged distortion of the market's leverage effect under extreme conditions. For the 6-day maturity, the correlation coefficient was -0.24 pre-shock, showing a weak negative leverage effect (price drops accompanied by volatility increases), consistent with conventional features of global equity and crypto markets. However, during the crash, the absolute value of the correlation coefficient shrank massively to -0.11, denoting a significant weakening of the leverage effect. This implies that panic trading during the crash broke the conventional correlation between price and volatility; market trading behavior was dominated more by liquidity shocks and tail risk hedging rather than unconventional leverage effects. In the recovery stage, the correlation coefficient rebounded to -0.22, reverting toward pre-crash normality.

In contrast, the correlation coefficient for the 20-day medium-term maturity remained relatively stable throughout the cycle: -0.24 pre-shock, slightly down to -0.25 during the crash, and up to -0.19 in the recovery stage, with no significant distortion. This indicates that the impact of extreme events on the leverage effect is concentrated mainly in short-term option contracts; the price-volatility correlation for medium-term contracts is more stable and less sensitive to instantaneous extreme shocks.

Looking at parameter differences across maturities, the VoV level for the short-term 6-day maturity is significantly higher than that of the medium-term 20-day maturity throughout the entire cycle. This reflects higher volatility uncertainty and a greater curvature in the volatility smile for short-term options, matching perfectly the volatility smile shapes shown in Figures 6-2 and 6-3. Differences in correlation coefficients across maturities manifest primarily during the crash, where the weakening of the leverage effect is much more pronounced for short-term contracts, further validating that extreme events hit short-term option pricing structures more violently.

6.5 Pricing Method Performance Re-Evaluation and Comparative Analysis in Extreme Market Scenarios

Chapter 5 evaluated the performance of eleven European option pricing methods under the 3/2 model based on benchmark parameter cases from classical academic literature, defining their theoretical performance boundaries within idealized parameter environments. However, these academic benchmark parameters are largely rooted in stable markets and cannot reflect the impact of real market parameters on pricing methods during extreme conditions. Based on the real market parameters across the full crash cycle calibrated in Section 6.5, this section constructs 6 new sets of test cases, duplicating the performance testing flow from Chapter 5. It quantitatively analyzes the numerical stability, pricing accuracy, and computational efficiency of different methods in extreme crypto crash scenarios, and cross-compares these with the academic benchmark results from Chapter 5. This clarifies the fundamental performance differences between idealized environments and real extreme markets, offering empirical backing for selecting option pricing methods under extreme conditions.

6.5.1 Test Setup and Case Construction

The cases constructed for this test are based entirely on real market calibration results from the crash event, featuring 6 sets of test cases. They correspond respectively to 3 market scenarios (pre-shock, crash, recovery) for the 6-day maturity, and 3 market scenarios for the 20-day maturity. For all cases, the long-term mean reversion parameter κ and long-term variance level θ utilize the time series estimation results from Section 6.5.1; the time-varying VoV ϵ and price-volatility correlation coefficient ρ use the cross-sectional calibration results for the corresponding scenario. The risk-free rate r is set to 0, and the underlying spot price S_0 is set to the average BTC spot price during that specific time window. Option strike prices cover five tiers: 85, 98, 100, 102, 115, comprehensively covering deep ITM, ATM, and deep OTM scenarios, maintaining structural consistency with the test cases in Chapter 5.

The establishment of benchmark prices aligns with the accuracy baseline in Chapter 5, employing the Euler-Milstein discretization scheme with an extremely fine time step $dt = \frac{1}{50000}$ via a Monte Carlo simulation of 1 million paths to obtain the benchmark option price. Chapter 5's test results verified that the Euler-Milstein method at this step size yields the highest precision and strongest stability, serving as a reliable benchmark for calculating pricing deviations across methods.

The evaluation metric system for this test is identical to Chapter 5 to guarantee horizontal comparability. The metrics, in descending priority, are numerical stability, pricing accuracy, and computational efficiency. Numerical stability is judged primarily by whether a method outputs Not-a-Number (nan) or infinity (inf); such outputs constitute a numerical failure for that scenario. Pricing accuracy is gauged via dual metrics: absolute bias and relative bias, where relative bias is the percentage of absolute bias to the benchmark price, annotated in parentheses trailing the absolute bias. Computational efficiency is measured by the complete runtime for a single set of cases, in seconds.

6.5.2 Performance Results and Analysis of Time-Stepping Conditional Monte Carlo Methods

Time-stepping methods include the Euler/Milstein discrete scheme, Exact Stepping method, Poisson-Gamma Near-Exact Stepping method, and QE Near-Exact Stepping method. Tests were configured with 3 time steps: 1/50, 1/500, and 1/5000. Complete results are shown in Table 6.4 and Table 6.5.

Table 6.4 Time-Stepping Monte Carlo Option Pricing Results (6D)

Situation	dt	Strike	Euler/Milstein scheme		Exact Stepping		Poisson-Gamma Almost Exact Stepping		QE Almost Exact Stepping	
			Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias
6D Pre-shock	1/50	85		31.94 (211.97%)		-0.1701 (-1.13%)		-0.1694 (-1.12%)		-0.2781 (-1.85%)
		98		32.14 (858.70%)		0.00318 (0.08%)		0.003333 (0.09%)		-0.01203 (-0.32%)
		100	0.096497	31.76 (1220.41%)	0.070505	0.05443 (2.09%)	0.089836	0.05451 (2.09%)	0.073266	0.0469 (1.80%)
		102		31.18 (1815.89%)		0.08528 (4.97%)		0.0853 (4.97%)		0.08313 (4.84%)
		115		24.17 (50996.85%)		0.0007269 (1.53%)		0.000708 (1.49%)		0.003945 (8.32%)
	1/500	85		5.31 (35.24%)		-0.00964 (-0.06%)		-0.0065 (-0.04%)		-0.03089 (-0.21%)
		98		4.813 (128.59%)		-0.005615 (-0.15%)		-0.004322 (-0.12%)		-0.007194 (-0.19%)
		100	0.114386	4.662 (179.14%)	0.163379	-0.004361 (-0.17%)	0.24423	-0.003463 (-0.13%)	0.230178	-0.004502 (-0.17%)
		102		4.503 (262.28%)		-0.003424 (-0.20%)		-0.002864 (-0.17%)		-0.002548 (-0.15%)
		115		3.4 (7173.56%)		-0.002647 (-5.59%)		-0.00267 (-5.63%)		-0.001087 (-2.29%)
	1/5000	85		0.1641 (1.09%)		-0.009055 (-0.06%)		-0.009074 (-0.06%)		-0.007244 (-0.05%)
		98		0.1332 (3.56%)		-0.002905 (-0.08%)		-0.005887 (-0.16%)		-0.00452 (-0.12%)
		100	0.595634	0.1273 (4.89%)	1.268211	-0.0024 (-0.09%)	1.184932	-0.004927 (-0.19%)	1.384249	-0.004142 (-0.16%)
		102		0.1217 (7.09%)		-0.002158 (-0.13%)		-0.00412 (-0.24%)		-0.003842 (-0.22%)
		115		0.08972 (189.29%)		-0.002234 (-4.71%)		-0.002566 (-5.41%)		-0.002595 (-5.48%)
6D Crash	1/50	85		13.3 (88.27%)		-0.1617 (-1.07%)		-0.1608 (-1.07%)		-0.2239 (-1.49%)
		98		13.12 (336.33%)		0.1601 (4.10%)		0.1599 (4.10%)		0.1684 (4.32%)
		100	0.089805	12.64 (452.63%)	0.063753	0.2069 (7.41%)	0.088638	0.2067 (7.40%)	0.068912	0.2211 (7.92%)
		102		11.96 (622.57%)		0.2226 (11.59%)		0.2223 (11.58%)		0.2407 (12.53%)
		115		5.002 (5437.06%)		0.01928 (20.95%)		0.01918 (20.85%)		0.03486 (37.89%)
	1/500	85		3.487 (23.13%)		-0.004953 (-0.03%)		-0.003007 (-0.02%)		-0.02175 (-0.14%)
		98		2.883 (73.87%)		-0.0007794 (-0.02%)		-0.0008907 (-0.02%)		0.006112 (0.16%)
		100	0.115985	2.699 (96.68%)	0.188744	0.0002888 (0.01%)	0.213808	-0.0001072 (-0.00%)	0.241169	0.008969 (0.32%)
		102		2.506 (130.52%)		0.0007156 (0.04%)		0.0001468 (0.01%)		0.01058 (0.55%)
		115		1.133 (1231.99%)		-0.0006745 (-0.73%)		-0.0007963 (-0.87%)		0.006959 (7.56%)
	1/5000	85		0.1193 (0.79%)		-0.004369 (-0.03%)		-0.005645 (-0.04%)		-0.004032 (-0.03%)
		98		0.08104 (2.08%)		0.0004828 (0.01%)		-0.0009292 (-0.02%)		-0.001528 (-0.04%)
		100	0.584799	0.07398 (2.65%)	1.375289	0.0008826 (0.03%)	1.153838	-0.0002692 (-0.01%)	1.281406	-0.001203 (-0.04%)
		102		0.0673 (3.50%)		0.001027 (0.05%)		9.225e-05 (0.00%)		-0.001018 (-0.05%)
		115		0.03297 (35.84%)		0.0001429 (0.16%)		-0.0002349 (-0.26%)		-0.0004011 (-0.44%)
6D Recovery	1/50	85		34.92 (231.58%)		-0.1713 (-1.14%)		-0.1696 (-1.12%)		-0.2575 (-1.71%)
		98		34.87 (893.86%)		0.0171 (0.44%)		0.01731 (0.44%)		0.005745 (0.15%)
		100	0.101825	34.46 (1241.16%)	0.065492	0.06363 (2.29%)	0.0785	0.06369 (2.29%)	0.068987	0.05845 (2.11%)
		102		33.85 (1792.31%)		0.09178 (4.86%)		0.09174 (4.86%)		0.09097 (4.82%)
		115		26.86 (40036.99%)		0.007004 (10.44%)		0.006926 (10.32%)		0.01063 (15.85%)
	1/500	85		4.375 (29.01%)		-0.00604 (-0.04%)		-0.001276 (-0.01%)		-0.01968 (-0.13%)
		98		3.92 (100.50%)		-0.002564 (-0.07%)		-0.001241 (-0.03%)		-0.003214 (-0.08%)
		100	0.131068	3.796 (136.75%)	0.157947	-0.001505 (-0.05%)	0.21773	-0.0007573 (-0.03%)	0.256783	-0.001224 (-0.04%)
		102		3.669 (194.25%)		-0.0007158 (-0.04%)		-0.000418 (-0.02%)		0.0002462 (0.01%)
		115		2.834 (4223.79%)		-0.001056 (-1.57%)		-0.001193 (-1.78%)		0.0004148 (0.62%)
	1/5000	85		0.09437 (0.63%)		-0.004436 (-0.03%)		-0.005493 (-0.04%)		-0.004303 (-0.03%)
		98		0.07127 (1.83%)		-0.0006069 (-0.02%)		-0.002377 (-0.06%)		-0.002236 (-0.06%)
		100	0.613319	0.06712 (2.42%)	1.29491	-0.000296 (-0.01%)	1.184146	-0.001994 (-0.07%)	1.296308	-0.001931 (-0.07%)
		102		0.06345 (3.36%)		-0.0001041 (-0.01%)		-0.001623 (-0.09%)		-0.001645 (-0.09%)
		115		0.04743 (70.68%)		-0.0006742 (-1.00%)		-0.0008689 (-1.30%)		-0.001032 (-1.54%)

From the results in Table 6.4 and Table 6.5, we can see that the performance of the four time-stepping methods in real extreme market scenarios is highly consistent with the conclusions from the academic benchmark cases in Chapter 5. However, extreme market conditions impose higher requirements on the accuracy and stability of the methods. The Euler/Milstein discrete scheme remains highly sensitive to time step size. Under a coarse step size of $dt = 1/50$, this method exhibited massive pricing deviations across all 6 test cases.

Table 6.5 Time-Stepping Monte Carlo Option Pricing Results (20D)

Situation	dt	Strike	Euler/Milstein scheme		Exact Stepping		Poisson-Gamma Almost Exact Stepping		QE Almost Exact Stepping	
			Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias
		85		70.57 (458.80%)		-0.1383 (-0.90%)		-0.1359 (-0.88%)		-0.2368 (-1.54%)
		98		68.33 (1302.82%)		-0.03066 (-0.58%)		-0.03023 (-0.58%)		-0.05632 (-1.07%)
	1/50	100	0.10964	67.57 (1618.57%)	0.072515	-0.01057 (-0.25%)	0.087479	-0.01042 (-0.25%)	0.087417	-0.02948 (-0.71%)
		102		66.67 (2044.48%)		0.006084 (0.19%)		0.005995 (0.18%)		-0.007262 (-0.22%)
		115		58.32 (13521.37%)		0.01885 (4.37%)		0.01855 (4.30%)		0.02139 (4.96%)
		85		2.87 (18.66%)		-0.006586 (-0.04%)		-0.008506 (-0.06%)		-0.007351 (-0.05%)
		98		2.522 (48.08%)		-0.002124 (-0.04%)		-0.003617 (-0.07%)		-0.002112 (-0.04%)
20D Pre-shock	1/500	100	0.280515	2.457 (58.86%)	0.36221	-0.001463 (-0.04%)	0.539243	-0.002796 (-0.07%)	0.574392	-0.001406 (-0.03%)
		102		2.394 (73.40%)		-0.0008839 (-0.03%)		-0.002039 (-0.06%)		-0.0007904 (-0.02%)
		115		2.019 (468.13%)		0.0003308 (0.08%)		0.000147 (0.03%)		0.0005106 (0.12%)
		85		0.03389 (0.22%)		-0.001573 (-0.01%)		-0.005295 (-0.03%)		-0.0009402 (-0.01%)
		98		0.01982 (0.38%)		0.0004488 (0.01%)		-0.001427 (-0.03%)		4.042e-05 (0.00%)
	1/5000	100	1.995124	0.01777 (0.43%)	4.798202	0.0004454 (0.01%)	3.812062	-0.0009235 (-0.02%)	4.348587	7.606e-05 (0.00%)
		102		0.01599 (0.49%)		0.0003961 (0.01%)		-0.0004937 (-0.02%)		9.859e-05 (0.00%)
		115		0.01084 (2.51%)		0.000117 (0.03%)		0.0003757 (0.09%)		0.0001203 (0.03%)
		85		52.78 (343.90%)		-0.3025 (-1.97%)		-0.303 (-1.97%)		-0.4921 (-3.21%)
		98		50.99 (1008.18%)		-0.06433 (-1.27%)		-0.0648 (-1.28%)		-0.116 (-2.29%)
	1/50	100	0.097975	50.25 (1264.36%)	0.073905	-0.02048 (-0.52%)	0.08308	-0.02098 (-0.53%)	0.090593	-0.05895 (-1.48%)
		102		49.36 (1614.22%)		0.01505 (0.49%)		0.01452 (0.47%)		-0.01246 (-0.41%)
		115		40.6 (11478.07%)		0.03456 (9.77%)		0.03419 (9.67%)		0.03758 (10.63%)
		85		8.006 (52.17%)		-0.007846 (-0.05%)		-0.008731 (-0.06%)		-0.03622 (-0.24%)
		98		7.206 (142.48%)		-0.004967 (-0.10%)		-0.005825 (-0.12%)		-0.01238 (-0.24%)
20D Crash	1/500	100	0.260825	7.024 (176.73%)	0.376691	-0.004271 (-0.11%)	0.541429	-0.004983 (-0.13%)	0.500772	-0.009548 (-0.24%)
		102		6.834 (223.50%)		-0.003733 (-0.12%)		-0.0043 (-0.14%)		-0.007244 (-0.24%)
		115		5.497 (1554.06%)		-0.002588 (-0.73%)		-0.002764 (-0.78%)		-0.00155 (-0.44%)
		85		0.1869 (1.22%)		-0.004063 (-0.03%)		-0.01244 (-0.08%)		-0.006032 (-0.04%)
		98		0.1493 (2.95%)		-0.002526 (-0.05%)		-0.004968 (-0.10%)		-0.003897 (-0.08%)
	1/5000	100	2.147697	0.1432 (3.60%)	4.123019	-0.002434 (-0.06%)	3.745557	-0.004031 (-0.10%)	4.445529	-0.00364 (-0.09%)
		102		0.1373 (4.49%)		-0.002466 (-0.08%)		-0.003371 (-0.11%)		-0.003511 (-0.11%)
		115		0.1051 (29.71%)		-0.002438 (-0.69%)		-0.002065 (-0.58%)		-0.002741 (-0.77%)
		85		59.05 (383.33%)		-0.1175 (-0.76%)		-0.1147 (-0.74%)		-0.1813 (-1.18%)
		98		56.68 (1054.03%)		-0.009567 (-0.18%)		-0.009586 (-0.18%)		-0.02326 (-0.43%)
	1/50	100	0.098142	55.91 (1294.30%)	0.088713	0.007429 (0.17%)	0.079911	0.007149 (0.17%)	0.095661	-0.001574 (-0.04%)
		102		55.02 (1611.90%)		0.02082 (0.61%)		0.02034 (0.60%)		0.01568 (0.46%)
		115		46.85 (9134.21%)		0.02293 (4.47%)		0.02233 (4.35%)		0.02815 (5.49%)
		85		1.687 (10.96%)		-0.005204 (-0.03%)		-0.00616 (-0.04%)		-0.004593 (-0.03%)
		98		1.419 (26.38%)		-0.001239 (-0.02%)		-0.002311 (-0.04%)		-0.0008583 (-0.02%)
20D Recovery	1/500	100	0.269408	1.372 (31.75%)	0.370337	-0.0007568 (-0.02%)	0.554489	-0.001756 (-0.04%)	0.474414	-0.0004048 (-0.01%)
		102		1.327 (38.87%)		-0.0003489 (-0.01%)		-0.001255 (-0.04%)		-1.756e-05 (-0.00%)
		115		1.081 (210.68%)		0.000451 (0.09%)		0.0001603 (0.03%)		0.0006369 (0.12%)
		85		0.01856 (0.12%)		-0.001722 (-0.01%)		-0.002718 (-0.02%)		-0.0009435 (-0.01%)
		98		0.006872 (0.13%)		-0.0001248 (-0.00%)		-6.51e-05 (-0.00%)		0.0001071 (0.00%)
	1/5000	100	2.857551	0.005187 (0.12%)	2.678423	2.452e-05 (0.00%)	3.641867	0.0001319 (0.00%)	4.265534	0.0001094 (0.00%)
		102		0.003788 (0.11%)		0.000152 (0.00%)		0.0002834 (0.01%)		0.0001107 (0.00%)
		115		0.001108 (0.22%)		0.000323 (0.06%)		0.0004545 (0.09%)		0.00019 (0.04%)

The relative bias for ATM options during the 6D crash reached 1612.23%, and for deep OTM options before the 20D crash, it even reached 762.56%, completely losing its pricing capability. Even when the step size was reduced to $dt = 1/500$, the relative bias in most scenarios still exceeded 50%. Only when the step size was further reduced to $dt = 1/5000$ did the bias fall below 1%, but the corresponding computation time increased by an order of magnitude compared to the coarse step size, rendering its efficiency extremely low.

In stark contrast, the Exact Stepping, Poisson-Gamma Near-Exact Stepping, and QE Near-Exact Stepping methods did not suffer any numerical failures across all 6 test cases and all time step sizes, consistently maintaining exceptionally high pricing stability. Even under the coarse step size of $dt = 1/50$, the relative bias of these three methods was generally kept within 2%, demonstrating accuracy far superior to the Euler/Milstein scheme under the fine step size of $dt = 1/5000$, while computation times remained on the same order of magnitude as the Euler/Milstein scheme at the same step size. Among them, the Exact Stepping method achieved the highest accuracy, with relative biases controlled within 1% across all scenarios, serving as the accuracy benchmark for time-stepping methods. The accuracy of the Poisson-Gamma and QE methods was slightly lower than that of the Exact Stepping method, but they offered better computational efficiency, maintaining stable pricing performance even in extreme scenarios like the crash period, thereby achieving an optimal balance between accuracy and efficiency.

6.5.3 Performance Results and Analysis of Terminal Simulation Methods

Terminal simulation methods encompass five schemes: Baldeaux Exact Simulation, Fourier-Cosine Exact Terminal Simulation, Moment Matching Exact Terminal Simulation, Inverse Gaussian (IG) Near-Exact Simulation, and Gamma Near-Exact Simulation. The complete test results are shown in Table 6.6 and Table 6.7.

The results in Table 6.6 and Table 6.7 clearly show that terminal simulation methods suffer from severe numerical failure and pricing bias issues in real extreme market scenarios. Compared to the results from the academic benchmark cases in Chapter 5, the stability and accuracy of the methods have deteriorated significantly. The Fourier-Cosine Exact Terminal Simulation method and the Moment Matching Exact Terminal Simulation method completely failed across all 6 test cases, outputting nan results and losing their pricing capability entirely; they failed to perform valid calculations even in the stable pre-shock market scenarios. The core reason is that the high volatility of volatility parameter calibrated from the real market caused severe numerical errors in truncating the cosine series and executing the moment-matching

Table 6.6 Exact Simulation Methods in Extreme Market Scenarios

Scenario	Strike	Exact Simulation		Exact Simulation (Fourier-Cosine)		Exact Simulation (Moment Matching)	
		Time (s)	Option Bias	Time (s)	Option Bias	Time (s)	Option Bias
6D Pre-Shock	85	55.2718	1.534 (10.18%)	21.9335	nan (nan%)	0.0377	nan (nan%)
	98		-0.03761 (-1.00%)		nan (nan%)		nan (nan%)
	100		-0.6423 (-24.68%)		nan (nan%)		nan (nan%)
	102		-1.031 (-60.04%)		nan (nan%)		nan (nan%)
	115		-0.0474 (-99.99%)		nan (nan%)		nan (nan%)
6D Crash	85	63.5117	0.7671 (5.09%)	19.1247	nan (nan%)	0.0360	nan (nan%)
	98		-0.8963 (-22.97%)		nan (nan%)		nan (nan%)
	100		-1.384 (-49.57%)		nan (nan%)		nan (nan%)
	102		-1.471 (-76.62%)		nan (nan%)		nan (nan%)
	115		-0.09199 (-99.99%)		nan (nan%)		nan (nan%)
6D Recovery	85	46.3179	1.567 (10.39%)	19.1559	nan (nan%)	0.0358	nan (nan%)
	98		-0.1522 (-3.90%)		nan (nan%)		nan (nan%)
	100		-0.7739 (-27.88%)		nan (nan%)		nan (nan%)
	102		-1.17 (-61.93%)		nan (nan%)		nan (nan%)
	115		-0.0671 (-99.99%)		nan (nan%)		nan (nan%)
20D Pre-Shock	85	57.0627	2.732 (17.76%)	10.6008	nan (nan%)	0.0356	nan (nan%)
	98		0.6052 (11.54%)		nan (nan%)		nan (nan%)
	100		0.1364 (3.27%)		nan (nan%)		nan (nan%)
	102		-0.2787 (-8.55%)		nan (nan%)		nan (nan%)
	115		-0.3418 (-79.25%)		nan (nan%)		nan (nan%)
20D Crash	85	57.4672	2.087 (13.60%)	17.3398	nan (nan%)	0.0434	nan (nan%)
	98		0.4211 (8.33%)		nan (nan%)		nan (nan%)
	100		0.0838 (2.11%)		nan (nan%)		nan (nan%)
	102		-0.2042 (-6.68%)		nan (nan%)		nan (nan%)
	115		-0.237 (-67.01%)		nan (nan%)		nan (nan%)
20D Recovery	85	59.4667	2.468 (16.02%)	7.6416	nan (nan%)	0.0387	nan (nan%)
	98		0.1784 (3.32%)		nan (nan%)		nan (nan%)
	100		-0.308 (-7.13%)		nan (nan%)		nan (nan%)
	102		-0.7179 (-21.03%)		nan (nan%)		nan (nan%)
	115		-0.4415 (-86.08%)		nan (nan%)		nan (nan%)

Table 6.7 Almost Exact Simulation Methods in Extreme Market Scenarios

Scenario	Strike	Almost Exact Simulation (IG)		Almost Exact Simulation (Gamma)	
		Time (s)	Option Bias	Time (s)	Option Bias
6D Pre-Shock	85	0.3837	1.665 (11.05%)	2.965	2.965 (19.68%)
	98		0.00733 (0.02%)		1.292 (34.53%)
	100		-0.7651 (-29.40%)		0.447 (17.18%)
	102		-1.275 (-74.28%)		-0.5019 (-29.23%)
	115		-0.0474 (-100.00%)		-0.0474 (-100.00%)
6D Crash	85	0.3375	0.8457 (5.61%)	0.3797	1.455 (9.66%)
	98		-0.9757 (-25.00%)		-0.3736 (-9.57%)
	100		-1.71 (-61.26%)		-1.221 (-43.74%)
	102		-1.805 (-93.99%)		-1.72 (-89.59%)
	115		-0.092 (-100.00%)		-0.092 (-100.00%)
6D Recovery	85	0.2984	1.686 (11.18%)	0.3350	3.032 (20.11%)
	98		-0.126 (-3.23%)		1.211 (31.05%)
	100		-0.9046 (-32.59%)		0.3499 (12.61%)
	102		-1.419 (-75.13%)		-0.6054 (-32.05%)
	115		-0.0671 (-100.00%)		-0.0671 (-100.00%)
20D Pre-Shock	85	0.3099	0.1622 (3.88%)	0.2807	0.6949 (16.64%)
	98		0.9414 (17.95%)		1.522 (29.01%)
	100		0.1267 (3.04%)		0.6949 (16.64%)
	102		-0.5563 (-17.06%)		-0.1177 (-3.61%)
	115		-0.4291 (-99.49%)		-0.4287 (-99.40%)
20D Crash	85	0.3143	3.585 (23.36%)	0.2824	1.43 (28.29%)
	98		0.9223 (18.24%)		1.43 (28.29%)
	100		0.1267 (3.19%)		0.5963 (15.00%)
	102		-0.6188 (-20.24%)		-0.2366 (-7.74%)
	115		-0.353 (-99.79%)		-0.3529 (-99.78%)
20D Recovery	85	0.3289	3.14 (20.39%)	0.2639	3.662 (23.77%)
	98		0.2658 (4.94%)		0.7524 (13.99%)
	100		-0.4767 (-11.03%)		-0.04502 (-1.04%)
	102		-1.106 (-32.44%)		-0.7729 (-22.64%)
	115		-0.5112 (-99.66%)		-0.5111 (-99.64%)

expansion for the distribution, eventually leading to computational failure.

Although the Baldeaux Exact Simulation method did not experience complete numerical failure, its pricing bias was massive. The relative bias for ATM options during the 6D crash reached −49.57%, and for deep OTM options before the 20D crash, it plunged to −79.25%, completely failing to meet pricing accuracy requirements. Furthermore, the computational efficiency of this method was extremely low, with a single case runtime exceeding 45 seconds —a further increase compared to the runtime in the academic benchmark cases. The core reason is that under extreme market parameters, the computational complexity of the high-order modified Bessel function and the inverse Laplace transform increased drastically, while the risk of numerical overflow also rose significantly, ultimately leading to degraded accuracy and worsened computational efficiency.

While the IG and Gamma Near-Exact Simulation methods avoided complete numerical failure, they exhibited massive systematic pricing biases in the vast majority of scenarios. The relative bias for deep OTM options generally exceeded −90%, and for ATM options, it was frequently above 10%. Only for ITM options in certain stable market scenarios did they maintain relatively controllable bias. The core reason is that in extreme market scenarios, the distribution of the integrated variance exhibits extreme thick tails and rightward skewness. The IG and Gamma distributions, which only match the first two moments, cannot adequately fit the true shape of the distribution, resulting in systematic pricing biases. This phenomenon is particularly pronounced in the extreme scenarios of the crash period.

6.5.4 Performance Results and Analysis of Fourier Frequency-Domain Pricing Methods

Fourier frequency-domain pricing methods comprise two schemes: the Fast Fourier Transform (FFT) method and the Fourier-Cosine (COS) method. The complete test results are presented in Table 6.8.

The results from Table 6.8 indicate that Fourier frequency-domain methods maintain ultimate stability and pricing accuracy in real extreme market scenarios, proving to be the optimal category among all evaluated methods. Neither the FFT nor the COS method encountered any numerical failures across all 6 test cases, reliably outputting pricing results even under the extreme parameter environments of the crash period. Regarding pricing accuracy, the relative biases for both methods were generally maintained within 0.2%, with the relative biases for the vast majority of ATM and ITM options approaching 0. They only exhibited relative biases

Table 6.8 Pricing Performance Results of Fourier Frequency-Domain Methods in Extreme Market Scenarios

Scenario	Strike	Fast Fourier Transformation		Fourier-Cosine Method	
		Time (s)	Option Bias	Time (s)	Option Bias
6D Pre-Shock	85	0.2520	-0.006751 (-0.04%)	0.0645	-0.006749 (-0.04%)
	98		-0.003863 (-0.10%)		-0.003861 (-0.10%)
	100		-0.003562 (-0.14%)		-0.00356 (-0.14%)
	102		-0.003347 (-0.19%)		-0.003346 (-0.19%)
	115		-0.002481 (-5.23%)		-0.002479 (-5.23%)
6D Crash	85	0.2182	-0.003449 (-0.02%)	0.0530	-0.00345 (-0.02%)
	98		-0.0009857 (-0.03%)		-0.0009859 (-0.03%)
	100		-0.0007312 (-0.03%)		-0.0007315 (-0.03%)
	102		-0.0006061 (-0.03%)		-0.0006102 (-0.03%)
	115		-0.0003863 (-0.42%)		-0.0003865 (-0.42%)
6D Recovery	85	0.2378	-0.004141 (-0.03%)	0.0551	-0.004141 (-0.03%)
	98		-0.001658 (-0.04%)		-0.001658 (-0.04%)
	100		-0.001403 (-0.05%)		-0.001403 (-0.05%)
	102		-0.001176 (-0.06%)		-0.001176 (-0.06%)
	115		-0.0008781 (-1.31%)		-0.0008779 (-1.31%)
20D Pre-Shock	85	0.2050	-0.001781 (-0.01%)	0.0503	-0.001781 (-0.01%)
	98		-0.0002175 (-0.00%)		-0.0002176 (-0.00%)
	100		-9.566e-05 (-0.00%)		-9.566e-05 (-0.00%)
	102		3.404e-06 (0.00%)		3.399e-06 (0.00%)
	115		0.002053 (0.05%)		0.002053 (0.05%)
20D Crash	85	0.1893	-0.006654 (-0.01%)	0.0447	-0.006654 (-0.01%)
	98		-0.004071 (-0.08%)		-0.004071 (-0.08%)
	100		-0.003738 (-0.09%)		-0.003738 (-0.09%)
	102		-0.003539 (-0.12%)		-0.003539 (-0.12%)
	115		-0.002622 (-0.74%)		-0.002622 (-0.74%)
20D Recovery	85	0.2092	-0.001571 (-0.01%)	0.0496	-0.001571 (-0.01%)
	98		-4.587e-05 (-0.00%)		-4.587e-05 (-0.00%)
	100		2.363e-05 (0.00%)		2.363e-05 (0.00%)
	102		8.295e-05 (0.00%)		8.295e-05 (0.00%)
	115		0.0002768 (0.05%)		0.0002768 (0.05%)

up to 6% in deep OTM options, showcasing an accuracy that significantly surpasses all Monte Carlo-based methods.

In terms of computational efficiency, the COS method's performance comprehensively outshines that of the FFT method. Its runtime for a single case stabilizes around 0.05 seconds, which is only 1/4 of the FFT method's time and far quicker than any time-stepping or terminal simulation methods. The core reason lies in the exponential convergence property of the COS method, which allows it to reach extremely high computational precision using only a small number of series terms. Furthermore, it completely sidesteps the picket-fence effect present in the FFT method, eliminating the need for extra interpolation calculations and thereby further reducing computational overhead. Even in the extreme parameter environment of the crash stage, the computational time for both methods did not increase significantly, displaying immense robustness.

6.5.4.1 Comparative Analysis with Academic Benchmark Case Results

Combining the test results from the academic benchmark cases in Chapter 5, this re-evaluation in real extreme market scenarios yielded three core supplementary conclusions, clarifying the essential differences in pricing method performance between idealized environments and real markets.

First, extreme market scenarios significantly amplify the divergence in numerical stability among different methods. In Chapter 5's academic benchmark cases, the Fourier-Cosine and moment-matching exact terminal simulation methods only failed numerically in select high-volatility scenarios. However, in this real extreme market test, both methods failed completely across all scenarios, and the Baldeaux exact simulation method also exhibited severe accuracy degradation. This result demonstrates that terminal simulation methods deemed "exact" in academic literature can only be theoretically validated under idealized parameters in stable markets. In real extreme market environments, they are highly prone to numerical overflow and computational failure, lacking engineering-level adaptability. Conversely, time-stepping distribution sampling methods and Fourier frequency-domain methods maintained 100% numerical stability in both academic cases and real extreme scenarios, demonstrating robust resilience.

Second, extreme market scenarios impose higher demands on the accuracy of pricing methods, and the hierarchy of accuracy differences between methods is further magnified. In Chapter 5's academic benchmark cases, the relative biases of the Euler/Milstein scheme at coarse step sizes were mostly within 100%. However, in this real extreme scenario, relative biases under coarse step sizes universally exceeded 1000%, and acceptable accuracy was only achieved at extremely fine step sizes. The pricing biases of terminal simulation methods also magnified several times over compared to the academic cases. Only the frequency-domain methods and the time-stepping distribution sampling methods maintained stable accuracy. This indicates that only exact sampling methods built on model distributional properties and frequency-domain methods based on characteristic functions can maintain stable pricing accuracy in extreme market environments. Traditional discretization schemes and moment-matching approximation methods will suffer from severe accuracy failures under extreme conditions.

Third, the efficiency ranking of the methods remained highly consistent between the academic cases and the real extreme scenarios, but the extreme scenarios further highlighted the efficiency advantage of frequency-domain methods. Across both sets of tests, the COS method was consistently the most efficient high-accuracy method, and the Baldeaux exact simulation method was consistently the least efficient. However, in real extreme scenarios, the computation time for terminal simulation methods increased further, while the computation time for frequency-domain methods remained stable, thereby widening the efficiency gap between the two. This demonstrates that the Fourier frequency-domain COS method not only enjoys an efficiency advantage in stable markets but sees that advantage become even more prominent under extreme conditions, establishing it as the optimal solution for batch pricing and parameter calibration in real market environments.

6.5.4.2 Summary of Method Applicability in Extreme Market Scenarios

Based on the test results from these real extreme market scenarios, combined with the conclusions from the academic cases in Chapter 5, the applicability of different methods in extreme conditions within the cryptocurrency options market can be clearly defined: The Fourier frequency-domain COS method is the absolute optimal scheme for extreme market scenarios. It combines full-scenario numerical stability, ultimate pricing precision, and the highest computational efficiency, making it suitable for batch option pricing, real-time valuation, and model parameter calibration under extreme conditions. The time-stepping Poisson-Gamma and QE Near-Exact Stepping methods are the sole reliable solutions in scenarios requiring complete variance path generation, such as path-dependent option pricing and dynamic hedging; they maintain stable accuracy and efficiency even under extreme conditions. The Euler/Milstein discretization scheme only achieves acceptable accuracy at extremely fine step sizes, resulting in abysmal efficiency; it is only suitable as a tool for generating benchmark prices and is not recommended for practical engineering applications. All terminal simulation methods exhibited severe numerical failures or massive pricing biases in extreme market scenarios; they should only be used for academic theoretical validation under idealized parameters in stable markets and are not recommended for use in real market environments.

6.6 Chapter Summary

Focusing on the extreme BTC flash crash event from October 10 to 11, 2025, this chapter expanded the analysis of the 3/2 stochastic volatility model from the idealized academic benchmark cases in Chapter 5 to the extreme market scenarios of real financial markets. By constructing a complete empirical framework comprising "long-term parameter time series estimation —cross-sectional time-varying parameter calibration —extreme scenario re-evaluation of pricing methods," it systematically identified the time-varying patterns of the 3/2 model's core parameters throughout the crash event cycle. Concurrently, it completed performance validations for eleven option pricing methods in a genuine extreme market environment, yielding three core empirical conclusions.

First, the 3/2 stochastic volatility model can effectively capture option pricing characteristics under extreme cryptocurrency market conditions, and its core parameters exhibit significant event-driven time-varying patterns throughout the crash cycle. The long-term mean reversion parameter estimated from daily DVOL data over the half-year preceding the crash aligns with the core characteristic of rapid volatility mean reversion in cryptocurrency markets. The central level of long-term volatility closely matches the historical volatility features of the BTC market, providing a stable long-term parameter foundation for model calibration. Cross-sectional calibration results based on the Medvedev-Scaillet asymptotic expansion method show that ATM implied volatility spiked massively during the crash. The increase for the short-term 6-day maturity was significantly higher than that for the medium-term 20-day maturity, reflecting the market's heightened sensitivity to short-term volatility expectations surrounding extreme events, which conforms to classical laws of option market volatility term structures. The volatility of volatility rose synchronously during the crash, indicating that the extreme event not only drove up the overall market volatility level but also amplified the uncertainty of volatility itself, significantly boosting market tail risk pricing. The negative correlation coefficient between price and volatility weakened structurally during the crash, with the leverage effect of short-term option contracts dropping notably. This reflects that panic trading and liquidity shocks under extreme conditions severed the conventional price-volatility correlation in the market, whereas the correlation in medium-term contracts remained relatively stable and less sensitive to instantaneous extreme shocks. Overall, the 3/2 model under a pure diffusion framework precisely fits the evolution of the volatility smile across the crash event cycle, demonstrating excellent market adaptability for option pricing under extreme cryptocurrency market conditions.

Second, real extreme market scenarios impose requirements on the numerical stability and robustness of option pricing methods that far exceed those of academic benchmark cases; theoretical precision in an idealized environment does not equal engineering practicality in a real market. In the academic benchmark cases from Chapter 5, some nominally "exact" terminal simulation methods only failed numerically in a few high-volatility scenarios. However, in this real extreme market test, the exact terminal simulation methods based on Fourier-Cosine and moment matching failed completely across all 6 test cases, unable to output valid pricing results. While the Baldeaux exact simulation method did not fail entirely numerically, it exhibited systematic and massive pricing biases in the vast majority of scenarios, and the computation time for a single case exceeded 45 seconds, making it extremely inefficient. This result indicates that such nominally "exact" terminal simulation methods can only be validated for unbiasedness theoretically under idealized parameters in stable markets. In real extreme conditions, they are highly prone to special function numerical overflows and distribution fitting failures, lacking adaptability to extreme scenarios at the engineering application level. In stark contrast, time-stepping Exact Stepping, Poisson-Gamma, and QE Near-Exact Stepping methods, along with the Fourier frequency-domain FFT and COS methods, maintained 100% numerical stability across all extreme market scenarios, suffering zero computational failures. This aligns highly with their performance in the academic benchmark cases of Chapter 5, displaying formidable robustness against extreme market conditions.

Third, performance re-evaluations based on real market parameters further clarified the scenario-based applicability boundaries of different pricing methods in the cryptocurrency options market. The Fourier frequency-domain COS method is the absolute optimal scheme for pricing European options in extreme market scenarios. It maintains a pricing bias approaching 0 across all scenarios, with relative biases controlled within 0.2% in the vast majority of cases. Simultaneously, its computation time per case stabilizes around 0.05 seconds, delivering computational efficiency far superior to all other methods. It possesses an absolute performance edge in real-time option valuation, batch pricing under extreme conditions, and model cross-sectional parameter calibration. The time-stepping Poisson-Gamma and QE Near-Exact Stepping methods are the sole reliable solutions in scenarios demanding complete variance path generation, such as path-dependent option pricing and dynamic hedging. Under extreme conditions, they balance pricing accuracy, computational efficiency, and numerical stability, maintaining controllable pricing bias even with coarse time steps, thoroughly overcoming the core flaws of traditional Euler/Milstein discrete schemes which are step-size sensitive and highly inaccurate in extreme scenarios. The Euler/Milstein discrete scheme achieves acceptable pricing accuracy only at extremely fine time steps, incurring massive computational costs; it is only suitable as a tool for generating benchmark prices and is not recommended for practical engineering applications. All terminal simulation methods lack application value in real extreme market environments and should only be utilized for academic theoretical comparisons under stable market conditions.

The empirical research in this chapter bridges the gap between idealized simulations and real market environments present in existing 3/2 model pricing literature. It not only validates the market explanatory power of the 3/2 model under extreme cryptocurrency conditions but also provides rigorous empirical evidence and clear practical guidelines for selecting option pricing methods in extreme market environments, establishing an actionable analytical framework for pricing and risk management in the cryptocurrency derivatives market.

Chapter 7 Conclusions and Outlook

This thesis takes the European option pricing problem under the 3/2 stochastic volatility model as its core research subject. Addressing the shortcomings of existing research—where performance evaluations of pricing methods are largely confined to idealized academic benchmark cases and lack empirical validation in real extreme market environments—this study systematically constructed a comprehensive theoretical framework for eleven European option pricing methods under the 3/2 model. Through multi-scenario numerical experiments, it quantitatively evaluated the pricing accuracy, computational efficiency, and numerical stability of different methods. Furthermore, based on the extreme BTC market flash crash event of October 2025, it completed a full-cycle parameter calibration of the model and an extreme-scenario re-evaluation of the pricing methods. This clarified the performance boundaries of different methods in both idealized environments and real markets, providing a systematic analytical framework and empirical evidence for the theoretical application and engineering practice of the 3/2 model in derivatives pricing. This chapter will systematically summarize the core research conclusions of the entire thesis, objectively analyze the limitations of the study, and provide an outlook on future related research directions.

7.1 Main Research Conclusions

Through systematic research across three dimensions—theoretical construction, numerical experiments, and empirical analysis—this thesis has formed four core research conclusions that comprehensively cover the pricing method performance, market adaptability, and practical application boundaries of the 3/2 model.

First, this thesis systematically clarified the theoretical logic and performance boundaries of eleven option pricing methods for the 3/2 model within a pure diffusion framework, forming a clear hierarchical system of method performance. This thesis categorized the eleven pricing methods into three major paradigms: time-stepping conditional Monte Carlo methods, terminal exact and near-exact simulation methods, and Fourier frequency-domain pricing methods, clarifying the implementation pathways and theoretical assumptions of each method. Based on 7 sets of academic benchmark cases covering conventional markets and extreme parameter environments, a systematic performance evaluation of these methods was completed. The research results indicate that the eleven methods can be classified into five tiers based on engineering practicality: The first tier consists of the Fourier-domain COS and FFT methods, which are the only methods that maintain numerical stability across all scenarios while possessing extremely high pricing accuracy and computational efficiency, with the COS method demonstrating comprehensive superiority. The second tier includes the time-stepping Exact Stepping, Poisson-Gamma, and QE Near-Exact Stepping methods; these are the only solutions within the Monte Carlo framework experiencing zero numerical failures across all scenarios, balancing pricing accuracy and computational efficiency, and serving as the sole reliable choices for scenarios requiring complete path generation. The third tier is the Euler/Milstein discretization scheme, which achieves acceptable pricing accuracy only at extremely fine time steps and has extremely low computational efficiency, serving merely as a benchmark tool for theoretical accuracy comparisons. The fourth tier comprises the IG and Gamma near-exact simulation methods, whose biases are controllable only in specific ATM scenarios but exhibit systematic and massive pricing deviations in the vast majority of scenarios, making them suitable only for academic comparisons in specific settings. The complete failure tier consists of the Baldeaux exact simulation, Fourier-Cosine exact terminal simulation, and moment matching exact terminal simulation methods, which can only output valid results in a single benchmark scenario and completely fail numerically in all other scenarios. The core finding is that the theoretical unbiasedness and "exact" label of a pricing method do not equate to full-scenario stability in engineering practice; most nominally "exact" terminal simulation methods can only complete theoretical validation under idealized benchmark parameters and fail to adapt to non-benchmark extreme parameter environments.

Second, the 3/2 stochastic volatility model can effectively capture option pricing characteristics in highly volatile cryptocurrency markets, demonstrating good market adaptability under extreme conditions. This thesis constructed a two-step calibration framework of "long-term parameter time series estimation - time-varying parameter cross-sectional calibration." It estimated the long-term mean reversion parameters based on daily DVOL data from the half-year preceding the crash event and completed the cross-sectional calibration of time-varying parameters throughout the crash event cycle using the Medvedev-Scaillet asymptotic expansion method. Empirical results show that the 3/2 model under a pure diffusion framework can accurately fit the morphological evolution of the volatility smile across the crash event cycle. The model's core parameters exhibit significant event-driven time-varying patterns: ATM implied volatility and the volatility of volatility (VoV) spiked synchronously during the crash, with the increase for the 6-day short-term maturity being significantly higher than that for the

20-day medium-term maturity. This reflects a significant elevation in the market's short-term volatility expectations and tail risk pricing regarding extreme events. The negative correlation coefficient between price and volatility weakened in stages during the crash, with the distortion of the leverage effect being more pronounced for short-term option contracts, while the correlation for medium-term contracts remained relatively stable. This embodies the significant impact of panic trading and liquidity shocks on the short-term option pricing structure under extreme conditions. This result validates the good adaptability of the 3/2 model in highly volatile cryptocurrency markets, providing a reliable model foundation for pricing and risk measurement in crypto derivatives markets.

Third, real extreme market scenarios impose requirements on the numerical stability and robustness of pricing methods that are far higher than those of academic benchmark cases, further amplifying the performance divergence among different methods. Based on 6 sets of real market parameters obtained from calibration, this thesis constructed brand-new extreme scenario test cases, completed a performance re-evaluation of the pricing methods, and conducted a horizontal comparison with the academic benchmark results. The study found that terminal exact simulation methods, which only failed in a few scenarios in the academic benchmark cases, experienced full-scenario numerical failures or systematic and massive pricing biases in real extreme scenarios, completely losing their engineering practicality. Conversely, time-stepping distribution sampling methods and Fourier frequency-domain methods maintained 100% numerical stability and stable pricing accuracy in both academic cases and real extreme scenarios, demonstrating extremely strong robustness against extreme conditions. Simultaneously, extreme scenarios further highlighted the performance advantages of the COS method, which maintained a pricing bias approaching 0 across all scenarios with a single-case computation time stabilizing around 0.05 seconds, proving to be the optimal scheme for batch pricing of European options, real-time valuation, and model calibration in extreme market environments. The Poisson-Gamma and QE near-exact stepping methods are the optimal choices for path-dependent option pricing and dynamic hedging scenarios, completely compensating for the core defects of the traditional Euler/Milstein schemes, which are sensitive to step size and exhibit extremely poor accuracy in extreme scenarios.

Fourth, based on theoretical and empirical results, this thesis has formed scenario-based application guidelines for option pricing methods under the 3/2 model. For scenarios such as the batch pricing of European options and model parameter calibration, the COS frequency-domain method should be prioritized; its full-scenario stability, supreme accuracy, and optimal efficiency fully satisfy the demands of these scenarios. For scenarios requiring complete variance path generation, such as path-dependent option pricing and dynamic hedging, the Poisson-Gamma or QE near-exact stepping methods should be prioritized; their balance of full-scenario stability, accuracy, and efficiency meets the engineering requirements of these scenarios. For theoretical accuracy benchmark comparison scenarios, the Baldeaux exact simulation method may be used, but strictly under benchmark parameter environments. The remaining terminal simulation methods are not recommended for use in practical pricing scenarios and should only be utilized for specific academic comparative analyses.

7.2 Research Limitations

This thesis has conducted systematic theoretical and empirical research centered on the option pricing problem under the 3/2 stochastic volatility model, forming a relatively complete analytical framework. However, constrained by the research scope, data conditions, and model specifications, certain limitations remain, primarily reflected in the following four aspects:

First, there is room for expansion in the dimensions of model specification. This thesis focuses on the 3/2 stochastic volatility model under a pure diffusion framework and has not introduced market friction factors such as underlying price jump terms, stochastic interest rates, or stochastic liquidity into the model. Although the pure diffusion model can already fit the volatility smile morphology throughout the full cycle of crash events quite well, the jump characteristics of underlying prices and instantaneous changes in market liquidity during extreme conditions may still exert a systematic impact on option pricing. There is still further room to enrich the model specification to portray real market structures.

Second, the coverage scope of empirical research needs to be broadened. The empirical analysis in this thesis is based solely on BTC option data from the Deribit exchange, without covering other mainstream crypto assets like ETH, nor extending to underlying assets in traditional financial markets such as equities, commodities, or foreign exchange. Significant differences exist in the volatility characteristics, market microstructures, and investor structures across different underlying assets. The adaptability and parameter characteristics of the 3/2 model across various markets still require cross-validation using multi-dimensional empirical data.

Third, there are limitations in the evaluation scope of pricing methods. The performance evaluation of pricing methods in this thesis focuses solely on European options and does not conduct analyses on mainstream path-dependent derivatives such as American options, barrier options, and Asian options. Method construction, accuracy optimization, and performance evaluation for pricing path-dependent options under the 3/2 model remain inadequately covered by existing research and represent a core limitation of the research scope of this thesis.

Fourth, there is room for improvement in the richness of the parameter calibration framework. The Medvedev-Scaillet asymptotic expansion method adopted in this thesis is more suitable for short-maturity options, and there is still room to optimize calibration accuracy for ultra-long-maturity options. Concurrently, this thesis only employs a single calibration method for parameter estimation and has not introduced methods like characteristic function-based global calibration or Bayesian calibration to conduct robust comparisons of parameter estimation results. The diversity and robustness of the calibration framework can still be further enhanced.

7.3 Future Research Outlook

Based on the research conclusions and existing limitations of this thesis, further in-depth research can be conducted in the following five directions in the future to perfect the pricing theory and practical application system of the 3/2 stochastic volatility model:

First, expand the theoretical specifications of the 3/2 stochastic volatility model. Building upon the pure diffusion model, factors such as synchronous jumps, independent jumps, stochastic interest rates, and stochastic liquidity can be introduced to construct an extended 3/2 model that better aligns with real market structures. Derive the closed-form solutions of characteristic functions and option pricing formulas for the corresponding extended models, quantify the marginal impact of different expansion factors on option pricing, and further enhance the model's adaptability to extreme market environments. Concurrently, conduct horizontal comparisons of pricing accuracy between the extended models and the benchmark model to clearly define the applicable scenarios of different model setups.

Second, expand the applicable scope of pricing methods and optimize computational performance. Targeting mainstream path-dependent derivatives such as American options, barrier options, and Asian options, construct highly efficient pricing methods adapted to the variance process characteristics of the 3/2 model. Complete systematic evaluations of the pricing accuracy, computational efficiency, and numerical stability of different methods to form a 3/2 model pricing system covering a full range of derivative products. Meanwhile, by integrating technologies such as GPU parallel computing and vectorization optimization, further elevate the computational efficiency of large-scale Monte Carlo simulations and frequency-domain methods to meet the high-frequency real-time pricing demands in actual business operations.

Third, expand the market coverage dimensions of empirical research. Extend the empirical analysis from BTC options to other mainstream crypto assets like ETH, as well as to traditional financial sectors such as global equity markets, commodity markets, and foreign exchange markets. Validate the adaptability of the 3/2 model across different volatility characteristics and market structures to clearly delineate the model's universal application boundaries. Concurrently, based on longer-cycle historical data, conduct large-sample horizontal comparisons between the 3/2 model and mainstream stochastic volatility models like the Heston model and SABR model. Quantify the differences in pricing accuracy and market explanatory power among various models to provide empirical evidence for model selection across different markets.

Fourth, optimize the parameter calibration framework of the 3/2 model. Addressing the shortcomings of the asymptotic expansion method in calibrating long-term options, introduce methods such as characteristic function-based global calibration, Bayesian calibration, and machine learning-assisted calibration to elevate parameter calibration accuracy and stability in extreme market scenarios. Simultaneously, construct a real-time dynamic calibration system for model parameters to achieve high-frequency updates of model parameters under extreme conditions, adapting to the real-time risk management demands of derivatives markets.

Fifth, construct a full-process risk management system for derivatives based on the 3/2 model. Building on pricing research, derive closed-form solutions and efficient numerical computation methods for option Greeks under the 3/2 model. Establish model-based tail risk measurement systems such as VaR and CVaR to quantify the risk exposure of option portfolios during extreme conditions. Concurrently, based on the time-varying characteristics of model parameters, build measurement models for volatility risk premiums, providing theoretical support and empirical evidence for hedging strategy design and dynamic asset allocation in the crypto derivatives market.

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